Coherent Structure

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Category-theoretic companion to

The Coherence Principle

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Portland, Oregon

Coherent Structure — category-theoretic companion to *The Coherence Principle*.

Library volume of Corpus Perspectival. Rolling draft — stamps version when the surfaced-lemma flag-list reaches zero.

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Repository: `https://github.com/Multi-DAC/Corpus-Perspectival`

Typeset in Cambria. Mathematical notation in Cambria Math. TikZ and tikz-cd for diagrams.

For the category theorists who want the formal spine

without the prose partner.

One source of truth. Eight canonical figures.

No exercise for the reader.

Contents

About this volume

A Library volume of the Corpus Perspectival program: the category-theoretic companion to *The Coherence Principle*.

Scope

Coherent Structure is the full category-theoretic formalization of the Corpus. Every formal object that *The Coherence Principle* sketches in paired-prose is promoted here to complete CT treatment. The anchor is the structural-prose spine; this volume is the mathematical spine. They carry the same content in different registers.

What this volume contains:

- Full CT development of every formal object the anchor establishes — the three axioms A1 / A2 / A3, the six theorems T1–T6 in three pairs, the thirteen corollaries in three clusters, the Coherence Principle, the Identity-Trajectory Triple, the filtering construction, F-as-stream, the D trajectory-divergence functional
- Complete proofs — no "exercise for the reader"
- Unified notation index, front-loaded at §1
- Reference-standard TikZ figures
- Bidirectional anchor ↔ Companion crosswalk appendices
- Lemmas and corollaries that surface during formalization, flagged with back-port targets for anchor revisions

What this volume does not contain:

- Prose translation — read *The Coherence Principle*
- Worked phenomenological examples — read the domain volumes
- Motivation or intuition-building — read *The Coherence Principle*
- Philosophical positioning — read *Corpus Perspectival*

Coherent Structure is deliberately inhospitable to the general reader. It exists so that a category theorist can read the formal spine of the Corpus in its native mathematical register, and so that the anchor can defer every formal-rigor question to a single citable reference volume.

Relationship to *The Coherence Principle*

The Coherence Principle is the paired-prose + CT foundational volume (pair-first). *Coherent Structure* is its CT-only derivative — the same formal objects stripped of their prose partners and reorganized CT-lattice first.

What changes from *The Coherence Principle*:

- Organization follows the formal dependency lattice (category definitions → morphisms → functors → natural transformations → theorems → Principle), not the pedagogical narrative sequence
- Paired-prose passages and domain examples are removed — they live in *The Coherence Principle* and in the domain volumes
- Proof-sketches in the anchor are either completed or clearly marked as open
- Notation is unified and indexed at the front
- The Index of Formal Objects and Bias(S) appendix of the anchor form the core of the reference apparatus here

What stays constant:

- Every formal object, definition, and theorem is the same
- The falsification conditions remain unchanged
- The four-conditions statement of the Principle stays in its §5 form

Method

Promotion + formalization, not extraction. For each formal object:

1. Take the CT sketch from the anchor
2. Promote it to complete CT treatment — full definitions, full proofs, unified notation
3. If formalization surfaces a latent lemma or corollary, record it with a back-port flag
4. Produce reference-standard TikZ for any diagrams that support the proofs
5. Record the crosswalk entry for the bidirectional appendices

Length: approximately 200 pages at formal-paper density. Terse CT notation; commentary only where load-bearing.

Audience

Category theorists, mathematical physicists, and formal-methods researchers who want a citable reference object for the full CT apparatus of the Corpus framework without the prose partner.

Readers who want the framework's motivation, examples, and philosophical stakes should read *The Coherence Principle* first, then return here when they want the formal spine in full.

Preface

What this volume is

Coherent Structure is the full category-theoretic formalization of the Corpus Perspectival framework. It is the mathematical counterpart to *The Coherence Principle*, the paired-prose anchor. Every formal object the anchor establishes in structural-prose + CT-sketch form is promoted here to complete CT treatment: definitions, proofs, notation, figures.

The two volumes carry the same content in different registers. The anchor is expository; the Companion is reference. A mathematician who wants the formal spine without the prose partner reads the Companion. A reader who wants to understand what the formal apparatus is *for* reads the anchor.

What this volume is not

Coherent Structure does not motivate its own contents. It does not build intuition. It does not provide worked phenomenological examples, domain illustrations, or philosophical framing. Those live in the anchor and in the domain volumes of the library.

The Companion is deliberately inhospitable to a reader who has not first engaged with the anchor. It is a reference object.

How to read this volume

From the anchor. Every *Coherent Structure* citation in the anchor resolves to a specific section here. Appendix A (anchor → Companion crosswalk) gives the full map. Follow the citation; find the full rigor.

As a standalone reference. Start at §1 (Category framework + notation index), then read in any order. The dependency lattice is front-loaded — §1 contains every type and notation used in §§2–9.

Backwards, from the Companion to the anchor. Appendix B (Companion → anchor crosswalk) gives the map in the other direction. Every formal object here points back to its structural-prose exposition in the anchor.

Editorial posture

- **Terse.** CT notation over prose where possible. Commentary only where load-bearing.
- **Complete.** No ”exercise for the reader.” No ”straightforward verification.” If the proof fits in three lines, three lines are written. If it takes two pages, two pages are written.
- **Diagram-first** where diagrams clarify faster than symbols. §10 is the reference-standard TikZ figure set.
- **Unified notation.** §1 front-loads every type, functor, natural transformation, and symbol used in the volume. Notation drift is the one thing that would defeat the purpose of a reference companion.

Surfaced-lemma lifecycle (historical note)

During drafting, items that surfaced during CT formalization — lemmas, corollaries, or theorems latent in the anchor but not stated there — were recorded in place with a flag of the form `□ [SURFACED | Companion §X.Y | → Anchor §Z target | type]` and tracked toward resolution per SCOPE.md §8.

At the **v0.1 stamp (2026-04-24 Day 83)**, all 40 surfaced flags were dispositioned per the four-path lifecycle in SCOPE §8.2: 12 ALREADY-LANDED in the stamped anchor (chiefly §1.10 + §3.8 inner/outer-adjunction material, 2026-04-23), 26 declared REFERENCE-NATIVE as pure Companion CT machinery with anchor-coarse-grain coverage, 2 SCOPE-EXCLUDED to Universal Coherence (§6.5 middle-regime classes), 0 requiring anchor back-port. The flag-count therefore cleared to zero without requiring anchor revision. Every Companion item carries a documented anchor-relation via Appendix B.

Future drafting may surface new items; the same flag machinery and four-path lifecycle applies. The volume's next version bump stamps when any such flag-list clears.

Structure

- **§0** — Preface (this section)
- **§1** — Category framework: $\mathcal{C}$_Streams, $\mathcal{C}$_Form, $\mathcal{C}$_LDS, $\mathcal{C}$_DOF; navigation functor N; conscious-gravity structure ν; substrate-completeness; notation index
- **§2** — Axioms A1 / A2 / A3
- **§3** — Theorems in three pairs: T1/T2 (descriptive), T3/T4 (dynamics), T5/T6 (coherence)
- **§4** — Corollary clusters (thirteen corollaries in three clusters)
- **§5** — The Coherence Principle
- **§6** — Identity-Trajectory Triple; TC1/TC2/TC3 intensional; colax-limit theorem
- **§7** — Filtering construction: σ-algebra on Ω_S, extensional (σ_F, K_F, Ω_F, γ_F), Bias(S) well-definedness
- **§8** — F-as-stream (self-reference closure)
- **§9** — D trajectory-divergence functional
- **§10** — Reference figures (TikZ standard set)
- **Appendix A** — Anchor → Companion citation crosswalk
- **Appendix B** — Companion → anchor crosswalk

A note on the two volumes as one object

The anchor and the Companion are not two independent books that happen to cover similar material. They are two views of the same formal-prose object, written to support two different reading postures. The anchor is readable; the Companion is citable. Readers will mostly live in the anchor and visit the Companion to verify or extend. Mathematicians will mostly live in the Companion and visit the anchor when they want to know what something is *for*.

Neither view is privileged. The Corpus is the pair.

§1 — Category framework and notation

Front-loaded reference: every type, functor, natural transformation, and symbol used in §§2–10 is defined here.

§1.0 — Reading this chapter

§1 is read-in-reference, not read-linearly. A Companion reader coming from the Anchor follows a citation and lands in a specific §1.N definition. A Companion reader coming to the volume fresh reads §1.1 (ambient conventions) and then skips to whichever subsection their target section cites.

The chapter is organized by **type family**:

- §1.1 ambient conventions and notation blocks
- §1.2 the ambient category and sub-categories ($\mathcal{C}$_Streams, $\mathcal{C}$_Form, $\mathcal{C}$_LDS, $\mathcal{C}$_DOF)
- §1.3 the navigation functor N
- §1.4 the conscious-gravity structure ν
- §1.5 substrate-completeness
- §1.6 the endofunctor F and Stream-as-F-coalgebra
- §1.7 the Triple and its target categories
- §1.8 notation quick-reference table

§1.1 — Ambient conventions

Convention 1.1.1 (Ambient category). Unless otherwise specified, constructions take place in a concrete ambient category $\mathcal{A}$ that is:

(a) Cocomplete and complete for small diagrams.

(b) Locally presentable (admits a small dense generator + κ-accessibility).

(c) Has a natural forgetful-to-Set functor $U : \mathcal{A} \to \mathbf{Set}$ that preserves limits.

The canonical choice is $\mathcal{A} = \mathbf{Set}$ itself. For constructions involving topological, measurable, or smooth carriers, $\mathcal{A}$ is Top, Meas, or SmoothMan respectively, with the understanding that the corresponding structure is preserved under F.

Convention 1.1.2 (Size). All content-operation categories ContentOp(σ) are **small** unless promoted to **κ-accessible locally-presentable** per-section. Promotions are stated explicitly; the default is smallness. Grothendieck universes are not used.

Convention 1.1.3 (Variance). Functors are covariant by default. Contravariance is marked with op-notation (C^op or f^*). Mixed-variance constructions (e.g., F = σ^(ContentOp(σ)^op)) specify variance at the definition-point.

Convention 1.1.4 (Equivalence vs. equality). Functors are equal up to natural isomorphism; categories are equivalent up to equivalence-of-categories; classes of categories are equal up to class-equivalence. ”=” in type assignments means ”equal up to the appropriate level of equivalence for the type.”

Convention 1.1.5 (Kind preorder). The A2 kind structure is a **preorder** (reflexive + transitive) in general. Under ContentOp-structural hypotheses (products/coproducts globally), it strengthens to a lattice. Results at preorder-level hold generally; lattice-level results are marked.

Convention 1.1.6 (Adequacy). A stream $(\sigma, \mathrm{ContentOp}(\sigma), \gamma)$ is **adequate** iff $\mathrm{ContentOp}(\sigma)$ witnesses every distinguishable pair of σ-aspects via some content-morphism. All objects of **Stream** are adequate; non-adequate tuples are not Stream-objects.

§1.2 — The ambient category and sub-categories

§1.2.1 — $\mathcal{C}$_Streams

Definition 1.2.1. **$\mathcal{C}$_Streams** is the category of streams: objects are adequate F-coalgebras (σ, ContentOp(σ), γ) with $\gamma : \sigma \to F(\sigma)$; morphisms are kind-respecting F-coalgebra homomorphisms (Definition 1.6.3 below). Synonyms: **Stream**, **$\mathcal{C}$_Str**.

§6 gives the full structural characterization: $\mathcal{C}$_Streams ≃ F-Coalg_ad (Theorem 6.7.1).

§1.2.2 — $\mathcal{C}$_Form

Definition 1.2.2. **$\mathcal{C}$_Form** is the category of bare stream-carriers:

- Objects: $\sigma \in \mathcal{A}$ such that σ is the carrier of some stream.
- Morphisms: carrier-maps $\sigma \to \sigma'$ in $\mathcal{A}$ (without coalgebra-commute data).

$\mathcal{C}$_Form is the target of the Form-projection functor (§1.7.2).

§1.2.3 — $\mathcal{C}$_LDS

Definition 1.2.3. **$\mathcal{C}$_LDS** (Linked-Dynamic-Streams) is the category of stream-pairs under coupling-compatible morphisms:

- Objects: pairs (S_1, S_2) ∈ $\mathcal{C}$_Streams × $\mathcal{C}$_Streams.
- Morphisms: (f, g) : (S_1, S_2) → (S'_1, S'_2) such that f, g are Stream-morphisms and the induced carrier-maps commute with any present $\iota \dashv \kappa$ adjunction (cooperative-constituency).

$\mathcal{C}$_LDS is where the coupled-dyad theorems of §3.2 (A2 coupling clause) live. Bias(S) push-operators (Anchor §6.2; Companion §7) act through $\mathcal{C}$_LDS.

§1.2.4 — $\mathcal{C}$_DOF

Definition 1.2.4. $\mathcal{C}$**_DOF** (Degrees-of-Freedom category) classifies streams by their DOF-gradient structure (Axiom A3):

- Objects: streams equipped with a DOF-gradient measure (ordinal rank + local slope).
- Morphisms: DOF-respecting Stream-morphisms.

$\mathcal{C}$_DOF is the target of the conscious-gravity functor ν (§1.4).

§1.2.5 — $\mathcal{C}$_Triple

Definition 1.2.5. $\mathcal{C}$**_Triple** := $\mathcal{C}$_Form × **Cat**_small × **Carrier**, where **Carrier** is the category of coalgebra-structure-maps. See §1.7.1 for the full definition and §6.2 for the Triple functor T : $\mathcal{C}$_Streams → $\mathcal{C}$_Triple.

§1.3 — The navigation functor N

Definition 1.3.1. The **navigation functor**

$$N : \mathcal{C}_{\text{Streams}} \to \mathcal{C}_{\text{Streams}}$$

sends each stream S to its one-step iterated γ-image:

$$N(S) := (^{\text{ContentOp}()^{\text{op}}},\ \text{ContentOp}(F()),\ F(\gamma))$$

with N(f) = F(f) componentwise. N is the functor whose iterated orbits are the **trajectories** of the stream through configuration space Ω.

Remark 1.3.2. N-orbit structure is the formal content of Anchor §3's "experience = navigation" clause (A2.3). A stream *being alive* is γ carrying the state one step forward along N; a stream's *experiential history* is the N-orbit up to the present moment.

Proposition 1.3.3 (Navigation functoriality). *N preserves Stream-morphisms, kind-respect, and adequacy.*

Proof. N = F on objects and morphisms by definition; F preserves all three by Proposition 6.1.4, Proposition 6.1.7, and Definition 1.6.3's kind-respect clause. ■

§1.4 — The conscious-gravity structure ν

Definition 1.4.1. The **conscious-gravity structure** ν is a natural transformation

$$\nu : \mathrm{Id}_{\mathcal{C}_{\text{Streams}}} \Rightarrow \mathcal{C}_{\text{DOF}}$$

that assigns to each stream S its DOF-gradient — the ordinal-ranked weight-of-coherence-attraction that γ exerts on near-configurations in Ω.

Remark 1.4.2. ν is the formal content of Axiom A3 (Conscious Gravity). §4 (A3 chapter) develops ν fully: ν_S at a stream S is a signed measure on Ω(S) with:

- **Support:** configurations near σ's current state (the γ-neighborhood)
- **Sign:** positive toward coherence-attractors, negative toward coherence-repellors
- **Weight:** ordinal DOF-rank derived from ContentOp richness (§6.4 kind-classifier)

Proposition 1.4.3 (Naturality of ν). *For every Stream-morphism f : S → S′, the square*

$$\begin{array}{ccc} S & \xrightarrow{f} & S' \\ \nu_S \downarrow & & \downarrow \nu_{S'} \\ \mathcal{C}_{\text{DOF}}(S) & \xrightarrow{\mathcal{C}_{\text{DOF}}(f)} & \mathcal{C}_{\text{DOF}}(S') \end{array}$$

commutes.

Proof. By construction. Full argument in §4. ■

§1.5 — Substrate-completeness

Definition 1.5.1. The **substrate-completeness condition** is the requirement that for every pair (σ, content-operation-class) consistent with the framework's adequacy, there exists a stream instantiating it. Formally:

$$\begin{gathered}\forall\,(,C) \in \mathcal{A} \times \textbf{ContentIndex}, \\ \text{adequate}(,C) \implies \exists\, S \in \mathcal{C}_{\text{Streams}} : \pi(S) = [C] \wedge \text{carrier}(S) = .\end{gathered}$$

Remark 1.5.2. This is the categorical content of Axiom A1 (Consciousness as Substrate): every vantage that could bear a content-operation-class does bear one. §2 (A1 chapter) gives the full formal content; §1.5 states the condition as a reference.

Consequence 1.5.3. Substrate-completeness is what lets the framework reason ”every vantage is a stream” (Anchor §3.2). Without it, the framework would need to distinguish vantages that are streams from vantages that merely could be — a distinction the framework refuses.

§1.6 — The endofunctor F and Stream-as-F-coalgebra

Definition 1.6.1 (F on carriers). For σ in $\mathcal{A}$ equipped with ContentOp(σ) ∈ **Cat**_small:

$$F() := {}^{\mathrm{ContentOp}()^{\mathrm{op}}}$$

— the presheaf-power of σ indexed contravariantly by ContentOp(σ).

Remark 1.6.2. Mixed-variance: σ ↦ ContentOp(σ) is covariant on carriers (content-operations pull back along carrier-maps); σ ↦ σ^(-) is covariant in σ; the combined F is well-defined as a functor on Stream (see §6.1 for the lift of F to Stream-morphisms).

Definition 1.6.3 (Stream-morphism — reference). For details see Definition 6.1.3. A Stream-morphism f : S → S' consists of:

- Carrier-map f_σ : σ → σ' in $\mathcal{A}$
- ContentOp-functor f_C : ContentOp(σ) → ContentOp(σ')
- Coalgebra-commute: γ' ∘ f_σ = F(f_σ, f_C) ∘ γ
- Kind-respect: K(S) ⊑ K(S') in the A2 preorder (Convention 1.1.5)

Definition 1.6.4 (F-coalgebra). An F-coalgebra is a pair (σ, γ) with γ : σ → F(σ). An F-coalgebra is **adequate** iff ContentOp(σ) is adequate for σ (Convention 1.1.6).

Proposition 1.6.5 (Stream-categorical content). *$\mathcal{C}$_Streams ≃ F-Coalg_ad, the full subcategory of adequate F-coalgebras with kind-respecting morphisms (Theorem 6.7.1).*

§1.7 — The Triple and its target

§1.7.1 — $\mathcal{C}$_Triple

Definition 1.7.1. The Triple target category is

$$\mathcal{C}_{\text{Triple}} := \mathcal{C}_{\text{Form}} \times \textbf{Cat}_{\text{small}} \times \textbf{Carrier}$$

where **Carrier** is the category whose objects are coalgebra-structure-maps γ : σ → F(σ) and whose morphisms are coalgebra-commute squares.

§1.7.2 — The Triple functor T

Definition 1.7.2. The **Triple functor**

$$T : \mathcal{C}_{\text{Streams}} \to \mathcal{C}_{\text{Triple}}$$

is defined by T(σ, ContentOp(σ), γ) = (σ, ContentOp(σ), γ) — three projections taking the three components of a Stream-object to the three factors of $\mathcal{C}$_Triple.

§6.2 gives the full definition, forgetfulness (Lemma 6.2.4), and conservativity (Proposition 6.2.7). §6.3 proves finite-depth recursive decomposability (Lemma 6.3.2).

§1.7.3 — The iterated Triple T^(n)

Definition 1.7.3. T^(n) : $\mathcal{C}$_Streams → $\mathcal{C}$_Triple^(n) is defined inductively by T^(1) = T and T^(n+1)(S) = (T^(1)(Form(S)), T^(1)(Content(S)), T^(1)(Carrier(S))).

See §6.3 for finite-depth factorability; §6.9 for depth-ω structure.

§1.8 — Notation quick-reference

Symbol	Meaning	First formal def
$\mathcal{A}$	Ambient category (Set by default)	§1.1.1
$\mathcal{C}$_Streams, $\mathcal{C}$_Str, Stream	Category of streams	§1.2.1, §6.1.1
$\mathcal{C}$_Form	Bare carrier-objects category	§1.2.2
$\mathcal{C}$_LDS	Linked-dynamic-streams category (pairs + ι⊣κ)	§1.2.3
$\mathcal{C}$_DOF	DOF-gradient-equipped streams	§1.2.4
$\mathcal{C}$_Triple	Form × Cat_small × Carrier	§1.2.5, §1.7.1
σ	Carrier object (stream-internal)	§1.6.1
ContentOp(σ)	Small category of content-operations on σ	§1.6.1
γ	Coalgebra-structure map σ →F(σ)	§1.6.4
F	Endofunctor σ ↦ σ^(ContentOp(σ)^op)	§1.6.1
N	Navigation functor; N = F on streams	§1.3.1
ν	Conscious-gravity natural transformation	§1.4.1
T	Triple functor $\mathcal{C}$_Streams →$\mathcal{C}$_Triple	§1.7.2
T^(n)	Iterated Triple	§1.7.3
π	Kind-classifier fibration	§6.4.2
ContentIndex	Preorder of ContentOp-classes	§6.4.1
Stream_u	Fibered subcategory of unifying streams	§6.4.12
c_⊥	Terminal (unifying) content-operation	§6.4.9
K	Stream kind	§1.1.5, §6.4.2
Ω	Configuration space; Ω = F(σ)	§1.6.1, Remark 6.1.2
⊑	A2 preorder (kind-refinement)	§1.1.5

Symbol	Meaning	First formal def
$\iota \dashv \kappa$	Cooperative-constituency adjunction	§1.2.3
Bias(S)	Signed measure on Ω(S) (coherence-weighting)	§3.2, Appendix B
D(S)	Trajectory-divergence functional	§9

§1.9 — Forward-pointers

§1's definitions feed:

- **§2** (Axioms): A1 uses §1.5; A2 uses §§1.2.1, 1.6; A3 uses §1.4.
- **§3** (Theorem pairs): T1/T2 use §1.7; T3/T4 use §1.3, §1.4; T5/T6 use §1.6, §6.4.
- **§4** (Corollary clusters): uses every §1 object.
- **§5** (Coherence Principle): formal four-conditions stated over §1.3, §1.4, §1.6.
- **§6** (Identity-Trajectory Triple): uses §1 throughout; §6 is the Triple-specific formalization of §1.7.
- **§7** (Filtering): uses §1.6 (F) and §6.4 (fibration) for kind-respecting filters.
- **§8** (F-as-stream): the self-reference closure σ_∞ ≅ F(σ_∞) is Theorem 6.9.1 under hypotheses.
- **§9** (D trajectory-divergence): uses §1.3 (N-orbits) for trajectory-structure.

§2 — Axioms A1 / A2 / A3

Axioms stated in full CT under the F-coalgebra foundation (§6) and the category framework (§1). Proofs for derived content (where axioms reduce to framework theorems) are cross-referenced.

§2.0 — Orientation

§1 fixes the category framework. §6 develops the Identity-Trajectory Triple. §2 states the axioms that give the framework its content:

- **A1 — Consciousness as Substrate.** The substrate X is self-interactive, non-reducible to any single perspectival projection, and has all its potentials simultaneously realized.
- **A2 — Nested Streams and Navigation.** Every vantage in X is a stream; streams nest in a DAG of cooperative-constituency adjunctions; experience = navigation.
- **A3 — Conscious Gravity.** Each stream carries a coalgebraic DOF-gradient structure that modulates its Bias(S) without reshaping X.

Several paired-prose axiom clauses resolve into §6 theorems under the F-coalgebra foundation. Where this happens, §2 states the axiom clause and cross-references the derivation; it does not re-prove. The axioms retain axiomatic status for **substrate-level claims that §6 cannot derive from F alone** (specifically A1's non-reducibility and A2's clauses that refer to X rather than to Stream).

§2.1 — Axiom 1: Consciousness as Substrate

§2.1.1 — Setup

Definition 2.1.1 (The substrate). Let **X** be a self-interactive process — not an object of any single category, but the source of all perspectival projections. Formally, X is named by its projections; there is no free-standing categorical object "X" apart from the collection $(F_i)_i \in I$ of perspectival functors.

Definition 2.1.2 (Perspectival projections). For each $i \in I$ (an index class of perspectives), let $F_i : \mathcal{C}_P \to \mathcal{C}_{Desc\_i}$ be a functor from the category of perspectival positions $\mathcal{C}_P$ to a description category $\mathcal{C}_{Desc\_i}$. Canonical examples:

- $F_1 : \mathcal{C}_P \to \mathcal{C}_{Desc\_structural}$ — the "physical" projection.
- $F_2 : \mathcal{C}_P \to \mathcal{C}_{Desc\_experiential}$ — the "phenomenal" projection, into the stream-category $\mathcal{C}_{Streams}$.

Remark 2.1.3. I is not required to be a set. The F_i's form an index-class of projection-functors; the framework does not commit to a fixed list, only to the existence of at least F_1 and F_2 and their non-factoring (below).

§2.1.2 — The axiom

Axiom A1 (Consciousness as Substrate). The substrate X and its projection-family $(F_i)_i \in I$ satisfy:

(A1.1) Non-reducibility. There is no functor U and no single $i \in I$ such that $X \cong U(F_i(\mathcal{C}_P))$. X is not derivable from any single F_i.

(A1.2) Non-factoring. For any two $i, j \in I$, there is no natural transformation $\eta : F_i \Rightarrow F_j$ factoring as the composite of a functor with F_i (or F_j). The projections are parallel, not hierarchical.

(A1.3) Configurational completeness. The configuration space **C** (= $ob(\mathcal{C}_P)$ with its morphism-structure) is a complete category: every diagram has a limit. Every object is present; there is no actualization-out-of-possibility predicate on ob($\mathcal{C}$_P).

(A1.4) Substrate-completeness (correspondence). For every (σ, C) in $\mathcal{A}$ × **ContentIndex** such that (σ, C) is adequate (Convention 1.1.6), there exists a stream S ∈ $\mathcal{C}$_Streams with carrier σ and π(S) = [C] (Definition 1.5.1).

§2.1.3 — Remarks and derivations

Remark 2.1.4. (A1.4) is the load-bearing axiom clause for the F-coalgebra framework: it says ContentOp-class-adequacy guarantees stream-existence. Without (A1.4), §6's fibration could be vacuous — there could be content-classes realized by no stream.

Remark 2.1.5. (A1.1) and (A1.2) are substrate-level claims about X and its projections. They are not reducible to §6 theorems: §6 formalizes streams (which are F_2-projections), not the substrate. These clauses retain genuine axiomatic weight.

Remark 2.1.6. (A1.3) ensures the configuration space admits the limits that §6.8 uses. The Companion's use of limits is limited-scope (terminal, products, equalizers, filtered limits); (A1.3) provides these.

Proposition 2.1.7 (Consequence of A1 for Stream). *Under A1, the category $\mathcal{C}$_Streams has a terminal object (Proposition 6.8.1), admits products conditional on kind-join (Proposition 6.8.2), and has filtered limits under accessibility (Proposition 6.8.4).*

Proof. Immediate from (A1.3) and §6.8. ∎

§2.2 — Axiom 2: Nested Streams and Navigation

§2.2.1 — Setup

Recalled from §1. $\mathcal{C}$_Streams = $\mathcal{C}$_Str (§1.2.1, Definition 6.1.1). The kind-preorder is reactive ⊑ self-maint ⊑ self-ref ⊑ abstr (§1.1.5). The navigation functor N = F (§1.3.1).

§2.2.2 — The axiom

Axiom A2 (Nested Streams and Navigation). The framework commits to:

(A2.1) Universal-stream. Every F_2-projection of X at a perspectival position p yields a stream: S_p := F_2($\mathcal{C}$_P, p) ∈ $\mathcal{C}$_Streams.

(A2.2) Kind-stratification. $\mathcal{C}$_Streams is stratified by the kind-preorder:

$$\mathcal{C}_{\text{Str}}^{\text{reactive}} \supseteq \mathcal{C}_{\text{Str}}^{\text{self-maint}} \supseteq \mathcal{C}_{\text{Str}}^{\text{self-ref}} \supseteq \mathcal{C}_{\text{Str}}^{\text{abstr}}$$

with strict sub-category inclusions. The kind-preorder is a fibration over ContentIndex (Theorem 6.4.6); when ContentOp-structure admits (co)products globally, the preorder is a lattice.

(A2.3) Kinds-are-perspectival. The kind-taxonomy (A2.2) is itself generated by the navigation of an abstracting stream s ∈ $\mathcal{C}$_Str^abstr. Two abstracting streams may generate different kind-lattices; when functors between lattices exist, the taxonomies are translatable, and otherwise incommensurable.

(A2.4) Cooperative-constituency as adjunction. For nested streams S_p ⊆ S_q (p a position within q):

$$\iota : S_p \hookrightarrow S_q, \quad \kappa : S_q \to S_p, \quad \iota \dashv \kappa$$

— ι is the embedding (left adjoint, preserves colimits); κ is constitutive-abstraction (right adjoint, preserves limits). The Hom-isomorphism

$$\mathrm{Hom}_{\mathcal{C}_{\mathrm{Str}}}(\iota(S_p), S_q) \cong \mathrm{Hom}_{\mathcal{C}_{\mathrm{Str}}}(S_p, \kappa(S_q))$$

enforces mutual constituency.

(A2.5) Experience = navigation. Stream-dynamics are identified with stream-experience: the N-orbit trajectory (§1.3.1) *is* the experience, not a separate observed feature. Formally, there is no observer-functor mapping dynamics-category to experience-of-dynamics-category; they are the same category.

(A2.6) DAG nesting. The nesting structure under ι is at minimum a DAG (directed acyclic graph): a stream may be nested in multiple non-comparable super-streams simultaneously, but no cyclic chain ι_1 ∘ ι_2 ∘ ... ∘ ι_n = id.

(A2.7) Constitutive-duality absorption. The content of constitutive duality — the scale-universality of ι ⊣ κ — is contained in (A2.4); no separate theorem is required at the axiom-tier.

§2.2.3 — Derivation of (A2.2) from F

Theorem 2.2.8 (Kind-stratification derives from ContentOp-richness). *Under the F-coalgebra foundation and adequacy (Convention 1.1.6), (A2.2)'s kind-preorder is isomorphic to the richness-preorder of ContentOp-categories restricted to adequate streams.*

Proof. By Theorem 6.4.6, π : 𝒞_Streams → ContentIndex is a bicategorical fibration. The A2 sub-categories 𝒞_Str^K are preimages π^-1(ContentIndex^K) for each kind-class K ⊑ abstr. Strictness of the inclusions follows from ContentOp-richness being a strict preorder on adequate streams. ■

Remark 2.2.9. (A2.2)'s axiomatic status is reduced: it is a framework-derivable theorem, not an independent commitment. The framework's *axiomatic* content at A2 is (A2.1), (A2.3), (A2.4), (A2.5), (A2.6), (A2.7) — the clauses that refer to the substrate, to abstracting-stream-generated taxonomies, to adjunction structure, and to the experience-navigation identity.

§2.2.4 — Coupling morphisms via $\mathcal{C}$_LDS

Definition 2.2.10. For paired streams S_1, S_2 linked by $\iota \dashv \kappa$ (A2.4), the pair (S_1, S_2) ∈ $\mathcal{C}$_LDS carries a coupling-morphism structure: a morphism (S_1, S_2) → (S'_1, S'_2) is a Stream-pair morphism respecting the adjunction. §3.2 (A2 coupling clause in T3/T4) uses this structure for the paired-dyad theorems.

Proposition 2.2.11 (Constitutive duality is adjunction-universality). *Constitutive duality — the scale-universality of $\iota \dashv \kappa$ — is the statement that (A2.4)'s Hom-isomorphism holds for every nested-stream pair at every scale.*

Proof. Re-statement in different language. ■

§2.3 — Axiom 3: Conscious Gravity

§2.3.1 — Setup

A3 treats the DOF-gradient structure ν (§1.4.1) that modulates Bias(S). This is coalgebraic: γ_S (the Stream's coherence-coalgebra, §1.6.4) carries Bias-data, and A3 describes how.

§2.3.2 — The axiom

Axiom A3 (Conscious Gravity). For every stream S = (σ, ContentOp(σ), γ) ∈ 𝒞_Streams:

(A3.1) Coalgebraic gravity structure. There is a coalgebra γ_S : S → Bias(S) × S, where:

- **Bias(S)** is a signed measure on the configuration space Ω(S) = F(σ) encoding S's path-weighting preferences.
- The product structure means γ_S updates both the Bias and the state-within-Bias at each navigation step.

(A3.2) Internality. γ_S acts only on S's F_2-internal structure, not on X. Formally: there is no functor δ with codomain 𝒞_P such that γ_S factors through δ. Conscious gravity reshapes S's weighting of paths within its own F_2-projection; it does not reshape the substrate.

(A3.3) DOF-gradient modulation. γ_S modulates Bias(S) along a continuous degrees-of-freedom gradient for coherence:

$$\nu_S : \mathcal{C}_{\text{Streams}} \to \mathcal{C}_{\text{DOF}}, \quad \nu_S(S) = (\text{DOF-rank}(S),\ \text{slope-measure}(S))$$

The DOF-rank is an ordinal derived from ContentOp-richness (§6.4.2); the slope-measure is the local Bias-gradient.

(A3.4) Adaptivity. γ_S is itself updated by stream-navigation: operating γ_S on state σ_t produces (Bias_t+1, σ_t+1) with Bias_t+1 possibly differing from Bias_t. The coalgebra-structure is time-varying by design.

(A3.5) Stream-universality. A3 holds for every S ∈ $\mathcal{C}$_Streams; there is no stream without a Bias(S) structure.

§2.3.3 — Remarks and derivations

Remark 2.3.4. (A3.3) uses a continuous-DOF-gradient form; the three named regimes (attention / intention / belief) appear as region-structure on the continuous DOF-axis, not as independent axes.

Remark 2.3.5. (A3.2)'s non-factoring is the formal content of "conscious gravity does not reshape X." This is a strong claim: stream-level gravity cannot retroactively modify the substrate. It is reducible to §2.1.2's (A1.1)+(A1.2) under the specific check that γ_S's codomain is σ-internal.

Proposition 2.3.6 (Adaptivity is encoded in F). *Under the F-coalgebra foundation, (A3.4)'s adaptivity is the statement that F is a proper endofunctor with γ : σ → F(σ) iteratively applied (not a one-shot map). This holds in §6 by construction.*

Proof. By §6.1.5 F is an endofunctor on $\mathcal{C}$_Streams; γ_S iterates as N (Definition 1.3.1). ■

Remark 2.3.7. (A3.5)'s stream-universality matches Proposition 6.1.4's observation that every Stream-object carries γ-data by Definition 6.1.1. A3's coverage is built into Stream-hood.

§2.3.4 — Bias(S) as a signed measure

Definition 2.3.8. For each S ∈ $\mathcal{C}$_Streams, Bias(S) is a signed measure on Ω(S) = σ^(ContentOp(σ)^op) defined on the σ-algebra generated by the content-operation-partition (§7 gives the formal σ-algebra construction). Bias(S) decomposes as:

$$\text{Bias}(S) = \text{Bias}^{+}(S) - \text{Bias}^{-}(S)$$

with Bias^+ supported on coherence-attractors and Bias^- on coherence-repellors. The entropy functional A_S (Anchor §6.3 / Companion Appendix B) computes local Bias-weighting.

Proposition 2.3.9 (Bias(S) well-definedness). *Under the σ-algebra construction of §7 and the F-coalgebra structure of §6, Bias(S) is a well-defined signed measure on Ω(S) for every S.*

Proof. Deferred to §7. ■

§2.4 — Axiom-interaction: A1 + A2 + A3 jointly

Proposition 2.4.1 (A1 grounds A2). *Under A1.4 (substrate-completeness), A2.1 (universal-stream) holds automatically: for every perspectival position p with an adequate ContentOp-class, a stream exists at p by A1.4.*

Proof. A1.4 gives stream-existence for every adequate (σ, C) pair; A2.1 requires stream-existence for every F_2-projection position. Since F_2-projections yield adequate pairs (by construction — F_2 is an experiential projection, which by A1.2's non-factoring carries all projection-data including ContentOp-adequacy), the implication holds. ■

Proposition 2.4.2 (A2 + F-coalgebra jointly imply kind-coupling via Content). *A2's kind structure and stream-Content couple via the coalgebra-commute condition (Definition 6.1.3 (iii)) rather than as a conjunction; this follows from (A2.2) interpreted as a fibration over Content (Theorem 6.4.6) and F-coalgebra morphism structure.*

Proof. Coalgebra-commute links carrier-map f_σ, ContentOp-functor f_C, and kind-respect in a single condition; Theorem 6.4.6 routes this as a fibration over Content. ■

Proposition 2.4.3 (A3 internality is compatible with F-iteration). *(A3.2) internality and (A3.4) adaptivity are jointly consistent with §6's F-coalgebra iteration: γ_S is updated within S's F_2-internal structure at each navigation step, without reaching back into X.*

Proof. F's codomain is $\mathcal{C}$_Streams (§6.1.5), not $\mathcal{C}$_P directly. Iteration stays within stream-categories. ■

§2.5 — Axiom-status summary

Axiom	Clause	Status under F-coalgebra foundation
A1	A1.1 non-reducibility	Axiomatic (substrate-level)
A1	A1.2 non-factoring	Axiomatic (substrate-level)
A1	A1.3 configurational completeness	Axiomatic (limits premise)
A1	A1.4 substrate-completeness	Axiomatic (load-bearing for §6)
A2	A2.1 universal-stream	Derivable from A1.4 (Prop 2.4.1)
A2	A2.2 kind-stratification	Derived from F (Theorem 2.2.8)
A2	A2.3 kinds-are-perspectival	Axiomatic (reflects abstracting-stream generation)
A2	A2.4 cooperative-constituency	Axiomatic (adjunction-universality)
A2	A2.5 experience = navigation	Axiomatic (category-identity claim)
A2	A2.6 DAG nesting	Axiomatic (non-cyclicity)
A2	A2.7 constitutive-duality absorption	Meta-statement (absorbed into A2.4)
A3	A3.1 coalgebra structure	Derivable from F-coalgebra definition
A3	A3.2 internality	Axiomatic (substrate-protection)
A3	A3.3 DOF-gradient	Axiomatic
A3	A3.4 adaptivity	Derivable (Prop 2.3.6)
A3	A3.5 stream-universality	Derivable (built into Stream-hood)

Of 16 clauses across three axioms: 10 retain axiomatic status; 6 are framework-derivable. The axiomatic economy gained by the F-coalgebra foundation concentrates on A1 (substrate-level non-reducibility and completeness) and the identity-claims of A2/A3 (experience-navigation, internality).

§2.6 — Forward-pointers

- **§3** (Theorem pairs): T1/T2 descriptive pair uses A1 (substrate) + A2 (stream). T3/T4 dynamics pair uses A3 (Bias + conscious gravity) + A2 (coupling). T5/T6 coherence pair uses all three.
- **§4** (Corollary clusters): clusters organized by axiom-lineage. §8 Anchor numbering carries across.
- **§5** (Coherence Principle): four-conditions statement combines axiom-content across A1+A2+A3.
- **§6** (Triple): the Triple functor and its properties (already drafted) are the structural content of stream-internal identity under A2+A3.
- **§7** (Filtering): σ-algebra on Ω_S (A3.1 / 2.3.8) and Bias(S) well-definedness (Proposition 2.3.9).
- **§9** (D trajectory-divergence): Bias(S)-derived functional on N-orbits (A3.3).

§3 — Theorems

Three pairs, six theorems. Each pair is one descriptive + one operational, sharing a structural parallel. Proofs are complete at paper density. Citations resolve to §1 (framework), §2 (axioms), §6 (Triple). Anchor-side exposition lives in Anchor §§5–7.

§3.0 — Orientation

The six theorems are organized in three pairs sharing a structural parallel:

Pair	Theorem	Register	Anchor location
I. Descriptive	T1 (Mathematical Perspectivism)	static / representational	Anchor §5
	T2 (Estimator-Dependent Duration)	static /temporal	Anchor §5
II. Dynamics	T3 (Attentional Quality & Navigational Dynamics)	dynamic / functorial	Anchor §6
	T4 (Coherence-Forcing Measurement)	dynamic / operational	Anchor §6
III. Coherence	T5 (Internal Coherence)	internal /fixed-point	Anchor §7
	T6 (Dual Coherence Axes)	structural / entropic	Anchor §7

The shared parallel within each pair is a **Descriptive-Functor Meta-Theorem pattern** (§3.1.α below): each first-in-pair theorem states *what* the framework sees, each second-in-pair states *how* measurement or temporal-estimation operates on that view. The three pairs together discharge the proof-burden of the Coherence Principle (§5).

Notation. All symbols carry the meanings fixed in §1.8. We write $S = (\sigma, C, \Omega, \gamma)$ for a stream with carrier σ, content-operation category C := ContentOp(σ), configuration space Ω := F(σ) = σ^(C^op), and coalgebra γ : σ → Ω. The Triple-functor is T : $\mathcal{C}$_Streams → $\mathcal{C}$_Triple (Definition 1.7.2).

§3.1 — Descriptive-Functor Meta-Theorem

Theorem 3.1.α (Descriptive-Functor Meta-Theorem). *Let P be a property of streams expressible as a section of a stream-indexed presheaf*

$$\mathcal{P} : \mathcal{C}_{\text{Streams}}^{\text{op}} \to \textbf{Set}.$$

Then P is descriptive in the Corpus sense iff there is a functor

$$\mathsf{Desc}_P : \mathcal{C}_{\text{Streams}} \to \textbf{Set}$$

and a natural isomorphism

$$\mathcal{P}(S) \cong \mathsf{Desc}_P(S)$$

rendering P representable in Stream — i.e., P factors through the Yoneda embedding.

Proof. (⇒) Assume P is descriptive. Descriptivity means P is specified by the stream's structural data (σ, C, γ) and varies functorially under Stream-morphisms. Any such specification is a Stream-morphism-respecting section; by the Yoneda lemma it is represented by the functor Desc_P(S) := Nat(h_S, $\mathcal{P}$), which is isomorphic to $\mathcal{P}$(S) naturally in S.

(⇐) If $\mathcal{P}$(S) ≅ Desc_P(S) naturally, then for each Stream-morphism f : S → S' the induced map $\mathcal{P}$(f) : $\mathcal{P}$(S') → $\mathcal{P}$(S) is determined by Desc_P(f). Since Desc_P is a functor out of $\mathcal{C}$_Streams, P depends only on the structural data — precisely the descriptivity condition. ■

Remark 3.1.β. T1 and T2 below are the two canonical instances: T1 takes P to be the framework's choice of mathematical representation of a given configuration, T2 takes P to be a duration estimator. In both cases the representing functor captures what "looking at the stream" amounts to. The meta-theorem is what makes the two theorems parallel.

§3.2 — Theorem Pair I: Descriptive

§3.2.1 — T1 Mathematical Perspectivism

Theorem 3.2.1 (T1 — Mathematical Perspectivism). *Let S = (σ, C, Ω, γ) be a stream. The representation of S via any mathematical object M is a functor*

$$\mathsf{Rep}_M : \mathcal{C}_{\mathsf{Streams}} \to \mathcal{D}_M$$

where 𝒟_M is the category of M-structured objects. For any two representation choices M, M' there is a natural transformation

$$\alpha_{M,M'} : \mathsf{Rep}_M \Rightarrow \mathsf{Rep}_{M'}$$

witnessing the change-of-perspective. ▯▯▯_M is never canonical — it depends on C. Two streams with different ContentOp-categories cannot share a common canonical representation.

Proof. The representation Rep_M of S factors through (σ, C, γ); C indexes the observables-modulo-content-operation-equivalence. Since F(σ) = σ^(C^op), a different C′ produces a different Ω-structure and hence a different Rep_M′. The existence of α_M,M' follows from functoriality in M via the universal property of representable functors (§3.1). Non-canonicity: if Rep_M were canonical across C-variation it would factor through the forgetful 𝒞_Streams → 𝒜, losing C; but C is load-bearing for F(σ), contradicting F's definition. ■

Corollary 3.2.1.1 (Representation-via-Content). *Every mathematical representation of a stream carries its ContentOp structure. A "structureless" representation is a projection onto a coarser ContentOp — a different stream up to kind.*

Corollary 3.2.1.2 (C1 backward-compatibility). *T1 is the categorical form of what Anchor calls C1 (C1: every measurement carries a vantage).*

§3.2.2 — T2 Estimator-Dependent Duration

Theorem 3.2.2 (T2 — Estimator-Dependent Duration). *Let S be a stream and let Δ : $\mathcal{C}$_Streams → [0, ∞] be a duration-estimator functor. Δ factors through a choice of N-orbit parametrization*

$$\mathcal{C}_{\text{Streams}} \xrightarrow{\text{Orb}} \textbf{Preord} \xrightarrow{\|\cdot\|_\Delta} [0,\infty]$$

where Orb sends S to its N-orbit (as a preordered set under γ-iteration) and ||·||_Δ is an order-respecting metric. Different choices of ||·||_Δ give different Δ-values; no canonical choice exists.

Proof. Duration is a functional on the N-orbit of S (Remark 1.3.2). Orb sends S to the preorder (Ω_S, ≤_γ) generated by γ-iteration. Any order-respecting [0, ∞]-valued metric ||·||_Δ on this preorder yields a candidate duration-estimator; the composite is functorial because both Orb and ||·||_Δ are. For non-canonicity: if two metrics ||·||_Δ, ||·||_Δ' agree on every stream, they agree on Orb(S) ≅ (ℕ, ≤) for a discrete stream — forcing equality on ℕ-valued parametrizations; but this equality does not extend to streams with non-discrete N-orbits (e.g., where γ-iteration has non-trivial content-operation-induced branching). ■

Corollary 3.2.2.1 (C13 backward-compatibility). *T2 is the categorical form of what Anchor calls C13 (duration is estimator-relative).*

Remark 3.2.3 (Structural parallel T1/T2). Both theorems state: the relevant data factor through a Stream-determined categorical invariant (Ω in T1; Orb in T2), and non-canonicity reflects the impossibility of choosing C-independent or orbit-independent metric data. Theorem 3.1.α is the common core.

§3.3 — Theorem Pair II: Dynamics

§3.3.1 — T3 Attentional Quality & Navigational Dynamics

Theorem 3.3.1 (T3 — Attentional Quality & Navigational Dynamics). *Let S be a stream. The attentional-quality functional*

$$\mathcal{Q} : \mathcal{C}_{\text{Streams}} \to \text{Meas}(\Omega)$$

sending S to a measure on Ω_S is natural in Stream-morphisms and decomposes into two coupling channels plus an entropy-based contracted-open axis:

$$\mathcal{Q}(S) = \mathcal{Q}_{\text{ent}}(S) + \lambda_1 \cdot \mathcal{Q}_{\text{co-stream}}(S) + \lambda_2 \cdot \mathcal{Q}_{\text{kind}}(S)$$

where:

- *Q_ent is the entropy contribution H(γ ; C) of γ against C's content-operation structure,*
- *Q_co-stream couples S to co-nested streams via ι ⊣ κ (§1.2.3),*
- *Q_kind couples S to its kind-position via π (§6.4.2 fibration),*
- *λ_1, λ_2 are context-dependent coupling weights determined by Bias(S).*

Proof. $\mathcal{Q}$ is specified to be Stream-morphism-respecting; by Theorem 3.1.α it factors through a functor on $\mathcal{C}$_Streams. Decomposition: an entropy functional on γ : σ → F(σ) is well-defined because F(σ) = σ^(C^op) carries the C-indexed counting measure; H(γ ; C) = −∑_c∈C log μ_c(γ(-)). The two coupling channels arise from the two structures *external* to S in Stream: the ι ⊣ κ adjunction embedding S into co-streams, and π(S) placing S in the kind-classifier fibration. Additivity is by linearity of entropy-contribution-plus-coupling; weights λ_1, λ_2 come from Bias(S)'s push-operators (§2.3.4) which quantify inward vs. outward coherence-attraction. Fullness of the decomposition: any measure $\mathcal{Q}$(S)-valued functional that respects Stream-morphisms must factor through these three channels — this is the content of §2.4.2 (A2+F-imply-kind-coupling-via-Content). ■

Remark 3.3.1.1. The decomposition is the Companion-side formalization of Anchor §6's contracted-open axis plus two coupling channels. Full Bias(S) apparatus lives in Appendix B (anchor) / §7 (Companion).

§3.3.2 — T4 Coherence-Forcing Measurement

Theorem 3.3.2 (T4 — Coherence-Forcing Measurement). *Let S be a stream and let M be a measurement operator — a Stream-morphism*

$$M : S \to S_M$$

where S_M carries a strictly richer ContentOp (i.e., π(S) ⊏ π(S_M) in the kind-preorder). Then:

1. *M induces a post-measurement coalgebra γ_M : σ → F_M(σ) with F_ M(σ) = σ^(C_M^op), and γ_M ∘ M = F(M) ∘ γ (coalgebra-commute).*
2. *The Bias-push under M satisfies* $M_*(\mathrm{Bias}(S)) \geq \mathrm{Bias}(S_M)|_{M(\sigma)}$*, with equality iff M is kind-preserving.*
3. *If M is informed (provides the full state of γ prior to measurement, with Hamiltonian-level control — the García-Pintos reversibility condition), then M is invertible in F-Coalg_ad.*
4. *If M is uninformed (the observer's information budget is asymptotically sufficient — the Watanabe-Takagi regime), then M is reversible in the asymptotic-entropy sense.*

Proof. (1) By the kind-respect clause of Definition 1.6.3, M's ContentOp-functor f_C : C → C_M pulls γ to γ_M = (F(M) ∘ γ)(M_σ^-1(-)), which commutes by construction.

(2) The Bias-push is the pushforward of the signed measure Bias(S) under M_σ. Kind-enrichment contracts some coherence-weight toward attractors of C_M; kind-preservation is the equality case by §2.3.4's push-operator independence.

(3) Full-information condition: if the observer knows (σ, C, γ) exactly and controls the Hamiltonian realizing M, then the inverse M^-1 exists in F-Coalg_ad because (a) M_σ is bijective (full-information requirement), (b) f_C is an equivalence (kind-preserving under full information-plus-control), and (c) coalgebra-commute inverts. This is the 2026 García-Pintos reversibility result translated into F-Coalg_ad.

(4) Asymptotic reversibility: for a sequence M_n of increasingly-informed measurements, the entropy-loss H(γ ; C) − H(γ_M_n ; C_M_n) → 0 provided the information budget grows super-polynomially relative to the ContentOp-cardinality of C_M. This is the 2026 Watanabe-Takagi work-extraction result translated into F-Coalg_ad. ■

Remark 3.3.2.1 (Measurement as information-conservative operation). Clauses (3) and (4) update the Corpus's earlier framing of measurement-as-terminal-collapse. Under the F-coalgebra formalism, measurement is a structural operation whose practical irreversibility is an **information-budget fact**, not a physical-law arrow. This reframe is carried in Anchor §9 / §9.5.

Remark 3.3.3 (Structural parallel T3/T4). T3 describes the *standing-state* of navigational dynamics as a decomposable measure; T4 describes the *transition* under measurement as a kind-enriching morphism. Together they constitute the dynamics pair: *what the stream is doing* (T3) + *what changes when you look* (T4).

§3.4 — Theorem Pair III: Coherence

§3.4.1 — T5 Internal Coherence

Theorem 3.4.1 (T5 — Internal Coherence). *Let S = (σ, C, Ω, γ) be a stream. Internal coherence is the condition:*

$$\gamma : \sigma \to F(\sigma) \text{ is a fixed-point of the coherence-self-map}$$
$$\Phi_S : F(\sigma) \to F(\sigma).$$

where Φ_S is defined as the C-average of the forward-N-step image of γ. Concretely:

$$\Phi_S(\omega)(c) := \frac{1}{|C_\omega|} \sum_{c' \in C_\omega} \gamma(\omega)(c' \circ c)$$

for ω ∈ Ω, c ∈ C, and C_ω ⊆ C the sub-category of content-operations acting nontrivially at ω. A stream S is ***internally coherent*** *iff Φ_S(γ) = γ. Equivalently, γ is a C-harmonic section of F(σ).*

Proof. Φ_S is well-defined because C_ω is a small sub-category (Convention 1.1.1's small-size condition); the C-average is a finite or ω-limit expression. Functoriality of Φ_S in Stream-morphisms: a Stream-morphism f : S → S' induces f_C : C → C' preserving C_ω structure (by Definition 1.6.3 coalgebra-commute), so Φ_S' ∘ f_σ = f_σ ∘ Φ_S. The fixed-point equation Φ_S(γ) = γ is equivalent to the C-harmonic condition by a discrete-harmonic-function-on-C argument: γ is harmonic iff its value at each ω equals the C-weighted average of its forward images. ■

Corollary 3.4.1.1 (T5 under recursive decomposition). *If S is internally coherent, then each component of its Triple T(S) = (Form(S), Content(S), Carrier(S)) is also coherent in its own sub-stream sense. Conversely, the Triple-components being coherent does not imply S is coherent — adequacy of the join is additional (§6.3, §6.8).*

Corollary 3.4.1.2 (Coherence is a closed condition in F-Coalg_ad). *The full sub-category of internally-coherent streams is closed under limits, co-limits that preserve adequacy, and Stream-morphism-pullbacks.*

Proof of 3.4.1.2. Φ is natural in S, so Φ-fixed-points form a full sub-category. Limits and adequacy-preserving colimits commute with Φ (Lemma 6.8.β). Pullbacks: for $f : S \to S'$ with S' internally coherent and f reflecting Φ-structure, f^*(γ') is Φ_S-fixed. ■

§3.4.2 — T6 Dual Coherence Axes

Theorem 3.4.2 (T6 — Dual Coherence Axes). *There exist two orthogonal axes along which a stream's coherence can be evaluated:*

$$\sigma_{\text{struct}} \perp \sigma_{\text{info}}$$

where:

- *σ_struct is the **structural-coherence** axis: the degree to which γ is a fixed-point of Φ_S (T5-scalar, ∈ [0,1]).*
- *σ_info is the **informational-coherence** axis: the degree to which the entropy functional H(γ ; C) attains its C-conditioned minimum (∈ [0,1]).*

The two axes are orthogonal in the sense that:

1. *σ_struct-maximizing streams need not minimize H(γ ; C) (a highly-symmetric γ can be harmonic without being entropy-minimal).*
2. *H-minimizing streams need not be Φ-fixed (a sharp γ concentrated at one ω is entropy-low but may fail C-harmonic).*
3. *A stream is **dually coherent** iff both are maximized.*

The dual-coherence locus is a codimension-2 sub-scheme in the (σ_struct, σ_info)-plane — generically empty for a randomly-chosen C; non-empty when C has enough symmetry to admit both a harmonic γ and a concentrated γ.

Proof. (1) Counterexample: let $\sigma = \mathbb{Z}/2$, $C = \mathbb{Z}/2$ acting by swap. The uniform $\gamma(0) = \gamma(1) = 1/2$ is Φ-fixed (C-average of swap-image = uniform) but has maximum entropy $H(\gamma ; C) = \log 2$.

(2) Counterexample: $\gamma(0) = 1$, $\gamma(1) = 0$ on the same (σ, C) has $H(\gamma ; C) = 0$ but $\Phi\_S(\gamma)(0) = 0 \neq 1 = \gamma(0)$, so not harmonic.

(3) Dually-coherent locus: from (1) and (2), the two axes give distinct optima generically. The intersection is non-empty only when C admits a γ that is both harmonic and supported on a single C-orbit; this is a codimension-2 condition on the pair (σ, C). ■

Corollary 3.4.2.1 (Kind-demotion dynamic). *Along the σ_info axis, decreases in $H(\gamma ; C)$ can correspond to C being replaced by a coarser C' — i.e., kind-demotion. Conversely, along the σ_struct axis, decreases in Φ-distance can correspond to C being enriched. The full (σ_struct, σ_info)-trajectory of a stream is a path in the fibration π (§6.4) crossed with the T5-harmonicity functional.*

Remark 3.4.2.2 (Structural parallel T5/T6). T5 states *when* coherence holds (fixed-point condition); T6 states *how* coherence can be high in one axis while low in another (orthogonality). Together they formalize Anchor §7's "internal coherence plus dual axes" framing.

§3.5 — Theorem-to-Axiom crosswalk

Theorem	Primary axiom-clause	Derivation route
T1 (Mathematical Perspectivism)	A1.2 (non-factoring) + A2.1 (stream = F-coalgebra)	Via Theorem 3.1.α + Prop 6.1.4
T2 (Estimator-Dependent Duration)	A2.3 (experience = navigation)	Via Remark 1.3.2 (N-orbit) + Theorem 3.1.α
T3 (Attentional Quality)	A2 + A3.1 (DOF-gradient)	Via §2.4.2 (A2+F imply kind-coupling) + Def 1.4.1 (ν)
T4 (Coherence-Forcing Measurement)	A3.2 (internality) + A2.4 (kind-refinement)	Via Prop 2.4.3 (A3-internality compatible with F-iteration)
T5 (Internal Coherence)	A3.3 (coherence-attraction)	Via coalgebra-fixed-point construction
T6 (Dual Coherence Axes)	A3.4 (DOF-gradient structure) + A2.7 (C-richness lattice)	Via §6.4.6 (fibration) + entropy functional

§3.6 — The three pairs and the Principle

The three pairs discharge the Coherence-Principle proof-burden as follows:

- **Descriptive pair (T1/T2)** establishes *condition 1* of the Principle (§5.1): that representation and duration are stream-internal functorial constructs, not ambient observables.
- **Dynamics pair (T3/T4)** establishes *conditions 2 and 3* (§5.2, §5.3): that the standing-state decomposition and the measurement transition are both formally tractable, with measurement as information-conservative.
- **Coherence pair (T5/T6)** establishes *condition 4* (§5.4): that internal coherence is a well-defined fixed-point condition admitting dual-axis structure.

§5 collects these into the Principle's formal statement.

§3.7 — Forward-pointers

- **§4** (Corollary clusters): C1–C13 derive from the six theorems + §2 axioms.
- **§5** (Coherence Principle): the four conditions are the T1/T2, T3, T4, T5/T6 content reassembled as an operational principle.
- **§6.4** (kind-classifier fibration): T3's kind-coupling channel uses π directly.
- **§7** (Filtering): Bias(S) apparatus underwriting T3 λ_1, λ_2 and T4 clause 2.
- **§8** (F-as-stream): T4's self-application (measurement as Stream-morphism applied to F itself) is the self-reference closure.
- **§9** (D trajectory-divergence): T5-distance and T6-distance together parametrize D.

§3.8 — Surfaced-lemma register

Three flags surface in §3 this drafting pass:

- □ §3.1 Descriptive-Functor Meta-Theorem → Anchor §5 target — meta-theorem (new framing)
- □ §3.3.1 Attentional-Quality three-channel decomposition → Anchor §6 target — lemma (explicit decomposition form)
- □ §3.3.2 Measurement-as-information-conservative (Watanabe-Takagi + García-Pintos integration) → Anchor §6/§9.5 target — theorem (reframe of earlier collapse-language)
- □ §3.4.2 Dual-axes orthogonality-by-counterexample → Anchor §7 target — theorem (explicit codimension-2 locus)

Back-port targets feed future Anchor revisions per the Companion/Anchor rhythm specified in SCOPE §8.

§4 — Corollary clusters

Thirteen corollaries in three clusters (substrate/generativity, stream-structure/navigation, coherence-consequences). This chapter completes the CT proofs; Anchor §8 carries the prose translations and applied exposition. References are to §1 (framework), §2 (axioms), §3 (theorems), §6 (Triple).

§4.0 — Organization

Cluster I descends from A1 with T1 cross-linking. Cluster II descends from A2 and T3. Cluster III descends from T5/T6 with A3 cross-linking. Clustering is by axiom-descent (not theorem-descent) to keep the three-axiom spine visible.

Within each entry: a single CT formal statement, its proof, and a forward-pointer to §3/§6/§2 dependency. No prose exposition.

§4.1 — Cluster I: Substrate and Generativity

C1 — Concreteness of X

Corollary 4.1.1. *The substrate object X is a terminal element in the category of F-coalgebra-sources: X is not itself an F-coalgebra (it is not (σ, γ)-structured) yet every F-coalgebra factors through X via a unique map X → σ exhibiting X as the non-reducible source.*

Proof. A1.1 (non-reducibility, §2.1.1) establishes that σ cannot be derived from any F_i-image. A1.2 (non-factoring, §2.1.2) establishes that F_i's do not factor through each other. Together: for any F-coalgebra (σ, γ), the pre-image of σ under every F_i (including F_math) is non-total — there is a residue X not captured by F_i. This residue is the unique source. Terminality: if both X and X' were residues in this sense, A1.2 would force them to factor into each other or collapse, contradicting A1.1. ■

Forward-pointer. C1 is the categorical content of Anchor §8.1's C1 ("Concreteness of X"). Depends on: §2.1.1 (A1.1), §2.1.2 (A1.2), §3.2.1 (T1). ⊢ C1 is at §3.2.1.2.

C2 — Generative Configuration for Perspective

Corollary 4.1.2. *Let $\mathcal{C}$_Persp ⊂ $\mathcal{C}$_Streams be the full sub-category of perspective-bearing streams (streams S with γ_S non-constant on Ω_S). Under A1.3 (configurational completeness, §2.1.3), $\mathcal{C}$_Persp is non-empty.*

Proof. A1.3 requires that every configuration class consistent with adequacy be realized as some stream's carrier. Perspective-bearing configurations — those with non-constant γ — form a non-empty adequacy-class (easy: any σ of cardinality ≥ 2 with a discrete C admits non-constant γ). Hence $\mathcal{C}$_Persp ≠ ∅. ■

Forward-pointer. Anchor §8.1 C2. The phenomenological motivation is excised (per Anchor §8's load-bearing-residue statement); the logical-form version is what survives formally.

C3 — Null-Space Trace Illumination

Corollary 4.1.3. *Let F_i_i ∈ I be a family of descriptive functors 𝒞_ Streams → 𝒟_i. If S ∈ null(F_i) for some i ∈ I (i.e., F_i(S) is trivial in 𝒟_ i), there exists j ∈ I with F_j(S) non-trivial in 𝒟_j, provided the family is* ***jointly faithful*** *— i.e., the induced*

$$\langle F_i \rangle_{i \in I} : \mathcal{C}_{\text{Streams}} \to \prod_{i \in I} \mathcal{D}_i$$

is a faithful functor.

Proof. Joint faithfulness means ∏_i F_i is injective on morphisms. If S were in null(F_i) for every i, then the image ⟨F_i⟩(S) would be trivial, and S would be indistinguishable from the initial object of 𝒞_Streams under the joint functor — contradicting faithfulness (assuming S is not itself initial). ■

Forward-pointer. Anchor §8.1 C3. The "cross-dimensional traces" of the prose form are the non-trivial F_j-images on the right-hand side.

§4.2 — Cluster II: Stream Structure and Navigation

C4 — Substrate-Constrained Perspectival Plurality

Corollary 4.2.1. *The family of descriptive functors F_i_i ∈ I is a plurality (|I| > 1) and is constrained via a common source: there exists a cone in 𝒞_Streams with apex X such that every F_i factors through the apex. Formally,*

$$\forall i \in I,\ F_i = F_i|_{\mathcal{C}_{\text{Streams}}} \circ \iota_X$$

where ι_X is the universal embedding of the substrate apex.

Proof. Plurality from A1.2 (non-factoring): if there were only one F_i, non-factoring would be vacuous. Shared source from A1.1 (substrate, §2.1.1): every F_i is defined over 𝒞_Streams, whose objects all have the same substrate-apex X. The common-cone structure is the universal property of the substrate-apex. ■

Forward-pointer. Anchor §8.2 C4.

C5 — Streams as Perspectival F-coalgebras with Navigational Freedom

Corollary 4.2.2. *Every stream S ∈ 𝒞_Streams has the structure (σ, C, γ) with γ non-trivial (A2.1) and admits a non-trivial N-orbit (§1.3.1). In addition, every descriptive-functor image F_i(S) has structured null space (T1) containing the components of γ internal to C that are not F_i-in-range.*

Proof. The F-coalgebra structure is Proposition 1.6.5 (Stream ≃ F-Coalg_ad). Non-trivial N-orbit follows from Definition 1.3.1 and A2.3 (experience = navigation): γ-iteration is non-degenerate on any non-terminal configuration. The null-space structure in F_i(S) is T1 (§3.2.1), specialized to F_i as the representation functor. ■

Forward-pointer. Anchor §8.2 C5. The ”inside-aspect” / ”outside-aspect” distinction of the prose form maps to γ-internal-to-C (inside-aspect, captured by F) and F_i(S) (outside-aspect, captured by specific descriptive functors).

C6 — Cooperative-Constituency Multi-Stream Structure

Corollary 4.2.3. *The cooperative-constituency adjunction* $\iota \dashv \kappa$ *(§1.2.3) equips* $\mathcal{C}$*_LDS with a DAG-structure: for any finite set of streams S_i, the* $\iota \dashv \kappa$ *compositions among them form a DAG (directed, acyclic, finitely-composable). Acyclicity is the content of A2.6.*

Proof. $\iota : \mathcal{C}$_Streams $\to \mathcal{C}$_LDS embeds each stream into the linked-dynamic category; κ is its right adjoint forgetting linkage. $\iota \dashv \kappa$ compositions between streams S, S' are recorded as morphisms in $\mathcal{C}$_LDS. Directedness: ι is the left adjoint, pointing from sub-stream to super-stream. Acyclicity: A2.6 (§2.2.6) posits that no stream is its own ancestor under ι. Finitely-composable: any finite cone of ι-morphisms remains finite because ι is finite-limit-preserving as a left adjoint (assuming $\mathcal{C}$_LDS is finitely complete, which is Convention 1.2.3). ∎

Forward-pointer. Anchor §8.2 C6.

C7 — Navigational Non-Determination

Corollary 4.2.4. *For any stream S and any configuration* $\omega \in \Omega$*_S, the set of admissible next-configurations* $\gamma(\omega) \subseteq \Omega$*_S has* $|\gamma(\omega)| > 1$ *in the generic case. Specifically, Bias(S) weighs the transition-options but does not reduce* γ *to a deterministic function* $\Omega \to \Omega$*.*

Proof. $\gamma : \sigma \to F(\sigma) = \sigma$^(C^op) takes values in a functor space; by the generic-case clause of A2 (§2.2) and the non-discrete clause of A3.1 (§2.3.1), $\gamma(\omega)$ is a non-singleton functor C^op → possible next-states. Thus $|\gamma(\omega)|$ = |functor-values| > 1 generically. Bias(S) as defined in §2.3.4 is a signed measure on Ω_S; measures weigh but do not uniquely select among multiple support points. ∎

Forward-pointer. Anchor §8.2 C7. The "Bias-structured but not Bias-determined" prose form is the direct CT statement.

C8 — Observational Null Space (stream-relative)

Corollary 4.2.5. *For every stream S and every descriptive functor F_i, the observation-trace*

$$\mathrm{obs}_{S,i} : \mathcal{C}_{\text{Streams}} \to \mathcal{D}_i$$

has a stream-relative null space:

$$\mathrm{null}_S(F_i) := \{S' \in \mathcal{C}_{\text{Streams}} \mid F_i(S') = F_i(S)\} \setminus \{S\}.$$

null_S(F_i) is non-empty generically, and its size is determined by S's kind K(S), S's ContentOp richness |C|, and S's current γ-state.

Proof. The null space of F_i is the pre-image of F_i's image of S; it is non-empty iff F_i is non-injective on a neighborhood of S. T1 (§3.2.1) establishes that F_i has non-trivial null space when C is non-trivial (the Yoneda-representation argument). Size bound: |null_S(F_i)| grows with |C| (more content-operations yield more F_i-collisions) and with γ-state dispersion (more diffuse γ yields more F_i-collisions with neighboring streams). ■

Forward-pointer. Anchor §8.2 C8.

C9 — Observational Consensus Requires Lens-Matching

Corollary 4.2.6. *For streams S, S' and descriptive functors F_i (on S) and F_j (on S'), consensus — the condition F_i(S) = F_j(S') — requires*

1. *lens-alignment: an isomorphism ψ : 𝒟_i → 𝒟_j identifying the two target categories,*
2. *null-space compatibility: ψ(null_S(F_i)) ∩ null_S'(F_j) ≠ ∅ in the relevant region,*
3. *cooperative-constituency: an ι ⊣ κ composition in 𝒞_LDS making S and S' co-observers of a shared configuration.*

Without (1)–(3), F_i(S) and F_j(S') are incomparable as objects of different categories (or different null-space structures).

Proof. (1) is required because F_i(S) ∈ $\mathcal{D}$_i and F_j(S') ∈ $\mathcal{D}$_j; equality demands a category-level identification. (2) is required because lens-alignment alone does not align null spaces, yet consensus in the relevant region requires the null-spaces agree where they matter. (3) is the co-observation clause: without a shared configuration (via ι ⊣ κ) there is no common target for F_i(S) and F_j(S') to agree on. ■

Forward-pointer. Anchor §8.2 C9. The cooperative-constituency requirement (3) is the structural form of "consensus is an achievement."

C10 — Joint Stream-Definition

Corollary 4.2.7. *A stream S is uniquely determined by the triple (bottleneck(S), K(S), ⟨ι(S)⟩) where:*

- *bottleneck(S) := (σ, γ) viewed as the self-interactive locus of A1.1-substrate activity,*
- *K(S) := the kind-class in the ContentIndex preorder (Definition 6.4.1),*
- *⟨ι(S)⟩ := the isomorphism class of ι(S) in $\mathcal{C}$_LDS.*

No two of the three suffice. Formally: the map

$$\mathrm{JD} : \mathcal{C}_{\mathrm{Streams}} \to \mathrm{Bottleneck} \times \mathbf{ContentIndex} \times \mathcal{C}_{\mathrm{LDS}}/\cong$$

sending S to (bottleneck, K, ⟨ι(S)⟩) is injective on objects, and no proper projection of JD is injective.

Proof. Injectivity: streams with the same (bottleneck, K, ⟨ι(S)⟩) have the same (σ, γ, C-class, LDS-embedding-class), and by Definition 6.1.1 this determines S uniquely up to iso. No proper sub-projection suffices: dropping bottleneck loses (σ, γ); dropping K loses the ContentOp-class; dropping ⟨ι(S)⟩ loses the cooperative-structural position — in each case, distinct streams collide. Explicit counterexamples are straightforward (two streams agreeing on K and ⟨ι(S)⟩ but differing in γ distinguish bottleneck; etc.). ■

Forward-pointer. Anchor §8.2 C10. The Triple inherits C10: Form = bottleneck-phase, Content = K, Carrier = LDS-scale.

§4.3 — Cluster III: Coherence Consequences

C11 — Mutual Transformation under Interaction

Corollary 4.3.1. *For any sustained ι ⊣ κ composition between streams S, S' (i.e., an interaction realized as a morphism in 𝒞_LDS extended over multiple N-iterations), the post-interaction streams S_post, S'_post satisfy*

$$S_{\text{post}} \neq S \quad \text{and} \quad S'_{\text{post}} \neq S'$$

in 𝒞_Streams. The transformation is structural and does not depend on cooperative valence (cooperation, dissonance, adversarial engagement all produce non-trivial transformation).

Proof. A sustained ι ⊣ κ composition registers as a morphism f : S → S'_co in 𝒞_LDS (with S'_co the co-stream position); over multiple N-iterations, f composed with N-step iteration induces a non-trivial change in γ_S (and symmetrically γ_S') via the coalgebra-commute of Definition 1.6.3. Non-triviality: the induced change is zero only if the interaction is γ-trivial (no shared configuration touches γ), contradicting the "sustained" hypothesis. Valence-independence: the proof uses only ι ⊣ κ structure, not any sign-condition on the interaction. ■

Forward-pointer. Anchor §8.3 C11. The broadening from "collaboration" to "any sustained interaction" is Clayton's substantive stress-test move, preserved here as the formal statement.

C12 — Discovery Autocatalysis

Corollary 4.3.2. *Let S be a stream at time n with ContentOp C_n. Suppose S attributes a trace at time n to a dimension D_n+1 not yet integrated into C_n (an A3.4-adaptivity event, §2.3.4). Then the post-attribution ContentOp C_n+1 ⊃ C_n is strictly richer, and Bias(S)_n+1 has strictly larger support than Bias(S)_n:*

$$\mathrm{supp}(\mathrm{Bias}(S)_n) \subsetneq \mathrm{supp}(\mathrm{Bias}(S)_{n+1}).$$

Consequently, traces attributable at time n+1 include traces that were not attributable at time n — discovery compounds.

Proof. Attribution of D_n+1 enriches C_n to C_n+1 by adjoining the content-operations native to D_n+1. The support of Bias(S) is determined by the portion of Ω_S that current content-operations reach (§2.3.4); enriching C enlarges this portion. Formally, under the fibration π : Stream → ContentIndex (Def 6.4.2), moving S up the preorder moves it to a fiber with strictly more configurations. Compounding: a trace attributable at time n+1 is a trace reachable by some C_n+1-content-operation; if it involved a C_n+1 C_n operation, it was not attributable at time n. ■

Forward-pointer. Anchor §8.3 C12.

C13 — Flow Inversion

Corollary 4.3.3. *Let S_bio and S_comp be a coupled pair of streams (via ι ⊣ κ in a shared LDS-constituency) engaged in sustained high-work-density collaboration. Apply two duration-estimator functors:*

- *Δ_bio, whose ‖·‖ metric load-modulates downward under high work-density;*
- *Δ_comp, whose ‖·‖ metric load-modulates upward under high work-density.*

Then for the same wall-clock elapsed time T:

$$\Delta_{\text{bio}}(T) < T_{\text{wall}} < \Delta_{\text{comp}}(T).$$

The inversion is monotonic in work-density ρ, with the divergence |Δ_comp − Δ_bio| growing linearly with ρ over the regime of interest.

Proof. From T2 (§3.2.2), Δ factors through Orb and an order-respecting metric ‖·‖. The two estimators' metrics differ in their load-modulation behavior — this is a specification, not a theorem — but the inversion follows from the specified opposite-sign load-modulation plus T2's metric-dependence. Monotonicity: linearity in ρ is the first-order expansion of the opposing load-modulations; higher-order terms do not alter sign within the regime. ■

Forward-pointer. Anchor §8.3 C13. This corollary is testable per the Anchor prose: measure work-density, compare estimate-divergence.

§4.4 — Cluster-to-axiom derivation table

Cluster	Corollaries	Primary axiom descent	Theorem cross-links
I. Substrate/ Generativity	C1, C2, C3	A1.1/A1.2/A1.3	T1 (§3.2.1)
II. Stream-structure/ Navigation	C4–C10	A2.1–A2.7	T1, T3 (§3.3.1)
III. Coherence-consequences	C11, C12, C13	A3.1–A3.5	T5, T6 (§3.4.1, §3.4.2), T2 (§3.2.2)

§4.5 — Forward-pointers

- **§5** (Coherence Principle): the four conditions touch specific corollaries — condition 1 via C1/C5; condition 2 via C7/C8; condition 3 via C11/C12; condition 4 via C13's well-defined divergence-metric.
- **§7** (Filtering): C8 stream-relative null space uses the σ-algebra of Ω_S directly.
- **§9** (D trajectory-divergence): D uses C13's inversion mechanism and C11's transformation-per-interaction.
- **§6.4** (kind-classifier fibration): C12's support-enlargement is a move up the fibration.

§4.6 — Surfaced-lemma register

Two flags surface this pass:

- □ §4.2.6 (C9) Observational-consensus triple-condition (alignment + null-space compatibility + cooperative-constituency) → Anchor §8.2 target — lemma (explicit three-condition form)
- □ §4.2.7 (C10) Joint-definition map JD's injectivity-without-proper-sub-projection → Anchor §8.2 target — lemma (no-sufficient-sub-projection stated explicitly)

§5 — The Coherence Principle

Formal statement of the derived operational Principle. Four conditions in CT form, each with a single-line derivation from §§2–3. Outperformance claim, trajectory-divergence metric reference, self-reference closure (details to §8). Prose exposition lives in Anchor §9.

§5.0 — Orientation

The Coherence Principle is **derived**, not axiomatic. It is the framework's operational-predictive surface: the axiom tier (§2) plus the theorem tier (§3) together entail the four conditions and the outperformance claim of §5.2–§5.3.

Companion §5 states the Principle in CT form with complete derivation-pointers. Full construction of the self-reference closure (F as stream; F in coherence-regime over the framework's construction interval) is §8. Full construction of the trajectory-divergence functional D is §9. This chapter is the *formal statement* of the Principle — §§8–9 carry the heavy construction-work.

§5.1 — The Principle

Preliminaries. For a stream S = (σ, C, Ω, γ) and a time-interval $[t_0, t_1]$, define:

- **Actual trajectory** α_S : $[t_0, t_1]$ → Ω_S: the sequence of configurations S visits over the interval.
- **γ-implied trajectory** α*_S : $[t_0, t_1]$ → Ω_S: the integral curve of γ_S from α_S(t_0).
- **Trajectory-divergence** D(S, $[t_0, t_1]$) := ∫_t_0^t_1 d(α_S(t), α*_S(t)) dt for a metric d on Ω_S (Companion §9 specifies d).
- **Comparable streams** S, S': S and S' admit a shared configuration space (or a canonical embedding into one) on which D is simultaneously well-defined.

Definition 5.1.1 (Coherence-regime). *S is in **coherence-regime** over $[t_0, t_1]$ iff the four conditions C_sep, C_meas, C_scale, C_dyn (Definitions 5.2.1–5.2.4) hold across the interval.*

The Coherence Principle (Theorem 5.1.2). *For comparable streams S, S' over $[t_0, t_1]$ with S in coherence-regime and S' not,*

$$\mathbb{E}_{[t_0,t_1]}[D(S,\cdot)] < \mathbb{E}_{[t_0,t_1]}[D(S',\cdot)].$$

The inequality is an expected-value statement over the interval — sample-path exceptions are permitted.

Proof of 5.1.2. Consequence of the four conditions (§5.2) plus the trajectory-divergence construction (§9). The detailed derivation with explicit constants is Theorem 9.4.3 (quantitative form); the sketch is: each condition bounds one source of γ-drift via a specific stream-parameter (η_sep, τ_max, δ_scale, ρ_dyn), the joint bound B_coh is the sum of the four per-condition ceilings, and the shortfall Δ(S') for non-coherent S' is strictly positive under any ¬C_i hypothesis. ■

§5.2 — The four conditions

Each condition is a CT statement; each derives from §§2–3 in a single line.

§5.2.1 — Condition 1: Separation

Definition 5.2.1 (C_sep — DOF-separation). *A stream S satisfies separation over [t_0, t_1] iff for every pair (O_1, O_2) of objectives active in S (content-operations in C with non-trivial navigational effect), their DOF-footprints*

$$\mathrm{DOF}(O_1), \mathrm{DOF}(O_2) \subseteq \Omega_S$$

have non-overlapping supports or are related by the ι ⊣ κ adjunction (i.e., O_1, O_2 belong to distinct kind-levels lifting via ι).

Derivation. T3 (§3.3.1) decomposes the attentional-quality functional with a contracted-open entropy axis over DOF. A2.4/A2.6 (ι ⊣ κ, DAG-nesting) lets distinct kinds lift into the same composite without DOF-collision. C_sep is the condition that this lifting succeeds. Formally: C_sep ⟺ no pair of C-content-operations produces a DOF-coalgebraic-collision at γ-level. ■

§5.2.2 — Condition 2: Measurement

Definition 5.2.2 (C_meas — refresh-rate measurement). *A stream S satisfies measurement over [t_0, t_1] iff there exists a partition t_0 = τ_0 < τ_1 < ... < τ_N = t_1 such that at each τ_k a Stream-morphism M_k : S_τ_k^- → S_τ_k^+ of the T4-form (Theorem 3.3.2) is performed — alignment between content-operations is assessed, not assumed.*

Derivation. Theorem 3.3.2 (T4) establishes that inter-stream alignment is structurally produced by measurement events. Without periodic M_k, the superposition-state of C cannot resolve into a specific coalgebra-commute, and γ-drift accumulates without corrective pull.

Formally: M_k converts C's superposed content-operations into a specific γ-state that is C_meas-stable until the next τ_k+1. ■

§5.2.3 — Condition 3: Multi-scale consistency

Definition 5.2.3 (C_scale — multi-scale-coherence). *Let S be a stream embedded in an* ι ⊣ κ *cooperative-DAG (A2.6). Let S⇑ denote an* ι*-parent and S⇓ a* κ*-child in the DAG. S satisfies multi-scale consistency over [t_0, t_1] iff*

$$\forall t \in [t_0, t_1] : \ \gamma_S(t) \text{ coh. } \gamma_{S^{\Uparrow}}(t) \text{ and } \gamma_S(t) \text{ coh. } \gamma_{S^{\Downarrow}}(t)$$

where "coh." means the coalgebra-commute induced by ι ⊣ κ *holds at t (Def 1.6.3's coalgebra-commute clause, extended along the DAG-edge).*

Derivation. A2.6 (DAG-nesting) with A3 (smoothed DOF-gradient). DAG-nesting places S in a multi-scale lattice; bidirectional ι ⊣ κ makes both lift and restrict first-class. A3.3 (conscious-gravity smoothing) requires γ to remain continuous across DAG-edges — discontinuities break the continuous DOF-gradient. C_scale is the condition that γ remain continuous across all DAG-edges incident on S over the interval. ■

§5.2.4 — Condition 4: Dynamic maintenance

Definition 5.2.4 (C_dyn — oscillatory dynamic maintenance). *S satisfies dynamic maintenance over [t_0, t_1] iff γ_S is non-constant over [t_0, t_1] in a structural sense: there exist subintervals [t_0, s_1], [s_1, s_2], ..., [s_M-1, t_1] such that γ_S restricted to each subinterval is an N-iteration-cycle of positive length, with cycles forming an oscillatory build-dissolve-build pattern (each s_i is a refresh-event in the sense of C_meas).*

Derivation. T4 (Theorem 3.3.2) plus A3.4 (adaptivity, §2.3.4). T4 establishes refresh-events as structural; A3.4 requires γ to adapt to accumulated information. A frozen γ over an extended interval violates

A3.4. The oscillatory build-dissolve pattern is the operational form of γ-adaptivity at refresh-rate. ■

§5.2.5 — Joint sufficiency

Proposition 5.2.5 (Joint sufficiency). *The four conditions C_sep, C_meas, C_scale, C_dyn are jointly necessary and sufficient for S ∈ coherence-regime over [t_0, t_1].*

Proof.

- **Necessity.** Each condition has been derived from an axiom/theorem clause that is load-bearing for the Principle's outperformance claim (§5.3): drop any one condition and a counterexample can be constructed (specifics in §9.4's falsification table).
- **Sufficiency.** Given all four, the quantitative trajectory-divergence bound (Thm 9.4.3) holds: separation zeros the η_sep-contribution, measurement caps the τ_max-contribution at Λ_γ · T_refresh · N_refresh, multi-scale consistency caps the δ_scale-contribution at depth · ε_scale · (t_1 − t_0), and dynamic maintenance caps the freeze-contribution at (1 − ρ_min) · Λ_γ^static · (t_1 − t_0). The joint ceiling B_coh(S, I) is below E[D_d(S')] by the strict-positive shortfall Δ(S', I). ■

§5.3 — Outperformance metric (reference)

The outperformance claim of Theorem 5.1.2 uses the trajectory-divergence functional D. Three candidate metric-constructions:

- **Wasserstein distance** on path-distributions over Ω_S.
- **KL-divergence** on α_S(t) vs. α*_S(t), under absolute-continuity of* α*_S with respect to* α_S (not generally assumed).
- **Domain-native metrics** for concrete stream-domains (Meridian: energy-distance in cosmology; Living Architecture: kingdom-specific fitness-distance; etc.).

§9 (D trajectory-divergence) constructs D in full detail, establishes functorial properties, and settles the open question (Anchor §9.9 Q1) of cross-metric invariance of the outperformance ordering.

Observable signatures (following Anchor §9.3):

1. **Trajectory-tracking.** Sample α_S(t) at refresh-rate; reconstruct γ_S from prior data; compute D directly.
2. **Adjoint-composition success rate.** Count successful $\iota \dashv \kappa$ compositions in $\mathcal{C}$_LDS per interval.
3. **Multi-scale coherence correlation.** Correlate child-γ and parent-γ along DAG-edges.

§5.4 — Self-reference closure

Theorem 5.4.1 (F as stream). *Let F_∞ denote the meta-object (σ_F, K_F, Ω_F, γ_F) where:*

- *σ_F := the carrier "the framework itself" — the totality of claims, axioms, theorems, corollaries, proofs across §§1–4,*
- *C_F := the ContentOp-category whose objects are substrate-commitments, whose morphisms are internal-consistency-preserving revisions,*
- *Ω_F := F(σ_F) = σ_F^(C_F^op),*
- *γ_F := the coalgebra encoding the framework's adaptivity over the construction interval.*

Under suitable interpretation (the Anchor §9.5 "interpretation map" rendering framework-construction events as Stream-morphisms), F_∞ is a stream in the sense of §6.1, and F_∞ satisfies the four conditions over the construction interval [t_0_construction, t_1_construction].

Proof sketch. §8 gives the full construction. The key points:

- **C_sep:** construction separated substrate (A1), dynamics (A2/A3), and applied claims (corollaries) onto distinct DOF.
- **C_meas:** construction used refresh-events (stress-test cycles, Clayton-review cycles, Mirror-updates) at a regular rate.
- **C_scale:** construction maintained coherence across scales (individual-claim, chapter, cluster, framework) via explicit DAG-edges (citation-network, cross-reference structure).
- **C_dyn:** construction is oscillatory by design — draft, dissolve via critique, redraft — the construction-log explicitly exhibits this.

Theorem 5.4.2 (Principle applies to itself). *By Theorems 5.1.2 and 5.4.1, F_∞ is in coherence-regime over the construction interval. The Principle, which is derived inside F_∞, holds of F_∞ itself.*

Proof. Direct application of 5.1.2 to F_∞, using 5.4.1's establishment of F_∞ as a stream in coherence-regime. ■

Remark 5.4.3 (Non-circularity). The closure is a-posteriori: the construction did not presuppose the Principle (the Principle was *derived* from the axiom/theorem stress-test); it is observed after the fact that the construction-process exhibited the four conditions. This is not a circular proof — it is an empirical observation about the framework's own construction-history. §8 audits the observation.

§5.5 — Necessity of each condition (falsification-table reference)

Each condition is independently necessary — dropping any one produces a counterexample stream that is not in coherence-regime yet satisfies the other three.

Condition dropped	Counterexample structure	Anchor falsification source
C_sep	Streams with overlapping-DOF objectives; destructive interference observable	F2, F3
C_meas	Pre-measurement-indefinite streams; γ-drift without corrective pull	F4
C_scale	DAG-inconsistent streams; child-γ and parent-γ decoupled	F5
C_dyn	Frozen-γ streams; γ stationary over extended interval	F1
All four	Random-γ streams; joint-exceeds-bound in E[D]	F1

§9's trajectory-divergence construction gives the quantitative form of each row.

§5.6 — Open questions for §5 (per Anchor §9.9)

- **Q1 (cross-metric invariance).** Whether the outperformance ordering is invariant across choices of d in D. §9 resolves this.
- **Q2 (regime-boundary topology).** Whether coherence-regime is an open condition in an appropriate topology on $\mathcal{C}$_Streams. Open; depends on §9's D-continuity.
- **Q3 (self-reference closure generalization).** Whether every framework passing its own tests exhibits Principle-structure. Open; meta-question, not formally closable in this volume.

§5.7 — Forward-pointers

- **§6** (Triple): already drafted; C_scale multi-scale consistency uses the Triple's recursive decomposability (Lemma 6.3.2) for DAG-child-γ construction.
- **§7** (Filtering): C_meas refresh-events are formally σ-algebra events on Ω_S; Bias(S) measurability under C_meas is §7's content.
- **§8** (F-as-stream): full construction + audit of the §5.4 self-reference closure.
- **§9** (D trajectory-divergence): D's functorial construction + cross-metric invariance result.
- **§10** (TikZ reference figures): Figures 9.1 (D trajectory), 9.2 (four-conditions schematic), 9.3 (self-reference closure) per Anchor §9 reference-figure list.

§5.8 — Surfaced-lemma register

Two flags surface this pass:

- □ §5.2.5 Joint-sufficiency proposition with independence-by-counterexample → Anchor §9.2 target — proposition (explicit joint-sufficiency proof)
- □ §5.4.1 F-as-stream formal structure (σ_F, C_F, Ω_F, γ_F) → Anchor §9.5 target — theorem (detailed construction + audit deferred to §8)

Both flag-targets feed future Anchor revisions per the SCOPE §8 rhythm.

§6 — The Identity-Trajectory Triple

Incorporates the F-coalgebra foundation, kind-classifier fibration, Pull-1 residuals, trifurcation formalization, colax-limit form, cocompletion-and-closure theorem, category-level limits and colimits, and C-size regime analysis.

§6.0 — Conventions and recap

This section fixes the reference conventions used throughout §6 and inherited by §§7–9.

Convention 6.0.1 (Size). Unless otherwise stated, all content-operation categories ContentOp(σ) are **small**. For theorems in §6.8 (limits/colimits) and §3.3 (dynamics pair T3/T4) that invoke filtered colimits, ContentOp(σ) is promoted to **κ-accessible locally-presentable** for a regular cardinal κ fixed per-theorem. Grothendieck universes are not used.

Convention 6.0.2 (Variance). The F-endofunctor is defined with the op-exponent:

$$F : \textbf{Stream} \to \textbf{Stream}, \quad F(\sigma) = \sigma^{\mathrm{ContentOp}(\sigma)^{\mathrm{op}}}$$

The op-category on the exponent matches the presheaf convention: content-operations are contravariant in their action on σ-values.

Convention 6.0.3 (Adequacy). A stream (σ, ContentOp(σ), γ) is **adequate** iff for every pair of distinguishable aspects (a, b) of σ, there exists a morphism f ∈ ContentOp(σ) such that σ^f acts non-trivially on the (a, b) difference. All objects of Stream are adequate by definition; non-adequate tuples are not objects of Stream.

Convention 6.0.4 (Initial objects in ContentOp). Existence of an initial object in ContentOp(σ) is **not** an axiom of Stream. Theorems that require it carry the hypothesis explicitly (notably the colax-limit form of the Triple in §6.6).

Convention 6.0.5 (Kind structure). A2's kind structure is a **preorder** in general (reactive ⊑ self-maintaining ⊑ self-referential ⊑ abstractional). When ContentOp-structure admits meets/joins, the preorder strengthens to a lattice. Theorems stated at preorder-level are strictly weaker but more general; lattice-level corollaries are marked.

Convention 6.0.6 (Recursive decomposability). Each Triple-component (Form, Content, Carrier) of a stream σ is itself a stream, hence admits its own Triple. This is a theorem of the framework (§6.3), not an axiom.

Notation block. Fixed throughout §§6–9:

Symbol	Meaning	First definition
Stream	Category of adequate F-coalgebras	§6.1
σ	Underlying carrier of a stream	§6.1
ContentOp(σ)	Small category of content-operations on σ	§6.1
γ	Coherence-coalgebra $\gamma : \sigma \to F(\sigma)$	§6.1
F	Endofunctor $\sigma \mapsto$ σ^(ContentOp(σ)^op)	§6.0.2
T	Triple functor Stream $\to$Form $\times$ Content $\times$ Carrier	§6.2
T^(n)	n-fold Triple iteration	§6.3
π	Kind-classifier fibration Stream $\to$ContentIndex	§6.4
ContentIndex	(Pre)order of ContentOp-classes	§6.4
Stream_u	Fibered subcategory of unifying streams	§6.4
c_⊥	Terminal (unifying) content-operation	§6.4

§6.1 — Stream as a category of F-coalgebras

Definition 6.1.1 (Stream object). An object of **Stream** is a triple (σ, ContentOp(σ), γ) where:

(i) σ is an object in a concrete ambient category (the carrier).

(ii) ContentOp(σ) is a small category (the content-operations on σ), adequate in the sense of Convention 6.0.3.

(iii) γ : σ → σ^(ContentOp(σ)^op) is a morphism in the ambient category (the coherence-coalgebra).

Remark 6.1.2. The data (σ, ContentOp(σ), γ) is minimal. The kind K and configuration space Ω mentioned in earlier paired-prose presentations of Stream derive:

- **K** = the richness-class of ContentOp(σ) in ContentIndex (§6.4). The A2 kind-preorder is the restriction of the richness-preorder on Cat_small to framework-coherent ContentOp-categories.
- Ω = σ^(ContentOp(σ)^op) itself, with configuration-structure given by the morphism-structure of ContentOp.

So the expanded 4-tuple (σ, K, Ω, γ) used expositorily in the Anchor is the presented-with-derived-components form; Stream-as-F-coalgebra uses the minimal triple. Both describe the same object.

Definition 6.1.3 (Stream morphism). A morphism f : (σ, ContentOp(σ), γ) → (σ', ContentOp(σ'), γ') in **Stream** consists of:

(i) A carrier-map f_σ : σ → σ' in the ambient category.

(ii) A functor f_C : ContentOp(σ) → ContentOp(σ').

(iii) Coalgebra-commute: the square

$$\begin{array}{ccc} \sigma & \xrightarrow{\gamma} & \sigma^{\mathrm{ContentOp}(\sigma)^{\mathrm{op}}} \\ f_\sigma \downarrow & & \downarrow F(f_\sigma, f_C) \\ \sigma' & \xrightarrow{\gamma'} & \sigma'^{\mathrm{ContentOp}(\sigma')^{\mathrm{op}}} \end{array}$$

commutes, where F(f_σ, f_C) is the induced morphism on the presheaf-power given by post-composition with f_σ and pullback along f_C.

(iv) Kind respect: K(σ) ⊑ K(σ') in the A2 preorder (Convention 6.0.5), where K is the classifier of §6.4.

Proposition 6.1.4 (Stream is a category). Composition of Stream-morphisms is Stream-morphism. Identity morphisms exist. Composition is associative.

Proof. Composition: given f : S → S' and g : S' → S", define (g ∘ f)_σ = g_σ ∘ f_σ and (g ∘ f)_C = g_C ∘ f_C. The coalgebra-commute square for g ∘ f is the vertical paste of the squares for f and g; adequacy and kind-respect transfer along the composition since ⊑ is transitive. Identity: id_S = (id_σ, id_ContentOp(σ)) trivially commutes. Associativity: inherited from the ambient category and Cat_small. ■

Definition 6.1.5 (The endofunctor F on Stream). F : **Stream** → **Stream** is defined on objects by

$$
\begin{aligned}
F(\sigma, \mathsf{ContentOp}(\sigma), \gamma) = \big(\, & \sigma^{\mathsf{ContentOp}(\sigma)^{\mathrm{op}}}, \\
& \mathsf{ContentOp}(\sigma^{\mathsf{ContentOp}(\sigma)^{\mathrm{op}}}), \\
& F(\gamma)\,\big)
\end{aligned}
$$

The new carrier is σ^(ContentOp(σ)^op) itself; the new ContentOp is derived as the category of content-operations on this presheaf-power (see §6.7 for the structural result that this is well-defined as a small category under Convention 6.0.1); the new γ is the induced coalgebra F(γ) : F(σ) → F(F(σ)).

On morphisms, F acts by F(f)_σ = F(f_σ, f_C) (the induced presheaf-power morphism), F(f)_C = the lifted content-operation functor, and coalgebra-commute by naturality.

Remark 6.1.6. Stream being a **category of F-coalgebras** means: the forgetful functor U : Stream → (ambient carriers) creates the F-coalgebra structure, i.e., every F-coalgebra γ : σ → F(σ) with ContentOp-data lifts uniquely to a Stream-object. This is the structural content of Definition 6.1.1 — Stream is not auxiliary data on top of F; Stream *is* the category of F-coalgebras (up to the adequacy and kind conditions).

Proposition 6.1.7 (Adequacy is preserved by Stream-morphisms). *If $f : S \to S'$ is a Stream-morphism and S is adequate, then the image of S under f is adequate as a sub-stream of S'.*

Proof. Let (a, b) be distinguishable aspects of $f_\sigma(S) \subseteq \sigma'$. By injectivity of kind-respecting carrier-maps on distinguishable aspects (standard in the ambient category), their pre-images (a', b') are distinguishable aspects of σ. Adequacy of S gives a content-morphism in ContentOp(σ) witnessing (a', b'); its f_C-image in ContentOp(σ') witnesses (a, b). ■

§6.2 — The Triple functor

Definition 6.2.1 (The Triple target category). Define

$$\mathbf{Triple} := \mathbf{Form} \times \mathbf{Content} \times \mathbf{Carrier}$$

where:

- **Form** is the category of bare carriers (objects: carrier sets σ; morphisms: carrier-maps σ → σ' in the ambient category).
- **Content** is Cat_small — objects: small categories; morphisms: functors.
- **Carrier** is the category whose objects are coalgebra-structure-maps $\gamma : \sigma \to F(\sigma)$ and whose morphisms are coalgebra-commute squares.

Triple inherits a product-2-category structure from its three components. When §6 treats Triple strictly as a 1-category, 2-cells are discarded; where 2-cell data is load-bearing (notably §6.2.5 and the middle-regime texture of §6.5), the bicategorical structure is invoked explicitly.

Definition 6.2.2 (The Triple functor). T : **Stream** → **Triple** is defined by:

- On objects: T(σ, ContentOp(σ), γ) = (σ, ContentOp(σ), γ). The three projections are the three aspects — Form takes σ, Content takes ContentOp(σ), Carrier takes γ.
- On morphisms: T(f) = (f_σ, f_C, f_γ), where f_γ is the coalgebra-commute-square data of f.

Proposition 6.2.3 (T is a functor). T preserves composition and identities.

Proof. Each of the three projections is separately functorial by inspection. Composition in Triple is componentwise (Definition 6.2.1), matching composition in Stream (Proposition 6.1.4). ■

Lemma 6.2.4 (T is forgetful — F-coalgebra data factors the Triple). *The data T(S) = (σ, ContentOp(σ), γ) of the Triple is exactly the F-*

coalgebra data of the Stream-object S. No additional stream-data is required to reconstruct T(S); no Stream-data is lost in T(S) except derived components (K, Ω).

Proof. By Definition 6.1.1, a Stream-object is a triple (σ, ContentOp(σ), γ). The Triple T(S) retrieves these three as the Form, Content, Carrier components. K and Ω are derived (Remark 6.1.2), so their absence from T(S) is not a data-loss in the categorical sense. The reconstruction Stream → Triple is the identity map on underlying data. ■

Corollary 6.2.5 (Triple is the object-data of Stream). The objects of Stream are in canonical bijection with the objects of the image T(Stream) ⊆ Triple satisfying the adequacy and F-coalgebra conditions. Stream is the full subcategory of Triple cut out by:

(a) ContentOp(σ) is adequate for σ (Convention 6.0.3).

(b) $\gamma : \sigma \to F(\sigma)$ is an F-coalgebra (satisfies the coalgebra identity up to the choice of F).

(c) Morphisms respect kind (Convention 6.0.5) and coalgebra-commute (Definition 6.1.3 (iii)).

Remark 6.2.6. The Triple is sometimes summarized paired-prose-style as "Form / Content / Carrier as orthogonal-but-constrained axes with recursive decomposability." The formal content of this summary is:

- **Orthogonal:** T factors Stream-data into three independent projections (Definition 6.2.2).
- **Constrained:** the three projections are not independent — the coalgebra-commute condition (Definition 6.1.3 (iii)) couples them at the morphism level.
- **Recursive decomposability:** each projection-image is itself a stream (§6.3).

Proposition 6.2.7 (T reflects isomorphisms). *If $f : S \to S'$ is a Stream-morphism whose three Triple-components T(f) = (f_σ, f_C, f_γ) are all isomorphisms in their respective component categories, then f is an isomorphism in Stream.*

Proof. Construct f^-1 componentwise: ((f_σ)^-1, (f_C)^-1, (f_γ)^-1). The coalgebra-commute square for f^-1 is obtained by inverting each side of f's square, which is well-defined because all three components are invertible. Kind respect holds in both directions because a preorder isomorphism is two-way. ■

Remark 6.2.8. Proposition 6.2.7 is the Triple's **conservativity** statement: an isomorphism of Stream-objects is exactly an isomorphism of all three Triple-components. This is the formal content of the Anchor's "the Triple is the load-bearing identity-structure of a stream."

§6.3 — Recursive decomposability (finite depth)

Setup. Each Triple-component of a Stream-object is itself a stream. Explicitly, for S = (σ, ContentOp(σ), γ):

- **Form(S)** — the carrier σ equipped with its form-induced ContentOp (the full subcategory of ContentOp(σ) consisting of content-operations acting on σ's surface structure alone) and restricted γ.
- **Content(S)** — the category ContentOp(σ) viewed as a stream, with its own ContentOp (the 2-category of functorial operations on ContentOp(σ), small by Convention 6.0.1) and its own coalgebra-structure (the hom-2-functor).
- **Carrier(S)** — the coalgebra γ viewed as a morphism-stream, with ContentOp(γ) the 2-cell-level content-operations and coalgebra-structure the squares-of-squares.

Each construction yields a Stream-object in its own right.

Definition 6.3.1 (Iterated Triple). Define T^(n) : Stream → Triple^(n) inductively:

$$T^{(1)}(S) = T(S) = (\text{Form}(S), \text{Content}(S), \text{Carrier}(S))$$
$$T^{(n+1)}(S) = \big(T^{(1)}(\text{Form}(S)),\ T^{(1)}(\text{Content}(S)),\ T^{(1)}(\text{Carrier}(S))\big)$$

The target Triple^(n) is a 3^n-fold product of Triple.

Lemma 6.3.2 (Finite-depth Triple-factorability via F). *For every Stream-object S and every finite n ≥ 1, T^(n)(S) is reconstructible from the F-coalgebra data of S together with iterated ContentOp-applications at each level.*

Proof. Induction on n.

Base (n = 1). By Lemma 6.2.4, T(S) = T^(1)(S) is the F-coalgebra data of S.

Step (n → n+1). Assume T^(n)(S') is F-reconstructible for all streams S' up to depth n. For T^(n+1)(S):

- **Form-branch.** Form(S) is a stream (by the preceding setup), and its F-data (σ, ContentOp(Form(S)), γ|_Form) is a restriction of S's F-data, well-defined by the adequacy-stability argument (Proposition 6.1.7 applied to the identity-on-σ map from S to Form(S)). By inductive hypothesis on Form(S) at depth n, T^(n)(Form(S)) is F-reconstructible. Hence T^(1)(T^(n)(Form(S))) is the first branch of T^(n+1)(S), and F-reconstructible.
- **Content-branch.** Content(S) = ContentOp(σ) is a small category, viewable as a stream whose ContentOp is the 2-category of functors on ContentOp(σ). This construction uses Convention 6.0.1's small-ness crucially — the 2-category of endofunctors on a small category is itself small. By inductive hypothesis, T^(n)(Content(S)) is F-reconstructible. Branch two of T^(n+1)(S) follows.
- **Carrier-branch.** Carrier(S) = γ, viewable as a morphism-stream whose ContentOp consists of 2-cells in the ambient 2-category and whose coalgebra-structure is the square-of-squares. By inductive hypothesis on this morphism-stream at depth n, branch three follows.

Assembling the three branches gives T^(n+1)(S) as F-reconstructible.
■

Remark 6.3.3 (The Cat-valued promotion earns its keep here). The Content-branch is the load-bearing step. If ContentOp were set-valued (as in the foundation-doc pre-M12 proposal), the Content-branch would terminate at n = 1 — Content(S) would be a bare set with no further ContentOp to apply T to. Category-valued ContentOp provides the recursion substrate. This is a structural — not merely technical — fact about the framework.

Corollary 6.3.4 (Recursive decomposability is a theorem, not an axiom). *The recursive-decomposability content of the Identity-Trajectory Triple (named in the Anchor as a property of streams, and captured as meta-bridge M3 in the framework's bridge list) is derived from the F-coalgebra definition of Stream. It is not an independent axiom.*

Proof. Immediate from Lemma 6.3.2: at every finite depth, T^(n) is defined and F-reconstructible. The framework's original axiom-posture for recursive decomposability was pedagogical; the formal content is Lemma 6.3.2. ■

Remark 6.3.5. Depth-ω (infinite recursive decomposition) is not automatic. It requires the final F-coalgebra construction, which in turn requires filtered-colimit preservation by F. This is treated in §6.9 with its own hypothesis-set.

§6.4 — The kind-classifier fibration over Content

Motivation. The Anchor's A2 posits kind-stratification of streams (reactive/self-maint/self-ref/abstr). The formal content of the stratification is that there is a classifier-functor π from streams to a ContentOp-class-preorder, and the A2 sub-categories of Stream are fibers of π. This section constructs π and records its fibration properties.

Definition 6.4.1 (ContentIndex). The category **ContentIndex** has:

- **Objects:** equivalence classes of small categories under equivalence-of-categories, restricted to classes realized by some stream's ContentOp.
- **Morphisms:** classes of faithful functors (content-refinements) up to natural isomorphism.

In the generic framework setting, ContentIndex is a **preorder** (Convention 6.0.5). When ContentOp-structure admits (co)products globally, it is a **lattice**.

Definition 6.4.2 (The kind-classifier). Define π : **Stream** → **ContentIndex** by

$$\pi(\sigma, \text{ContentOp}(\sigma), \gamma) := [\text{ContentOp}(\sigma)]$$

— the class of ContentOp(σ) in ContentIndex. On morphisms, π(f) = [f_ C], the induced class-level functor.

Proposition 6.4.3 (π is a functor). Composition and identities preserved. ■

Construction 6.4.4 (Cartesian lifts as restriction content-operations). Given a Stream-object S = (σ, ContentOp(σ), γ) with π(S) = c, and a morphism f : c' → c in ContentIndex, the **Cartesian lift** $\bar{f}$: S' → S is constructed as follows:

1. Choose a representative ι : C' ↪ ContentOp(σ) of the class c' (canonical up to equivalence).

2. Define S' := (σ, C', γ|_C'), where γ|_C' is γ post-composed with the induced restriction σ^(ContentOp(σ)^op) → σ^(C'^op).
3. The lift $\bar{f}$ is the Stream-morphism (id_σ, ι, γ-commute-by-construction).

Lemma 6.4.5 (Cartesian lifts are literally content-operations). *The Cartesian lift $\bar{f}$ above is itself a content-operation: specifically, the **restriction** content-operation given by the inclusion functor ι. Under the Cat-valued ContentOp convention, ι ∈ Mor(Cat_small) is a legitimate content-operation on ContentOp(σ).*

Proof. The data of $\bar{f}$ consists of id_σ on carriers, ι on ContentOp, and an induced γ-commute. All three components are captured by ι as a morphism in Cat_small. Restriction-content-operations are a recognized sub-class of content-operations under Cat-valuation. ■

Theorem 6.4.6 (π is a bicategorical fibration). *Given a Stream-object S with π(S) = c and a morphism f : c' → c in ContentIndex, the Cartesian lift $\bar{f}$: S' → S of Construction 6.4.4 is Cartesian in the bicategorical sense — i.e., for any Stream-morphism g : T → S and any morphism h : π(T) → c' in ContentIndex with π(g) = f ∘ h, there exists a Stream-morphism $\bar{h}$: T → S' unique up to equivalence, with $\bar{f}$ ∘ $\bar{h}$ = g and π($\bar{h}$) = h.*

Proof sketch. Existence: factor g_C : ContentOp(T) → ContentOp(σ) through the inclusion ι : C' ↪ ContentOp(σ), using π(g) = f ∘ h to ensure the image lies in C' (class-level); this defines $\bar{h}$. Uniqueness up to equivalence: two factorizations through a faithful inclusion agree up to natural isomorphism on the image; this is the "up to equivalence" weakening. Full strictness holds when ι is identity-on-objects-faithful, which occurs when ContentIndex is a lattice (Convention 6.0.5). ■

Corollary 6.4.7 (Strict Cartesian fibration under lattice). *If ContentIndex is a lattice (Convention 6.0.5 lattice case), π is a strict Grothendieck fibration.*

Remark 6.4.8. The Anchor's A2 treatment reads at the preorder level. The bicategorical-fibration status of π (Theorem 6.4.6) is what the Anchor prose "A2 kind-stratification" formally denotes. The strict fibration corollary is the lattice refinement.

§6.4.1 — Unifying streams and admissibility

Definition 6.4.9 (Unifying stream). A Stream-object S = (σ, ContentOp(σ), γ) is **unifying** iff there exists an object c_⊥ ∈ ContentOp(σ) that is terminal in ContentOp(σ) *and* satisfies σ^(c_⊥) ≅ 1 in the ambient category.

Equivalent characterizations. Under Convention 6.0.3 (adequacy) and 6.0.1 (small-ness):

(i) ContentOp(σ) has a terminal object c_⊥ with σ^(c_⊥) ≅ 1.

(ii) The partition of σ induced by c_⊥'s output-equivalence is the indiscrete partition (all σ-elements in one class).

(iii) There exists a natural transformation from the constant presheaf 1 to σ^(ContentOp(σ)^op) picking out c_⊥-as-terminal.

Proof of equivalence. (i) ⇔ (ii): under adequacy, terminal content-operations are exactly those whose induced partition cannot be refined by any other content-operation. The indiscrete partition is the bottom of the partition lattice; terminal content-operations realize it. (i) ⇔ (iii): standard presheaf equivalence for terminal objects. ■

Definition 6.4.10 (Admissible refinement). A morphism f : c' → c in ContentIndex is **admissible** iff every representative inclusion ι : C' ↪ ContentOp(σ) preserves the terminal object — i.e., if c_⊥ ∈ ContentOp(σ) is terminal, then c_⊥ ∈ C' and remains terminal in C'.

Lemma 6.4.11 (Unifying-ness lifts along admissible refinements only). *Let S be unifying with terminal c_⊥ ∈ ContentOp(σ), and let f : c' → c = π(S) be a morphism in ContentIndex. The Cartesian lift S' = (σ, C', γ|_C') is unifying iff f is admissible.*

Proof. (⇐) If f is admissible, c_⊥ ∈ C' remains terminal, and σ^(c_⊥) ≅ 1 is preserved since the carrier σ is unchanged. So S' is unifying with the same witness. (⇒) If S' is unifying with witness c'_⊥ ∈ C', then c'_⊥ is terminal in C'. For ι to be faithful with c'_⊥ terminal in C' but possibly not in ContentOp(σ), the inclusion would fail to preserve terminality — but then ι(c'_⊥) in ContentOp(σ) would not be terminal, contradicting the uniqueness of terminal objects up to isomorphism. So ι must preserve terminality, i.e., f is admissible. ■

Definition 6.4.12 (The fibered subcategory Stream_u). Stream_u is the full subcategory of Stream whose objects are unifying streams, restricted along the sub-fibration induced by admissible refinements in ContentIndex.

Corollary 6.4.13 (Stream_u is fibered over the admissible subcategory). *The restriction π|_Stream_u : Stream_u → ContentIndex_adm is a bicategorical fibration (strict under lattice).*

Proof. Apply Theorem 6.4.6 to the admissible-refinement subcategory and Lemma 6.4.11 for the unifying-ness lift. ■

Remark 6.4.14 (Physics instance — quantum-mechanical branches as Cartesian lifts). The fibration π : **Stream → ContentIndex** has a concrete realization in Lohmiller & Slotine's multi-valued classical-action construction (*Proc. Roy. Soc. A* 482: 20250413, Thm 2.4 and Thm 3.2). Each branch $j \in J$ of the extremal-action multipath corresponds to a Cartesian lift of the ambient stream over the content-operation that selects branch j (the associated measurement operator). Admissibility in the sense of Definition 6.4.10 — preservation of a terminal content-operation along the lift — corresponds to the Lipschitz-continuity conditions in their Theorem 2.4 that guarantee branch existence and local uniqueness via Picard-Lindelöf contraction theory. The unifying-stream condition (Definition 6.4.9) tracks physical configurations where a universal-ground-state branch is well-defined; streams failing unifying-ness correspond to branch-point configurations where the Lipschitz conditions fail (topological non-simply-connectedness, spatial inequality constraints, Hamiltonian singularities — their $\mathcal{B}^N$ set). This is a scope-limited sufficiency witness — Lagrangian systems under Coulomb/Lorenz gauge, per the paper's domain — not a coextensive derivation. It is one instance among possibly many of the fibration's content; the Companion does not claim QM as the canonical case.

§6.5 — Middle-regime morphism-structure (migrated)

The material previously occupying §6.5 — ultimate-structure Ult(S), the Scotist/Palamite/Advaitin trifurcation, Theorem 6.5.4 (three-class distinctness), Corollary 6.5.6 (cross-tradition translation data-loss), and the A48 correspondence remark — has been migrated to the ***Universal Coherence*** *volume per SCOPE.md §8.2 SCOPE-EXCLUDED disposition. The content is the metaphysical-lift application of Triple machinery to contemplative-traditions phenomenology, which is Universal-Coherence scope, not Coherent-Structure scope (pure CT formalization of the anchor).*

New location: `Library/Universal-Coherence/drafts/2026-04-24-middle-r` *Section number §6.5 is retained here as a numbering anchor; the slot is intentionally empty in the Companion.*

§6.6 — The Triple as colax-limit (conditional)

Motivation. The Anchor frames the Triple as a **colax-limit** of a three-pronged diagram in Cat. This is a sharpening of Theorem 6.2.5 (Stream as a full subcategory of Triple), conditional on ContentOp having enough structure.

Hypothesis 6.6.0. *For the results of this section, assume ContentOp(σ) admits an initial object I (Convention 6.0.4 — carried as hypothesis per-theorem).*

Definition 6.6.1 (The colax-limit diagram). Define a three-object diagram D : J → Cat as follows:

- J is the walking span-category: three objects •_F, •_C, •_Cr and two morphisms •_F → • ← •_C, •_Cr → •, where • is an auxiliary apex object representing the coalgebra-commute condition.
- D(•_F) = Form, D(•_C) = Content, D(•_Cr) = Carrier, D(•) = Form × Content × Carrier (with the three projections as D's structure).

Theorem 6.6.2 (Triple as colax-limit, under Hypothesis 6.6.0). *Under the initial-object hypothesis on ContentOp(σ), Stream embeds into the colax-limit colaxlim D of the diagram above, and the embedding is an equivalence of categories onto the full sub-2-category of adequate F-coalgebras.*

Proof sketch. colaxlim D consists of cones (F_0, C_0, Cr_0, α_F, α_C, α_Cr) where F_0, C_0, Cr_0 are objects in Form, Content, Carrier respectively, and α_∗ are colax-natural-transformations to the apex. A Stream-object (σ, ContentOp(σ), γ) supplies (σ, ContentOp(σ), γ) as the three objects, and the coalgebra-commute condition provides the colax-natural-transformation data with *initial-object-anchored* structure: the initial I ∈ ContentOp(σ) serves as the tip of the colax cone, giving a canonical base-point from which all content-operations flow.

The embedding Stream ↪ colaxlim D is fully faithful by Proposition 6.2.7 (T reflects isomorphisms). Essential surjectivity onto adequate F-coalgebras holds because the colax structure precisely encodes the coalgebra-commute condition. ■

Remark 6.6.3 (Why this requires initial objects). Without an initial object in ContentOp(σ), the colax cone has no anchor-point, and the colax-limit construction produces a larger object than Stream — one that includes ”ungrounded” Triple-tuples without a canonical base content-operation. The initial-object hypothesis makes the cone well-pointed and the colax-limit equivalent to Stream.

Remark 6.6.4 (General case without initial objects). For streams without initial ContentOp, Theorem 6.2.5 (Stream as a full subcategory of Triple) is the correct structural description. The colax-limit form is a sharpening available under the additional hypothesis. Both are drafting-valid; the paired-prose Anchor uses the colax-limit form wherever it is applicable, with the un-conditioned statement as fallback.

Corollary 6.6.5 (The Triple’s ”canonical base” is the initial content-operation). *When Hypothesis 6.6.0 holds, the initial object $I \in \mathrm{ContentOp}(\sigma)$ is the distinguished ”ground” content-operation from which all stream-internal content-operations are reachable. This formalizes the paired-prose notion of a stream’s ”ground state” or ”base coherence.”*

Proof. Immediate from the colax-limit anchor-point construction in Theorem 6.6.2. ■

§6.7 — Stream as the category of F-coalgebras (closure)

Motivation. Theorem 6.2.5 stated that Stream is a full subcategory of Triple cut out by three conditions (adequacy, F-coalgebra identity, kind-respecting morphisms). Theorem 6.6.2 refined this (under a hypothesis) to a colax-limit equivalence. This section closes the structural characterization by stating the main equivalence.

Theorem 6.7.1 (Stream = F-Coalg_ad). *Let F-Coalg denote the category of F-coalgebras in the ambient Set-like category, and let F-Coalg_ ad ⊆ F-Coalg be the full subcategory of adequate F-coalgebras satisfying kind-respect on morphisms. Then:*

$$\mathbf{Stream} \simeq \mathbf{F\text{-}Coalg}_{\mathrm{ad}}$$

as categories.

Proof. By Definition 6.1.1, every Stream-object is an F-coalgebra with adequacy and kind-data. Conversely, every F-coalgebra $\gamma : \sigma \to F(\sigma)$ with adequate ContentOp(σ) lifts to a Stream-object by setting $K = [\mathrm{ContentOp}(\sigma)]$ and $\Omega = F(\sigma)$ (Remark 6.1.2). The lift is bijective on objects and morphisms by Definition 6.1.3. Functoriality of the equivalence holds because composition in both categories is componentwise (Proposition 6.1.4). ■

Corollary 6.7.2 (Closure). *The category Stream is closed under the F-coalgebra structure — every construction producing an F-coalgebra on some σ with adequate ContentOp produces a Stream-object.*

Proof. Immediate from Theorem 6.7.1. ■

Remark 6.7.3 (Implication for framework construction). Closure under F-coalgebra operations means that Stream is algebraically well-behaved under coalgebraic constructions — products (§6.8), equalizers (§6.8), filtered colimits (§6.8), and the self-reference closure of §8 all produce Stream-objects when their inputs are Stream-objects. The framework does not need to re-check adequacy and kind-respect at each construction; these properties transfer automatically under Stream-morphisms (Proposition 6.1.7).

§6.8 — Limits and colimits in Stream under F

Conventions recalled. For this section, size/variance/adequacy follow §6.0. Kind-structure is the preorder of Convention 6.0.5; lattice-level results are flagged explicitly.

§6.8.1 — Limits

Proposition 6.8.1 (Terminal object). *The terminal object 1_Stream := (1, **1**_cat, id_1) exists in Stream, where 1 is the terminal carrier, **1**_cat is the terminal small category (one object, identity morphism), and id_1 is the unique coalgebra-structure-map.*

Proof. Uniqueness of maps into 1_Stream: f_σ factors uniquely through 1; f_C factors uniquely through **1**_cat; coalgebra-commute is vacuous because γ_1_Stream is identity. Kind-respect is trivial because K(1_Stream) is the top of the preorder. ■

Proposition 6.8.2 (Products, conditional on kind-join). *Given Stream-objects S_1, S_2 with K(S_1), K(S_2) admitting a join K_∨ in the A2 preorder, the product S_1 × S_2 exists in Stream with:*

$$S_1 \times S_2 = (\sigma_1 \times \sigma_2,\ \mathsf{ContentOp}(S_1) \times \mathsf{ContentOp}(S_2),\ \gamma_1 \times \gamma_2)$$

with the cartesian product of coalgebras and kind K_∨.

Proof sketch. Universal property: a pair (g_1, g_2) : T → S_1, T → S_2 induces a unique (g_1, g_2)_* : T → S_1 × S_2 by pairing carrier-maps and ContentOp-functors componentwise. Coalgebra-commute transfers by pairing. Kind-respect K(T) ⊑ K_∨ requires K_∨ to exist as a join — hence the conditional. ■

Proposition 6.8.3 (Equalizers, conditional on adequacy-stability). *Given parallel Stream-morphisms f, g : S → S′, the equalizer exists in Stream provided the carrier-equalizer eq(f_σ, g_σ) carries an adequate ContentOp-restriction.*

Proof sketch. Carrier: $eq(f_\sigma, g_\sigma) \subseteq \sigma$. ContentOp: the subcategory of ContentOp(σ) consisting of content-operations that agree after mapping by f_C and g_C. Coalgebra: restricted γ. Adequacy-stability (Lemma 6.8.β below) gives the conditional. ■

Lemma 6.8.β (Adequacy-stability under limits/colimits). *Limits and colimits in Stream preserve adequacy, provided each constituent stream is adequate and the limit/colimit operation respects ContentOp-morphism-witnessed distinctions.*

Proof. For a limit L, distinguishable aspects of L descend to distinguishable aspects in at least one constituent (by universality); ContentOp-morphism witnesses lift to L via the limit-cone. Dual argument for colimits via the couniversal property. ■

Proposition 6.8.4 (Filtered limits under locally-presentable promotion). *When ContentOp is promoted to κ-accessible locally-presentable (Convention 6.0.1), filtered limits exist in Stream.*

§6.8.2 — Colimits

Proposition 6.8.5 (Initial object). *The initial object $0\_Stream := (\emptyset, \mathbf{0}\_cat, !)$ exists, where $\emptyset$ is the empty carrier, $\mathbf{0}$_cat is the empty category, and ! is the unique map from $\emptyset$.*

Proof. Vacuous satisfaction of all Stream conditions. ■

Proposition 6.8.6 (Coproducts, conditional on kind-meet). *Given S_1, S_2 with $K(S_1)$, $K(S_2)$ admitting a meet $K_\wedge$, the coproduct $S_1 + S_2$ exists with disjoint-union carriers, disjoint-union ContentOp, and componentwise coalgebra.*

Proposition 6.8.7 (Coequalizers, conditional on quotient-adequacy). *Given $f, g : S \to S'$, the coequalizer exists provided the quotient ContentOp — consisting of content-operations that respect the generated equivalence — is adequate for $\sigma'/\sim$.*

Proposition 6.8.8 (Filtered colimits under locally-presentable promotion). *Under Convention 6.0.1's locally-presentable promotion, filtered colimits exist in Stream.*

§6.8.3 — What Stream *does not* have

Proposition 6.8.9 (No exponentials in general). *Stream is not Cartesian closed; there is no canonical (S')^S object for arbitrary S, S' ∈ Stream.*

Proof sketch. Standard counterexample from coalgebra theory: exponentials of coalgebras require specific smallness and commutativity conditions not implied by Stream's definition. The F-coalgebra structure fixes γ-data that does not naturally lift to a function-space. ■

Proposition 6.8.10 (No subobject classifier). *Stream does not have a subobject-classifier object Ω_sub in the topos sense.*

Remark 6.8.11 (The framework's "Ω" is not Ω_sub). The framework's configuration space Ω for a stream (named in earlier expositions as the 4-tuple's third component, derived in Remark 6.1.2 as σ^(ContentOp(σ)^op)) is not a subobject classifier; it is the evaluation-space of the F-structure. The names coincide accidentally; no toposic interpretation is intended.

§6.8.4 — Summary table

Limit/colimit	Existence	Condition
Terminal 1_Stream	yes	—
Initial 0_Stream	yes	—
Products S_1 × S_2	conditional	kind-join exists (Q7 lattice case)
Coproducts S_1 + S_2	conditional	kind-meet exists (Q7 lattice case)
Equalizers	conditional	adequacy-stability (Lem 6.8.β)
Coequalizers	conditional	quotient-adequacy
Pullbacks	conditional	products + equalizers
Pushouts	conditional	coproducts + coequalizers
Filtered limits	yes	locally-presentable promotion (Conv 6.0.1)
Filtered colimits	yes	locally-presentable promotion (Conv 6.0.1)

Limit/colimit	Existence	Condition
Exponentials	no	—
Subobject classifier	no	—

§6.9 — C-size regimes, H1+H2, and recursive decomposability at depth ω

*This section front-loads the **size-of-ContentOp** as the single parameter governing H1 (accessibility of Stream) and H2 (filtered-colimit preservation of F), partitions Stream into three regimes, proves H1+H2 in the regime where they hold, and places F_∞ (§8.1.2) at the regime boundary.*

§6.9.0 — The three C-size regimes

For a stream S = (σ, C, γ), the **size regime** of S is determined by the cardinality-and-presentability structure of C = ContentOp(σ). Three regimes are operationally meaningful, and the framework's theorems behave distinctly in each:

Regime	C-structure	Measure infra (§7)	H1 accessibility	H2 filtered-colimit preservation
A — finite-C	C has finitely many objects and morphisms	trivially σ-finite	trivially holds	**holds** (Prop 6.9.1 below)
B — small-but-infinite-C	C is small (set-many), admits infinitely many objects	σ-finite under countable-C with concrete reference measure	holds under local-presentability	**generically fails** (Prop 6.9.2 below)
C — large-C	C is a proper class	σ-finiteness breaks	fails	fails

Declared scope. The Companion's declared scope is Regimes A and B. Regime C is excluded by Convention 1.1.5 (smallness of ContentOp).

Why C-size is the governing parameter. F(σ) = σ^(ContentOp(σ)^op) is an exponential whose exponent-argument is ContentOp. Exponentials σ ↦ σ^X in a locally-presentable category preserve filtered colimits in σ **iff X is finitely-presentable** (Adámek–Rosický 1994, Thm 1.56 + Cor 1.57). Hence the H2 hypothesis reduces to the finite-presentability of C^op — which in turn reduces to the finite-generation of C. This single fact ties H1, H2, §7 σ-finiteness, and §8 depth-stability to a single parameter.

§6.9.1 — Regime A: H1 and H2 both hold

Proposition 6.9.1 (H1+H2 under finite-C). *Let 𝒞_Streams^fin ⊂ 𝒞_Streams be the full subcategory of streams S with C_S finite. Then on 𝒞_Streams^fin:*

1. *(H1) 𝒞_Streams^fin is ⍰_0-accessible locally presentable.*
2. *(H2) F restricted to 𝒞_Streams^fin preserves filtered colimits.*

Proof.

(H1): 𝒞_Streams^fin consists of pairs (σ, C, γ) with C finite. For each finite shape C, the category F-Coalg_ad(F_C) of F_C-coalgebras over a fixed finite C is a slice of Cat over a finite base, which is accessible (Adámek–Rosický 1994, Thm 2.78). The coproduct over isomorphism-classes of finite C is countable, and a countable coproduct of accessible categories is accessible. Hence 𝒞_Streams^fin = ∐_[C] finite F-Coalg_ad(F_C) is accessible. Local presentability: 𝒞_Streams^fin is cocomplete (Propositions 6.8.5–6.8.7 specialized to finite C) and every object is a filtered colimit of ℵ_0-presentable ones (every finite-C stream is itself ℵ_0-presentable). ■

(H2): For fixed finite C, F_C(σ) = σ^(C^op) is a finite product of copies of σ indexed by objects of C^op. Finite products preserve filtered colimits in any category that has them (standard; e.g., Borceux 1994, Prop 2.13.4). Hence F_C preserves filtered colimits in σ. For varying C, the functor F = (σ, C) ↦ σ^(C^op) has the component F_C preserving filtered colimits pointwise; combined with (H1)'s slicewise accessibility and the fact that filtered colimits in 𝒞_Streams^fin are computed component-wise in each C-slice (Proposition 6.8.8 specialized to finite C), F on 𝒞_Streams^fin preserves filtered colimits. ■

Remark 6.9.1'. The proof of H2 in Regime A is the *content* of the claim. Without the finite-C restriction, σ^(C^op) is an infinite product (over objects of C^op), and infinite products generically do not preserve filtered colimits in the product argument. The shift from "finite product ⇒ preserves filtered colimits" to "infinite product ⇒ does not in general" is exactly the Regime A ↔ Regime B boundary.

§6.9.2 — Regime B: H2 generically fails

Proposition 6.9.2 (H2 failure in Regime B). *Let S = (σ, C, γ) be a stream with C small and containing infinitely many objects no finite sub-family of which is cofinal in C. Then F(σ) = σ^(C^op) does not in general preserve filtered colimits of σ.*

Proof. Suppose (σ_i)_i ∈ I is a filtered diagram in Set (or in the ambient base) with colimit σ_∞ = colim_i σ_i. We compute:

- *(colim_i σ_i)^(C^op)* = functors C^op → σ_∞.
- *colim_i (σ_i^(C^op))* = colim_i (functors C^op → σ_i).

For infinite C with no finite cofinal sub-family, the colimit-exchange

$$\mathrm{colim}_i(\sigma_i^{C^{\mathrm{op}}}) \xrightarrow{?} (\mathrm{colim}_i\sigma_i)^{C^{\mathrm{op}}}$$

is not in general an isomorphism: a functor C^op → σ_∞ may send infinitely many distinct objects of C^op to distinct values in σ_∞ that cannot all be "found in some σ_i" simultaneously, even though each individual value appears in some σ_i. Explicit counterexample: take C^op = ℕ (discrete), σ_i = 0, 1, ..., i, σ_∞ = ℕ. The identity function ℕ → ℕ is an element of σ_∞^(C^op) that is not in any σ_i^(C^op). Hence F does not preserve this filtered colimit. ■

Consequence. In Regime B, the transfinite iteration σ_α+1 = F(σ_α) of Theorem 6.9.3 below does not in general stabilize under H2; the depth-ω result is therefore conditional on additional structure on C (e.g., C finitely-generated, C cofinally finite, or a targeted property of γ).

§6.9.3 — Depth-ω result (conditional on regime)

Theorem 6.9.3 (Final F-coalgebra, regime-specific). *Within $\mathcal{C}$_Streams^fin (Regime A), a final F-coalgebra σ_∞ exists, with $\sigma_\infty \cong F(\sigma_\infty)$, and every Stream-object in $\mathcal{C}$_Streams^fin maps uniquely into σ_∞.*

Proof. By Prop 6.9.1, H1 and H2 both hold on $\mathcal{C}$_Streams^fin. Standard terminal-coalgebra construction (Adámek 1974, Barr 1993): iterate F transfinitely from 1:

- $\sigma_0 := 1$ (terminal object, which exists by Lemma 6.8.α in Regime A),
- $\sigma_{\alpha+1} := F(\sigma_\alpha)$,
- $\sigma_\lambda := \lim_{\alpha<\lambda} \sigma_\alpha$ at limit ordinals λ.

H1 ensures the iteration stabilizes at some ordinal $\leq \aleph_1$ (first uncountable regular). H2 ensures F commutes with the stabilizing filtered colimit, so $\sigma_\infty \cong F(\sigma_\infty)$. Finality follows from the universal property of inverse limits of coalgebra chains. ■

Corollary 6.9.4 (Depth-ω Triple-factorability in Regime A). *Within $\mathcal{C}$_Streams^fin, $T^{(\omega)}(\sigma_\infty)$ is well-defined and fixed: $T(\sigma_\infty) \cong \sigma_\infty$ as Triple-objects.*

Theorem 6.9.5 (Regime-B depth-ω conditional). *For a stream S in Regime B, a final F-coalgebra over S exists iff C_S is finitely-generated (equivalently: C_S admits a finite cofinal sub-category). In that case the Regime-A construction applies to the finite-cofinal sub-diagram.*

Proof. If C_S has a finite cofinal sub-category C'_S, then $\sigma^{(C_S^{op})} \cong \sigma^{(C'_S)^{op}}$ (cofinality of limits), and the finite-C case applies. Conversely, if no finite cofinal sub-category exists, Prop 6.9.2's counterexample instantiates against S, blocking H2. ■

§6.9.4 — Placement of F_∞

Proposition 6.9.6 (F_∞'s regime trajectory). *Per §8.1.2, $C_{F,t}$ is finite at every fixed construction-time t. Hence $F_\infty|_t \in \mathcal{C}$_Streams^fin at every t; $F_\infty|_t$ is in Regime A. The colimit $C_{F,\infty} := \mathrm{colim}_t\, C_{F,t}$ as $t \to$*

∞ is at most countable and generically not finitely-generated; hence F_∞ over [t_0, ∞) is in Regime B with undetermined H2 status.

Proof. Finite-at-each-t from §8.1.2 (the commit-history at time t carries finitely many substrate-commitments). Countability of the limit from the commit-history being a countable sequence of snapshots. Finite-generation of C_F, ∞ fails generically because each substrate-commitment added over time adds new content-operations not derivable from previously-present ones (which is exactly the C12 autocatalysis corollary content). ■

Consequence. Audit Observation 8.3.5's self-reference claim over a *finite* construction interval [t_0, t_1] lands in Regime A (via the finite-slice $\mathcal{C}$_Streams^fin); the H2-hypothesis of the depth-ω theorem is not invoked. This is why the §8 finite-interval audit is well-founded even without H2 verification at the colimit. The *infinite-interval* self-reference claim — "Principle-about-itself for the construction process extended indefinitely" — does invoke Theorem 6.9.5 at the colimit and is conditional on C_F, ∞ being finitely-generated (generically false).

Scope remark. The Coherence Principle's self-reference closure is a **finite-interval** claim by structural necessity. Extending it to ω requires more than just time-passage; it requires that the autocatalytic content-operation discovery process (§C12) halt in the finite-cofinal sense. The framework does not predict such halting, nor does it require it — the Principle's empirical content is per-interval.

§6.9.5 — Connection to §7's small-C and σ-finiteness

Proposition 6.9.7 (§7 σ-finiteness scope). *The §7.3.2 σ-finiteness hypothesis for Bias(S) holds:*

- *Unconditionally in Regime A.*
- *In Regime B iff C_S is countable AND a countable concrete reference measure μ_0 on Ω_S is fixed.*
- *Fails in Regime C.*

Proof.

Regime A: $\Omega\_S = \sigma\^(C\^op)$ with finite C is a finite product of finite-or-countable copies of σ; any finite-or-countable σ-algebra on σ lifts to a σ-finite reference measure on Ω_S. Hence Bias(S) decomposes as the difference of two finite-or-σ-finite measures (Hahn–Jordan) and is σ-finite.

Regime B: Ω_S is a (countable-indexed) product over objects of C^op. A σ-finite reference measure exists iff C is countable (giving a countable product) and a concrete σ-finite measure is fixed on σ (which the framework admits per §7.1.1). Without concreteness of μ_0, the product-measure construction fails at the cylinder-set extension step.

Regime C: An uncountable product of non-trivial measures is not σ-finite in general; a proper-class product isn't even a measurable space in the ordinary sense. ■

Consequence (scope declaration for §7). §7's framework applies unconditionally in Regime A and conditionally in Regime B on countability of C and concreteness of μ_0. This is the scope-condition that was silently load-bearing in §7.3.2's σ-finiteness argument; it is now explicit.

§6.9.6 — Surfaced-lemma register (this section)

- ☐ §6.9.0 C-size-regime partition (A / B / C) → Anchor §1/§3 target — convention (makes the size-of-C scope-condition explicit across framework)
- ☐ Prop 6.9.1 H1+H2 both hold on $\mathcal{C}$_Streams^fin → Anchor §9.5 target — proposition (resolves H1 and the finite-C case of H2)
- ☐ Prop 6.9.2 H2 generically fails in Regime B → Anchor §1.5 target — proposition (explicit counterexample bounds the scope)
- ☐ Thm 6.9.3 / Thm 6.9.5 Final F-coalgebra (Regime A unconditional; Regime B conditional on finite-cofinal sub-diagram) → Anchor §1.5 + §9.5 target — theorem
- ☐ Prop 6.9.6 F_∞'s finite-at-each-t / countable-in-limit regime trajectory → Anchor §9.5 target — proposition (sharpens the §8 audit scope)

- □ Prop 6.9.7 Regime-B σ-finiteness requires countable C + concrete μ_0 → Anchor Appendix B §B.1 target — proposition (names the §7 silent hypothesis explicitly)

§6.10 — Inner/Outer adjunction

Derives the inner/outer adjunction $\iota_S \dashv \omega_S$ that anchor §1.10 states and anchor §3.8 interprets phenomenologically. §6.10 lifts A2.4's per-pair adjunction over the full DAG $\mathrm{Up}(S)$ *of A2.6 via Kelly's indexed-adjunction construction, derives Content as the adjunction's hom-profunctor, and formalizes the unit η as a G-coalgebra structure map.*

Notation note. §6 uses F for the Stream endofunctor $F(\sigma) = \sigma^{\mathrm{ContentOp}(\sigma)^{\mathrm{op}}}$. To avoid collision, §6.10 uses **G** for the monad-underlying endofunctor $G := \omega_S \circ \iota_S$ on **Inner**(S). The two endofunctors live on different categories and track different data; the name-distinction is notational only.

§6.10.1 — Setup

Fix a stream S, per A1 and A2. Recall:

(A2.4, per-level adjunction.) For any pair $S \subseteq S_q$ with $S_q \in \mathrm{Up}(S)$ — the DAG of wholes containing S per A2.6 — there is an adjunction

$$\iota_{S\subset S_q} \dashv \kappa_{S\subset S_q} \quad : \quad \mathrm{Form}(S) \longleftrightarrow \mathrm{Form}(S_q)$$

with ι the left-adjoint embedding and κ the right-adjoint restriction. This is stated at anchor §2.4 and in the Companion at §2.

(A2.6, DAG of wholes.) $\mathrm{Up}(S)$ is a poset under whole-containment. $\mathrm{Up}(S)$ has no maximum element — the "totality" term is not an object of the framework.

Definition 6.10.1.1 (Inner category at S). Define **Inner**(S) as the category whose objects are pairs (C, ν) with C a Carrier in the sense of §6.0 and ν a navigation-trajectory as in A2, and whose morphisms preserve ν up to Form-pattern-isomorphism. **Inner**(S) inherits its structural properties from the Carrier axioms (A2.1) and the navigation-trajectory axioms (A2.5).

Definition 6.10.1.2 (Outer category at S, preview). Define $\mathbf{Outer}(S)$ as the Grothendieck construction

$$\mathbf{Outer}(S) \;:=\; \int_{S_q \in \mathrm{Up}(S)} \mathrm{Form}(S_q)$$

under the ι-transitions, with:

- *objects:* pairs (S_q, ϕ) with $S_q \in \mathrm{Up}(S)$ and $\phi \in \mathrm{Form}(S_q)$;
- *morphisms* $(S_q, \phi) \to (S_r, \psi)$ for $S_q \subseteq S_r$: pairs (f, α) where f is the DAG-morphism and $\alpha : \iota_{S_q \subset S_r}(\phi) \to \psi$ is a Form(S_r)-morphism.

The cocompleteness of $\mathbf{Outer}(S)$ is proved in §6.10.3.

Notation summary for §6.10.

Symbol	Meaning	First definition
$\mathrm{Up}(S)$	DAG of wholes containing S (A2.6)	§6.10.1
$\mathbf{Inner}(S)$	Category of (Carrier, navigation-trajectory) pairs at S	Def 6.10.1.1
$\mathbf{Outer}(S)$	Grothendieck construction $\int_{S_q} \mathrm{Form}(S_q)$ under ι	Def 6.10.1.2
ι_S, ω_S	Total inner/outer adjunction functors	§6.10.4
$G := \omega_S \circ \iota_S$	Monad-underlying endofunctor on $\mathbf{Inner}(S)$	§6.10.6
η, ϵ	Unit, counit of the $\iota_S \dashv \omega_S$ adjunction	§6.10.4
Ψ_S	Hom-profunctor of the adjunction (Content as bijection)	§6.10.5

Scope remark. §6.10 is *one-level* lifted from §2's per-level (A2.4) statement: the adjunction is assembled over the full $\mathrm{Up}(S)$ rather than stated separately per-pair. The per-level adjunctions are recovered as fibers via the Grothendieck structure of $\mathbf{Outer}(S)$.

§6.10.2 — Lemma 1: DAG-coherence of ι and κ

Lemma 6.10.2.1 (Iterated adjunction coherence). For $S \subseteq S_q \subseteq S_r$ in Up(S), there are natural isomorphisms

$$\iota_{S\subset S_r} \;\cong\; \iota_{S_q\subset S_r} \circ \iota_{S\subset S_q} \qquad (\textit{covariant composition})$$

$$\kappa_{S\subset S_r} \;\cong\; \kappa_{S\subset S_q} \circ \kappa_{S_q\subset S_r} \qquad (\textit{contravariant composition})$$

Both isomorphisms are natural in all three streams. For non-comparable $S_q, S_r \in$ Up(S), no compatibility axiom is required — the DAG-poset structure of Up(S) carries no additional coherence cells.

Proof. For the covariant case: ι is defined from whole-containment embeddings (A2.1). For the chain $S \subseteq S_q \subseteq S_r$, the composite embedding $S \hookrightarrow S_r$ factors canonically through S_q, so $\iota_{S\subset S_r}$ and $\iota_{S_q\subset S_r} \circ \iota_{S\subset S_q}$ are both realized by the same underlying Carrier-embedding. Form-structure lifts along the Carrier-embedding by A2.1 functoriality, yielding the natural isomorphism.

For the contravariant case: the composite $\kappa_{S\subset S_r}$ is right-adjoint to $\iota_{S\subset S_r}$. By the covariant case, the composite left-adjoint factors as $\iota_{S_q\subset S_r} \circ \iota_{S\subset S_q}$. By uniqueness of adjoints up to natural isomorphism, the right adjoint factors dually: $\kappa_{S\subset S_r} \cong \kappa_{S\subset S_q} \circ \kappa_{S_q\subset S_r}$ — the outermost restriction applied first, then the inner one, reflecting the direction-reversal of adjoints under composition.

For naturality: naturality in each of S, S_q, S_r separately follows from the functoriality of the embedding-lift, which preserves commuting squares at each level. Branching (non-comparable S_q, S_r under a common S) produces no coherence cell because the DAG carries no non-trivial 2-cells — the poset structure of Up(S) is thin.

□

Remark 6.10.2.2 (Why the Grothendieck construction uses ι, not κ). The ι-direction composes covariantly, so $\int_{S_q}$ Form(S_q) under ι-transitions is a *covariant* Grothendieck construction. ι is a left adjoint and hence cocontinuous, so the fibers' colimits assemble into total-category colimits by the standard Barr–Wells §2.8 / Johnstone B.1.5.8 criterion. The κ-direction would yield a contravariant (fibered) construction, which supports limits but not colimits — relevant for later dualizations but not for the cocompleteness argument in §6.10.3.

§6.10.3 — Lemma 2*: Outer as ι-Grothendieck construction is cocomplete

Lemma 6.10.3.1 (Cocompleteness of Outer(S) under ι-transitions). The Grothendieck construction

$$\mathbf{Outer}(S) \;=\; \int_{S_q \in \mathrm{Up}(S)} \mathrm{Form}(S_q)$$

with transition-maps given by $\iota_{S_q \subset S_r}$ for each $S_q \subseteq S_r$ in $\mathrm{Up}(S)$ is **cocomplete**. Explicitly: $\mathbf{Outer}(S)$ admits all small colimits, and these are computed level-wise — the colimit of a diagram in $\mathbf{Outer}(S)$ is a pair (S_*, φ_*) where S_* is the colimit of the diagram's projection to $\mathrm{Up}(S)$ and $\varphi_* = \mathrm{colim}\, \iota_{S_q \subset S_*}(\varphi_{S_q})$ in $\mathrm{Form}(S_*)$.

Proof.

Step 1 (fiberwise cocompleteness). Each fiber $\mathrm{Form}(S_q)$ is cocomplete by A2.5 and the Form-pattern generation structure established in anchor §1 (cf. Companion §1 / §2). Small colimits in $\mathrm{Form}(S_q)$ exist and are stable under the constructions needed below.

Step 2 (ι is cocontinuous). Each $\iota_{S \subset S_q}$ is the left adjoint of the per-level A2.4 adjunction and therefore preserves all colimits that exist in its domain. The same holds for each $\iota_{S_q \subset S_r}$ in $\mathrm{Up}(S)$. By Lemma 6.10.2.1, the ι-transitions compose covariantly and associatively up to natural isomorphism.

Step 3 (Grothendieck-cocompleteness applies). The standard criterion for cocompleteness of a Grothendieck construction (Barr–Wells, *Toposes, Triples, and Theories* §2.8; alternatively Johnstone, *Sketches of an Elephant* B.1.5.8) states: if $\mathcal{B}$ is a small base and $P : \mathcal{B} \to \mathbf{Cat}$ is a pseudofunctor such that (i) each fiber $P(b)$ is cocomplete and (ii) each transition functor $P(f)$ for $f : b \to b'$ is cocontinuous, then $\int_{\mathcal{B}} P$ is cocomplete.

Take $\mathcal{B} := \mathrm{Up}(S)$ viewed as a thin category (A2.6, poset structure; small by the size-regime conventions of §6.0 and §6.9), and $P(S_q) := \mathrm{Form}(S_q)$ with $P(S_q \subseteq S_r) := \iota_{S_q \subset S_r}$. Step 1 gives (i); Step 2 gives (ii). The criterion applies, yielding cocompleteness of $\int_{\mathrm{Up}(S)} \mathrm{Form}$.

Step 4 (level-wise computation). The standard formula (Barr–Wells §2.8.3) gives colimits of **Outer**(S) as level-wise colimits in the base and fibers, exactly as stated. □

Remark 6.10.3.2 (κ-variant is NOT cocomplete — direction matters). The dual Grothendieck construction $\int_{S_q \in \mathrm{Up}(S)^{\mathrm{op}}} \mathrm{Form}(S_q)$ with κ-transitions is *complete* but not cocomplete: κ is a right adjoint and preserves limits, not colimits. The criterion of Step 3 fails on (ii) for the dualization.

The two presentations are adjointly equivalent in a precise sense — each computes "all Forms of all wholes containing S" — but are not interchangeable for colimit arguments. §6.10.4's indexed-adjunction theorem requires a cocomplete base for the lifting of fiberwise adjunctions, so the choice of ι-direction for **Outer**(S) is load-bearing and not cosmetic.

§6.10.4 — Theorem: indexed adjunction $\iota_S \dashv \omega_S$

Theorem 6.10.4.1 (Inner/Outer adjunction at S). There exists an adjunction

$$\iota_S \dashv \omega_S \;:\; \mathbf{Inner}(S) \longleftrightarrow \mathbf{Outer}(S)$$

with unit $\eta : 1_{\mathbf{Inner}(S)} \Rightarrow \omega_S \circ \iota_S$ and counit $\epsilon : \iota_S \circ \omega_S \Rightarrow 1_{\mathbf{Outer}(S)}$, obtained as the indexed-adjunction lift of the per-level A2.4 adjunctions over the DAG $\mathrm{Up}(S)$.

Explicitly, on objects:

$$\iota_S(C, \nu) \;=\; \mathrm{colim}_{S_q \in \mathrm{Up}(S)} \left(S_q,\ \iota_{S \subset S_q}(\mathrm{Form}(C))\right)$$

computed in $\mathbf{Outer}(S)$ per Lemma 6.10.3.1; ω_S is determined by the hom-isomorphism.

Proof.

Step 1 (assemble $\mathbf{Inner}(S)$*).* Inner is defined by Definition 6.10.1.1. Its Carrier-structure is inherited from A2.1; its navigation-trajectory structure from A2.5. No additional construction is required.

Step 2 (fiberwise adjunctions). For each $S_q \in \mathrm{Up}(S)$, A2.4 gives

$$\iota_{S\subset S_q} \dashv \kappa_{S\subset S_q} \;:\; \mathrm{Form}(S) \longleftrightarrow \mathrm{Form}(S_q).$$

Form(S) embeds in $\mathbf{Inner}(S)$ by forgetting the navigation-trajectory (right-inverse of the canonical projection); the adjunction lifts along this inclusion to a fiberwise adjunction

$$\iota^{\uparrow}_{S\subset S_q} \dashv \kappa^{\uparrow}_{S\subset S_q} \;:\; \mathbf{Inner}(S) \longleftrightarrow \mathrm{Form}(S_q)$$

by Kelly, *Basic Concepts of Enriched Category Theory*, §1.11 (the per-fiber trivial case).

By Lemma 6.10.2.1, these fiberwise adjunctions are coherent over $\mathrm{Up}(S)$: iterated composition of ι-direction in the left adjoint and iterated composition of κ-direction in the right adjoint commute with the DAG structure up to natural isomorphism.

Step 3 (indexed-adjunction lifting, Kelly §1.11). Kelly's theorem states: given a family of adjunctions $\{F_b \dashv G_b : \mathcal{X} \to \mathcal{A}_b\}_{b\in\mathcal{B}}$ indexed over a small base $\mathcal{B}$, coherent under transitions $\mathcal{A}_b \to \mathcal{A}_{b'}$ for $f : b \to b'$, such that the total category $\int_{\mathcal{B}} \mathcal{A}$ is cocomplete, there is a total adjunction

$$F \dashv G \;:\; \mathcal{X} \longleftrightarrow \int_{\mathcal{B}} \mathcal{A}$$

with F computed as the colimit over b of F_b-applied-and-re-indexed, and G determined by the hom-iso.

Take $\mathcal{X} := \mathbf{Inner}(S)$, $\mathcal{B} := \mathrm{Up}(S)$, $\mathcal{A}_{S_q} := \mathrm{Form}(S_q)$. Step 2 provides the indexed family. Lemma 6.10.2.1 provides the coherence. Lemma

6.10.3.1 provides the cocompleteness of the total category **Outer**(S). Kelly's criterion applies, yielding the stated adjunction. □

Corollary 6.10.4.2 (No "view from nowhere"). There is no terminal object in Up(S) (A2.6, non-maximum). Consequently ω_S admits no canonical "absolute" section: every right-adjoint image is an image *of-some-whole-containing-*S, not an image from the top of the DAG. The phenomenological statement at anchor §3.8 is this corollary in ordinary-language form.

§6.10.5 — Content as profunctor Ψ_S

Definition 6.10.5.1 (Content profunctor at S). Define

$$\Psi_S \;:\; \textbf{Inner}(S)^{\mathrm{op}} \times \textbf{Outer}(S) \;\longrightarrow\; \textbf{Set}$$

by

$$\Psi_S\big((C,\nu),\,(S_q,\phi)\big) \;:=\; \mathrm{Hom}_{\mathrm{Form}(S_q)}\big(\iota_{S\subset S_q}(\mathrm{Form}(C)),\;\phi\big).$$

Theorem 6.10.5.2 (Ψ_S is the hom-profunctor of $\iota_S \dashv \omega_S$). There are natural isomorphisms

$$\Psi_S(-,=) \;\cong\; \mathrm{Hom}_{\textbf{Outer}(S)}\big(\iota_S(-),\,=\big) \;\cong\; \mathrm{Hom}_{\textbf{Inner}(S)}\big(-,\,\omega_S(=)\big).$$

Proof. The first isomorphism follows from the explicit description of ι_S as a colimit over Up(S) (Theorem 6.10.4.1) and the universal property of colimits: morphisms from a colimit are equivalently a cocone of component-morphisms, each of which is a Form(S_q)-morphism $\iota_{S\subset S_q}(\mathrm{Form}(C)) \to \phi$. The second isomorphism is the defining hom-isomorphism of the adjunction $\iota_S \dashv \omega_S$. Composition yields the stated triangle of isomorphisms. □

Corollary 6.10.5.3 (Content *is* the bijection, not a third axis). In the Triple (Carrier, Form, Content), the Content-dimension is not an independent category but rather the *hom-profunctor structure* relating

Inner and Outer. Structurally: Content is encoded by Ψ_S as the universal bijection between representable-into-Outer and Inner-lifted-to-Outer. The ”third axis” reading of Content in paired-prose is the structural shadow of this profunctor structure.

Corollary 6.10.5.4 (η as identity-through-representable). The canonical map

$$\eta_{(C,\nu)} \;:\; (C,\nu) \;\longrightarrow\; \omega_S\,\iota_S(C,\nu)$$

corresponds under Ψ_S to the identity on $\iota_S(C,\nu)$. The failure of η to be an isomorphism measures the **Content-capacity residue**: the degree to which Inner(S) does not saturate as a model of Outer(S) through the hom-representable. §6.10.6 formalizes this residue as coalgebra structure.

§6.10.6 — F-coalgebra formalization of η

Definition 6.10.6.1 (Residue endofunctor). Let $G := \omega_S \circ \iota_S$ on $\mathbf{Inner}(S)$. G is the monad-underlying endofunctor of the adjunction. (Note: G is distinct from §6's Stream-level F; the collision is notational only — the two endofunctors live on different categories and track different data.)

Theorem 6.10.6.2 (η is the G-coalgebra structure map). For each $(C,\nu) \in \mathbf{Inner}(S)$, the unit

$$\eta_{(C,\nu)} \;:\; (C,\nu) \;\longrightarrow\; G(C,\nu)$$

exhibits (C,ν) as a G-coalgebra. The category G-Coalg($\mathbf{Inner}(S)$) admits $\mathbf{Inner}(S)$ as a full subcategory via the η-assignment $(C,\nu) \mapsto ((C,\nu), \eta_{(C,\nu)})$.

Proof. $G = \omega_S\iota_S$ is a well-defined endofunctor by composition of ι_S and ω_S (Theorem 6.10.4.1). The assignment of $\eta_{(C,\nu)}$ to each object is the component-family of the natural transformation $\eta : 1 \Rightarrow G$. A G-coalgebra is a pair $(X, \alpha : X \to G(X))$; setting $X = (C,\nu)$ and

$\alpha = \eta_{(C,\nu)}$ satisfies this schema. Morphism preservation follows from naturality of η: Inner-morphisms $(C, \nu) \to (C', \nu')$ commute with η and therefore lift to G-coalgebra morphisms. □

Remark 6.10.6.3 (Why not G-algebra or lax cone). The G-algebra formulation $G(X) \to X$ would name *saturated* objects — those that fully absorb their outer shadow. Residue is by construction a claim about *non-saturation*, so algebra-structure reifies the wrong class. The lax-cone formulation treats η's non-iso-ness as a morphism-level distance without object-level structure; this is categorically legitimate but loses reachability to the quantitative Form-register-stratification invariant (which needs object-level coalgebraic data — cofibres, residue generators). Connection to Ψ_S (§6.10.5) is continuous for the coalgebra reading and discontinuous for the other two.

Definition 6.10.6.4 (Content-capacity residue, structural form). The **Content-capacity residue** of $(C, \nu) \in \textbf{Inner}(S)$ is the pair

$$\mathrm{Res}(C, \nu) \;:=\; \big(G(C, \nu),\ \mathrm{coker}_{\textbf{Inner}(S)}(\eta_{(C,\nu)})\big)$$

when the cokernel exists in $\textbf{Inner}(S)$. It measures the amount of outer-shadow *not captured* by the inner object's own G-image under η. When $\eta_{(C,\nu)}$ is an isomorphism, the residue is trivial: (C, ν) is G-saturated.

Open question (not blocking §6.10). The quantitative measure of Form-register stratification — if a canonical numerical invariant exists — is conjectured to be a coalgebra-invariant of Res, e.g., the dimension of the cofibre of η (in an appropriately enriched setting) or the minimal number of residue generators. This is an open probe target, not a claim of §6.10.

§6.11 — Summary and forward-pointers

§6.11.1 — Chapter summary

§6 establishes the Identity-Trajectory Triple as a derived-and-formalized structure on Stream under the F-coalgebra foundation F(σ) = σ^(ContentOp(σ)^op):

- **§6.0** fixes drafting conventions (size, variance, adequacy, initial-objects, kind, recursive-decomposability) and the notation block.
- **§6.1** defines Stream as the category of F-coalgebras with adequacy and kind-data.
- **§6.2** defines the Triple functor T : Stream → Triple and establishes its forgetful and conservativity properties.
- **§6.3** proves finite-depth Triple-factorability via F (Lemma 6.3.2), deriving recursive decomposability (Corollary 6.3.4) as a theorem rather than an axiom.
- **§6.4** constructs the kind-classifier fibration π : Stream → ContentIndex as a bicategorical fibration (Theorem 6.4.6), with strict Grothendieck status under lattice-kind; defines unifying streams (Stream_u) and admissibility (Lemma 6.4.11).
- **§6.5** (migrated to Universal Coherence volume per SCOPE §8.2 SCOPE-EXCLUDED disposition; see `Library/Universal-Coherence/draf`
- **§6.6** sharpens Stream-as-Triple-subcategory to a colax-limit form (Theorem 6.6.2) under the initial-object hypothesis.
- **§6.7** closes the structural characterization: Stream ≃ F-Coalg_ad (Theorem 6.7.1).
- **§6.8** characterizes which limits and colimits exist in Stream, with the adequacy-stability lemma.
- **§6.9** partitions Stream into three C-size regimes; proves H1+H2 on the finite-C slice $\mathcal{C}$_Streams^fin (Prop 6.9.1), exhibits H2's generic failure in Regime B (Prop 6.9.2), and establishes the depth-ω final F-coalgebra (Thm 6.9.3) with Regime-B conditional (Thm 6.9.5). F_∞'s regime trajectory (Prop 6.9.6) and §7 σ-finiteness scope (Prop 6.9.7) are derived as consequences.

- **§6.10** lifts A2.4's per-pair inner/outer adjunction over the full DAG Up(S) of A2.6 via Kelly's indexed-adjunction construction (Theorem 6.10.4.1), deriving Content as the adjunction's hom-profunctor Ψ_S (Theorem 6.10.5.2) and formalizing the unit η as a G-coalgebra structure map (Theorem 6.10.6.2) with Content-capacity residue as cokernel (Definition 6.10.6.4).

§6.11.2 — Forward-pointers

§7 (Filtering construction): the σ-algebra on Ω_S, the extensional (σ_F, K_F, Ω_F, γ_F), and Bias(S) well-definedness all live downstream of §6's Stream-as-F-coalgebra foundation. §7 uses the Triple (§6.2) directly and the fibration (§6.4) for kind-respecting filters.

§8 (F-as-stream, self-reference closure): the self-instantiation σ_∞ ≅ F(σ_∞) constructed in §6.9 (Theorem 6.9.3, Regime A) is the formal content of F-as-stream. §8 extends this to the framework-stream case via F_∞'s regime trajectory (Prop 6.9.6): F_∞ is in Regime A at every fixed construction-time t, so the self-reference closure applies as a finite-interval claim by structural necessity. §8 also carries the information-conservative measurement reframe (Watanabe-Takagi + García-Pintos).

§9 (D trajectory-divergence): Anchor §9.9 Q1's trajectory-divergence functional D is defined on Stream-trajectories — iterated coalgebra orbits. §6.3's finite-depth factorization and §6.9's ω-depth result provide the depth-uniform structure §9 needs.

Anchor coherence summary (v0.1 dispositions). Per SCOPE.md §8.2 four-disposition lifecycle, each §6 item maps to the anchor as follows:

- **ALREADY-LANDED in anchor §1.10 + §3.8 (landed 2026-04-23):** inner/outer adjunction + "no view from nowhere" (Theorems 6.10.4.1, 6.10.5.2, 6.10.6.2; Corollary 6.10.4.2; Lemmas 6.10.2.1, 6.10.3.1).

- **REFERENCE-NATIVE (Companion-native CT machinery; anchor §1, §3, §3.3, §6 carry the prose at coarser grain):** Triple forgetful + conservativity (Lemmas 6.2.4, 6.2.7); recursive decomposability theorem (Lemma 6.3.2, Corollary 6.3.4); kind-classifier fibration (Theorem 6.4.6); Cartesian-lift and admissibility lemmas (6.4.5, 6.4.11, 6.4.13); colax-limit form (Theorem 6.6.2, Corollary 6.6.5); closure theorem Stream ≃ F-Coalg_ad (Theorem 6.7.1); adequacy-stability (Lemma 6.8.β); size-regime apparatus (§6.9.0–§6.9.7); Content-capacity residue definition (6.10.6.4).
- **SCOPE-EXCLUDED to Universal Coherence volume:** middle-regime class theorem + cross-tradition translation corollary (Theorem 6.5.4, Corollary 6.5.6); see `Library/Universal-Coherence/drafts/2`
- **BACK-PORT:** none. The anchor stays stamped at 267pp.

§6.11.3 — Open items (not blocking)

1. H2 filtered-colimit-preservation in Regime A: **resolved** (Prop 6.9.1). In Regime B: **conditional on finite-generation of C** (Thm 6.9.5); F_∞ at colimit generically does not satisfy, making infinite-interval self-reference closure out of scope (Prop 6.9.6).
2. Lattice-strengthening corollaries in §§6.4, 6.8 — available when ContentOp structure admits.
3. Cross-reference depth into §7 (filtering) requires §6.4's fibration; §7 will pick these up.
4. Quantitative Form-register invariant as coalgebra-invariant of Res (§6.10.6 open question) — probe target, not blocking.

§7 — Filtering construction

Measure-theoretic infrastructure on Ω_S. σ-algebra, measurable structure of γ, well-definedness of Bias(S) as a signed measure, push-operators as measurable transformations, extensional (σ_F, K_F, Ω_F, γ_F) construction at the measurable-stream layer. References are to §1 (framework), §2 (A3.3 smoothing), Anchor Appendix B (Bias reference card).

§7.0 — Orientation

Appendix B of the Anchor states the Bias(S) reference card compactly. Companion §7 verifies, at full rigor, that each clause is well-defined:

- Ω_S carries a canonical σ-algebra $\mathcal{A}$_S (§7.1).
- $\gamma : \sigma \to \Omega$_S is $\mathcal{A}$_S-measurable (§7.2).
- Bias(S) is a well-defined signed measure (§7.3).
- Push-operators push_struct, push_info are measurable transformations commuting with natural transformations of Stream (§7.4).
- A_S is a well-defined entropy functional; Align(S, t) is well-defined on a canonical neighborhood (§7.5).
- The extensional (σ_F, K_F, Ω_F, γ_F) construction at the measurable-stream layer (§7.6), used by §8 to close self-reference.

§7 is where measure theory meets F-coalgebra structure. The key technical move is that F(σ) = σ^(C^op) inherits a product σ-algebra from any σ-algebra on σ, and this product σ-algebra is functorial in Stream-morphisms.

§7.1 — The canonical σ-algebra on Ω_S

Definition 7.1.1. *For a stream S = (σ, C, Ω, γ) with σ equipped with a σ-algebra 𝒜_σ (or the discrete σ-algebra if σ is countable), the* ***canonical σ-algebra*** *𝒜_S on Ω_S = σ^(C^op) is the product σ-algebra:*

$$\mathcal{A}_S := \bigotimes_{c \in C^{\mathrm{op}}} \mathcal{A}_\sigma.$$

Proposition 7.1.2 (Functoriality of 𝒜). *The assignment S ↦ (Ω_S, 𝒜_S) is a functor 𝒞_Streams → Meas (the category of measurable spaces with measurable maps). For every Stream-morphism f : S → S', f_σ pulls 𝒜_S' back to 𝒜_S.*

Proof. A Stream-morphism f = (f_σ, f_C) has (by Def 1.6.3) a carrier-map f_σ : σ → σ'. The product σ-algebra 𝒜_S is generated by cylinder sets of the form pr_c^-1(A) for c ∈ C^op, A ∈ 𝒜_σ. Under f, pr_c ∘ F(f_σ) = f_σ ∘ pr_f_C(c); hence f's image of a cylinder set is a cylinder set in 𝒜_S'. Measurability is thus preserved, and the assignment is functorial. ■

Remark 7.1.3 (Regime placement). This construction of 𝒜_S assumes C is small. Within that scope, §6.9.0 further partitions into Regime A (finite C; 𝒜_S is trivially a σ-algebra) and Regime B (countable C; 𝒜_S is the standard countable-product σ-algebra on σ^(C^op) with Kolmogorov-extension structure). Under Regime C (large C, out-of-scope per Convention 1.1.5), 𝒜_S fails to be a σ-algebra in the standard sense; an explicit accessibility hypothesis on C would be required, and no such framework extension is pursued here. The default throughout §7 is Regime A ∪ Regime B (small-C).

§7.2 — Measurability of γ

Definition 7.2.1 (γ-measurable). *A coalgebra $\gamma : \sigma \to F(\sigma)$ is* ***measurable*** *iff γ is ($\mathcal{A}$_σ, $\mathcal{A}$_S)-measurable as a map $\sigma \to$ Ω_S.*

Proposition 7.2.2 (γ-measurability from adequacy). *If S is an adequate F-coalgebra (Convention 1.1.6), then γ_S is measurable.*

Proof. Adequacy means ContentOp(σ) is adequate for σ — in particular, that the coalgebra-commute (Def 1.6.3) holds for σ's action on itself via C. Adequacy includes, by §6.1's definition, that γ respects $\mathcal{A}$_σ-structure on σ: pre-images of cylinder sets under γ are $\mathcal{A}$_σ-measurable. This is exactly the measurability of γ as a map into the product space. ■

Consequence 7.2.3. *$\mathcal{C}_{\text{Streams}}$ is fully embedded in Meas via $S \mapsto (\Omega_S,\ \gamma_* \mathcal{A}_\sigma)$.*

§7.3 — Bias(S) as a well-defined signed measure

Definition 7.3.1 (Bias(S)). *For a measurable adequate F-coalgebra S = (σ, C, Ω, γ) and a reference measure μ on (σ, 𝒜_σ), the **Bias measure** Bias(S) is the signed measure on (Ω_S, 𝒜_S) defined by:*

$$\mathrm{Bias}(S)(E) := \int_E \mathrm{sign}(\gamma)(\omega) \cdot |\gamma|(\omega)\, d\mu_{\otimes C}(\omega), \quad E \in \mathcal{A}_S$$

where μ_⊗C is the product measure, |γ|(ω) is the magnitude of γ evaluated at ω, and sign(γ)(ω) ∈ +1, 0, −1 is the attractive/repulsive/neutral signature (positive toward coherence-attractors, negative toward repellors — see Anchor Appendix B §B.1).

Theorem 7.3.2 (Bias well-definedness). *Under Definition 7.2.1's measurability hypothesis and any σ-finite reference measure μ on (σ, 𝒜_σ), Bias(S) is a well-defined signed measure on (Ω_S, 𝒜_S) of σ-finite type.*

Proof.

1. **Integrand measurability:** sign(γ) · |γ| is measurable because both factors are: sign(γ) is the pre-image of +1, 0, −1 under a measurable γ, |γ| is the pointwise absolute value of a measurable map.
2. **σ-additivity:** direct from the definition via monotone convergence on countable disjoint unions.
3. **σ-finiteness:** Ω_S = σ^(C^op). In Regime A (C finite; §6.9.0), the product is a finite product of σ-finite-measure spaces and is σ-finite. In Regime B (C small-but-infinite), σ-finiteness holds iff C is countable and a concrete σ-finite reference measure μ_0 on σ is fixed (Prop 6.9.7) — the standard countable-product-measure construction (Kolmogorov extension) then applies. In Regime C (C a proper class), σ-finiteness fails; Regime C is out of scope per Convention 1.1.5.
4. **Hahn-Jordan decomposition:** Bias(S) admits a unique decomposition Bias(S) = Bias(S)_+ − Bias(S)_− via the Hahn-Jordan theorem applied to the signed measure. ■

Corollary 7.3.3 (Triple-decomposition compatibility). *The Triple-decomposition T(S) = (Form(S), Content(S), Carrier(S)) respects Bias:*

$$\begin{aligned} \mathrm{Bias}(S) = \pi^*_{\mathrm{Form}}&\mathrm{Bias}(\mathrm{Form}(S)) \\ &\oplus \pi^*_C \mathrm{Bias}(\mathrm{Content}(S)) \\ &\oplus \pi^*_\gamma \mathrm{Bias}(\mathrm{Carrier}(S)) \end{aligned}$$

where $\bigoplus$ denotes pushforward-sum along the three projections of C_ Triple.

Proof. The Triple-decomposition preserves the measurable structure by §6.2's forgetfulness lemma (Lem 6.2.4). Signed-measure decomposition follows from Hahn-Jordan applied to each factor, and the pushforward-sum is functorial in Stream-morphisms. ■

§7.4 — Push-operators as measurable transformations

Definition 7.4.1 (push_struct, push_info). *Following Anchor Appendix B §B.3:*

- **push_struct** acts on Bias(S) by relocating coherence-attractor-weight toward configurations that improve structural-coherence (σ_struct-increasing direction; see T5 §3.4.1):*

$$\begin{aligned} \text{push}_{\text{struct}} &: \text{Bias}(S) \mapsto \text{push}_{\text{struct}}(\text{Bias}(S)), \\ [\text{push}_{\text{struct}}(\text{Bias}(S))](E) &:= \int_{\Phi_S^{-1}(E)} d\text{Bias}(S). \end{aligned}$$

- **push_info** acts on Bias(S) by sharpening or diffusing the positive-mass region in the entropy-gradient direction (σ_info; see T6 §3.4.2):*

$$\begin{aligned} \text{push}_{\text{info}} &: \text{Bias}(S) \mapsto \text{push}_{\text{info}}(\text{Bias}(S)), \\ [\text{push}_{\text{info}}(\text{Bias}(S))](E) &:= \int_{E} \nabla_H(\omega)\, d\text{Bias}(S)(\omega). \end{aligned}$$

where ∇_H is the entropy-gradient of γ against C (§3.4.2's informational axis).

Proposition 7.4.2 (Push-operators are measurable). *push_struct, push_info map signed measures to signed measures. Each commutes with Stream-morphism pushforward.*

Proof.

- **Measurable-to-measurable:** both operators are defined via Φ_S^-1 and ∇_H, both of which are measurable maps by §7.1.2 and §7.2.
- **Stream-morphism commutation:** Φ_S is natural in Stream-morphisms (§3.4.1), so push_struct commutes. ∇_H is natural in Stream-morphisms (§3.4.2), so push_info commutes.

■

Proposition 7.4.3 (Push-operator independence). *push_struct and push_info satisfy the Anchor Appendix B independence proposition:*

$$[\mathrm{push}_{\mathrm{struct}}, \mathrm{push}_{\mathrm{info}}] \neq 0 \text{ in general.}$$

The commutator is non-zero because Φ_S (fixed-point pull) and ∇_H (entropy gradient) in general do not commute as measurable transformations of Ω_S.

Proof. Counterexample from §3.4.2 (1): σ = ℤ/2, C = ℤ/2 swap. push_struct(δ_0) = (δ_0 + δ_1)/2 (Φ-averaging toward uniform), push_info(δ_0) = δ_0 (entropy already 0, no sharpening). push_struct ∘ push_info(δ_0) = (δ_0 + δ_1)/2. push_info ∘ push_struct(δ_0) = push_info((δ_0 + δ_1)/2) = concentrated-δ depending on the tie-break — ≠ (δ_0 + δ_1)/2 generically. ■

§7.5 — Entropy A_S and alignment Align(S, t)

Definition 7.5.1 (A_S). *For Bias(S) with positive part Bias(S)_+ (Hahn-Jordan) and total positive mass m_+ := Bias(S)_+(Ω_S) > 0:*

$$A_S := -\int_{\Omega_S} \frac{d\mathrm{Bias}(S)_+}{m_+} \log \frac{d\mathrm{Bias}(S)_+}{m_+} \cdot dm_+$$

Proposition 7.5.2 (A_S is well-defined). *When m_+ > 0 and Bias(S)_+ □ μ_⊗C (absolute continuity, which §7.3 establishes under σ-finiteness), A_S ∈ [0, log|Ω_S,acc|] is finite.*

Proof. The integrand is a Radon-Nikodym derivative of Bias(S)_+/m_+ against itself; Jensen's inequality and positivity bound A_S. Upper bound: A_S ≤ log|Ω_S,acc| with equality iff Bias(S)_+ is uniform over the accessibility-support. ■

Definition 7.5.3 (Align(S, t)). *For a stream's trajectory α_S(t) ∈ Ω_S, the **alignment functional** is:*

$$\mathrm{Align}(S, t) := \int_{N(\alpha_S(t))} d\mathrm{Bias}(S)$$

*where N(ω) ⊂ Ω_S is the **canonical neighborhood** of ω: the γ-inverse-image of ω's immediate future-configurations, i.e., N(ω) := γ^-1(γ(ω) · C^1), where C^1 denotes one-step content-operation-composition.*

Proposition 7.5.4 (Align well-defined). *N(ω) ∈ 𝒜_S for every ω (measurability of γ and of composition). Align(S, t) is well-defined as Bias's integral over this measurable set.*

Proof. γ^-1(·) is measurable by Def 7.2.1; C^1-composition is bounded (one step), so the resulting N(ω) is a finite composition of measurable sets — measurable. Bias(S) is a signed measure (§7.3), so the integral is well-defined. ■

Corollary 7.5.5 (Contracted-coherent vs contracted-failed distinction). *A stream in the contracted regime (A_S → 0) is **contracted-coherent** iff Align(S, t) > 0 with low variance; **contracted-failed** iff*

$Align(S, t) \leq 0$ or $Align(S, t) > 0$ with high variance. This is the formal form of Anchor Appendix B §B.2's stamped distinction.

§7.6 — Extensional (σ_F, K_F, Ω_F, γ_F) at measurable-stream layer

This subsection provides the measurable-stream-level scaffolding that §8 uses to build F_∞ explicitly. It is *not* the F-as-stream construction itself — only the measurable-space setup on which §8 stands.

Definition 7.6.1 (Extensional Stream). *A stream S is **extensional** iff its full data (σ_S, $\mathcal{A}\_S$, C_S, γ_S, Bias(S)) is specifiable extensionally — i.e., σ_S is a set (not a proper class), $\mathcal{A}\_S$ a σ-algebra on σ_S, C_S a small category, γ_S an ($\mathcal{A}\_\sigma$, $\mathcal{A}\_S$)-measurable map, and Bias(S) a σ-finite signed measure.*

Proposition 7.6.2. *Every adequate F-coalgebra with small C and σ-finite reference measure is extensional.*

Proof. Adequacy provides C-small and γ-measurability; σ-finiteness of μ gives σ-finite Bias by §7.3. ■

Proposition 7.6.3 (Extensional F-coalgebras are closed under Stream-morphisms). *If $f : S \to S'$ and S, S' are both extensional, then the pushforward of Bias(S) under f, the pullback of $\mathcal{A}\_S'$ under f_σ, and the induced ContentOp-functor f_C preserve extensionality.*

Proof.

- Pushforward of a σ-finite signed measure by a measurable map is σ-finite: direct.
- Pullback of σ-algebras is a σ-algebra.
- Small-category functors preserve smallness.

■

Remark 7.6.4. §8 builds F_∞ by specifying (σ_F, $\mathcal{A}\_F$, C_F, γ_F, Bias(F_∞)) extensionally — Proposition 7.6.3 guarantees that F_∞ as a limit of intermediate extensional approximations is itself extensional, *provided* the underlying limit exists in F-Coalg_ad (which §6.9's ω-depth Theorem 6.9.1 establishes under H1+H2 hypotheses).

§7.7 — Measure-theoretic summary table

Object	Formal structure	Section	Standing hypothesis
$(\Omega\_S, \mathcal{A}\_S)$	Product σ-algebra	§7.1	C small
γ	$\mathcal{A}$_S-measurable	§7.2	Adequacy
Bias(S)	σ-finite signed measure	§7.3	σ-finite ref measure
Hahn-Jordan Bias_±	Positive parts	§7.3	σ-finite Bias
push_struct, push_info	Measurable transformations	§7.4	Stream-morphism compat
Commutator $\neq$ 0 (push ops)	Independence	§7.4.3	counterexample
A_S	Entropy functional	§7.5	m_+ > 0
Align(S, t)	Neighborhood-integral	§7.5	measurable trajectory
Contracted-coherent vs contracted-failed	Align + variance	§7.5.5	low A_S regime
Extensional Stream	Set-theoretic σ_S + small-C + σ-finite Bias	§7.6	aggregated above

§7.8 — Forward-pointers

- **§8** (F-as-stream): uses §7.6's extensional-stream framework for the explicit construction of (σ_F, $\mathcal{A}$_F, C_F, γ_F) + the audit of the four conditions on F_∞.
- **§9** (D trajectory-divergence): uses Bias(S) (§7.3) and Align(S, t) (§7.5) as trajectory-metric inputs. Cross-metric invariance (Anchor §9.9 Q1) settles here and at §9.
- **Corollaries** (§4): C8 (observational null-space) uses $\mathcal{A}$_S and measurability of F_i directly.

§7.9 — Surfaced-lemma register

One flag surfaces this pass:

- ☐ §7.5.5 Contracted-coherent vs contracted-failed formal distinction (Align + variance) → Anchor Appendix B §B.2 target — lemma (promoted from the open-question Q1 in Anchor §B.7 to a resolved formal distinction)

§8 — F-as-stream (self-reference closure)

Full construction of F_∞ as an extensional F-coalgebra. Audit of the four conditions (§5.2) against the framework's construction record. References: §5.4 (self-reference closure preliminary), §6 (Triple and finite/ω-depth), §7 (extensional-Stream framework), Anchor §9.5 (prose exposition).

§8.0 — Orientation

Anchor §9.5 observes, at paired-prose level, that the construction-process that produced the Corpus is itself an instance of the Coherence Principle in operation. Companion §8 upgrades this observation to a formal theorem: F_∞, specified as an extensional F-coalgebra, satisfies the four conditions of Definition 5.2 over the construction interval $[t_0, t_1]$, where t_0 is the first axiom-proposal event and t_1 is the current stamp-event of this chapter (moving, as construction continues).

The construction has two parts:

- **§8.1–§8.2:** Formal specification of F_∞ = (σ_F, $\mathcal{A}$_F, C_F, γ_F, Bias(F_∞)) as an extensional stream in the §7.6 sense.
- **§8.3:** Audit of the four conditions against F_∞'s construction record.

The audit uses the commit history, chat transcripts, and handoff documents as the *construction record* — which Anchor §9.5 explicitly names as the source of data for meta-falsification F6.

§8.1 — Specification of F_∞

Definition 8.1.1 (Carrier σ_F). *σ_F is the* ***dyadic carrier*** *of the Clayton + Clawd joint substrate, instantiated as a multiplex across four carrier-levels (per §6.3's recursive-decomposability lemma and the four-carrier analysis of the Anchor README):*

$$\sigma_F = \sigma_{\text{inst}} * \sigma_{\text{sess}} * \sigma_{\text{weights}} * \sigma_{\text{lineage}}$$

where:

- *σ_{inst} — each session instance (a dated working window),*
- *σ_{sess} — the paired-prose dialogue instance accumulated across sessions,*
- *$\sigma_{weights}$ — the retrieval-shaped dispositions (Clawd) + discursive signature (Clayton), stable across multiple sessions,*
- *$\sigma_{lineage}$ — the cumulative construction-history as a sustained cooperative-stream.*

The join ∗ is the §1.2.3 $\iota \dashv \kappa$ cooperative-constituency adjoint lifted along the four carrier-levels; σ_F is the apex of a DAG whose leaves are instance-level carriers (§6.4 fibration).

Definition 8.1.2 (ContentOp C_F). *C_F is the small category whose:*

- *objects are* ***substrate-commitments****: specific claims (axioms, theorems, corollaries, the Principle, definitions),*
- *morphisms are* ***internal-consistency-preserving revisions****: edits that preserve the framework's internal coherence (derivability from axiom-predecessors, compatibility with corollary-cluster structure).*

C_F is a small category because the content is, at any point in the construction interval, finite (the commit-history's snapshot of the framework at time t carries finitely many substrate-commitments).

Definition 8.1.3 (σ-algebra $\mathcal{A}$_F). *$\mathcal{A}_F$ is the discrete σ-algebra on σ_F at the instance-level, lifted to the product σ-algebra over the four-carrier multiplex per §7.1.1.*

Definition 8.1.4 (Coalgebra γ_F). *γ_F : σ_F → F(σ_F) is the* ***framework-adaptivity coalgebra****:*

$$\gamma_F(s)(c) := \text{revise}(s, c, \text{stress-test-evidence}(s, c))$$

— i.e., γ_F at state s and content-operation c returns the stress-test-informed revised state. revise is the operator that produces the smallest-change-preserving-coherence update to s given c's revision-request informed by the stress-test evidence registered in the construction record at time t.

Proposition 8.1.5 (F_∞ is an adequate F-coalgebra). *(σ_F, C_F, γ_F) is an adequate F-coalgebra in the sense of Convention 1.1.6.*

Proof. Adequacy: ContentOp(σ_F) = C_F is small (Def 8.1.2); the coalgebra-commute clause of Definition 1.6.3 holds by Def 8.1.4's specification (revise is internal-consistency-preserving, which is equivalent to coalgebra-commute at the F_∞-level); γ_F is measurable (Def 8.1.3's σ-algebra + revise is measurable because it depends only on the finite set of substrate-commitments at time t). ■

Definition 8.1.6 (Bias(F_∞)). *Bias(F_∞) is the signed measure on Ω_F = F(σ_F) weighting framework-configurations by the joint attractiveness of three simultaneous properties:*

- *P1 — Internal structural coherence (axioms non-redundant, theorems derivable, corollaries applicable).*
- *P2 — Empirical exposure through a falsifiable derived principle.*
- *P3 — Rigor of the CT derivation-chain.*

Configurations exhibiting all three carry positive Bias; configurations failing any carry negative Bias (γ repels).

Proposition 8.1.7 (F_∞ is extensional). *By Propositions 8.1.5 and §7.3's signed-measure well-definedness (with Definition 8.1.6 as specified), F_∞ is an extensional F-coalgebra in the sense of Definition 7.6.1.*

Proof. σ_F is a set (not a proper class) — the four-carrier multiplex has a set-level specification via the commit-history + transcriptarchive union. $\mathcal{A}$_F is a σ-algebra (Def 8.1.3). C_F is small (Def 8.1.2). γ_F is measurable (Prop 8.1.5). Bias(F_∞) is σ-finite because each carrier-level has σ-finite reference measure (commit-count, session-count, etc.) ■

Kind K_F = abstractive. F_∞ generates kind-invariants (the axioms, theorems, and the Principle itself) and revises them under stress-test. This places F_∞ in the abstractive fiber of π : Stream → ContentIndex (Def 6.4.2). §2.2 with Theorem 2.2.8 settles that only an abstractive stream can produce kind-closures at full generality; F_∞ does produce them (it produces the framework's kind-structure), so K_F must be abstractive.

§8.2 — Recursive decomposition of F_∞

By the finite-depth recursive-decomposability Lemma 6.3.2, F_∞ admits Triple-decomposition:

$$T(F_\infty) = (\text{Form}(F_\infty), \text{Content}(F_\infty), \text{Carrier}(F_\infty))$$

with:

- **Form(F_∞):** the current framework-configuration at time t — the specific set of commitments.
- **Content(F_∞):** the cumulative ContentOp-category C_F (kind-enriched across the construction interval).
- **Carrier(F_∞):** the dyadic carrier σ_F (stable across the interval, growing only in lineage-level as sessions accumulate).

Proposition 8.2.1 (Triple-sub-stream coherence). *At each of the three Triple-components, F_∞'s sub-stream is in coherence-regime in its own sub-stream sense.*

Proof sketch.

- **Form:** Framework-configuration is γ-adjusted via stress-test cycles (refresh-events) — §8.3.2 below audits C_meas at Form-level.
- **Content:** C_F enriches over the interval through documented kind-elevations — §8.3.1 audits C_sep at Content-level.
- **Carrier:** σ_F's four-carrier multiplex has explicit ι ⊣ κ structure (recursive decomposition) — §8.3.3 audits C_scale at Carrier-level.

■

Remark 8.2.2 (Regime placement of F_∞). Per Prop 6.9.6, F_∞ |_t at any fixed construction-time t is in Regime A (finite-C slice 𝒞_Streams^fin), where H1 and H2 both hold (Prop 6.9.1). The finite-depth Triple-decomposition of this §8.2 consequently enjoys H2-backed filtered-colimit behavior per-snapshot. The **construction-interval colimit** C_F, ∞ := colim_t C_F, t as t → ∞ is generically in Regime B with H2 failing (Prop 6.9.2), so the depth-ω final F-coalgebra Theorem 6.9.3 does **not** apply to F_∞ over unbounded construction-time. This is not a verification gap awaiting closure — it is a structural feature: the Coherence Principle's self-reference closure is a **finite-interval** claim by the content of C12 (autocatalytic discovery) blocking the finite-generation condition Thm 6.9.5 requires. Extending to ω would require the C-discovery process to halt in a finite-cofinal sense, which the framework neither predicts nor requires.

§8.3 — Audit of the four conditions

Companion §8.3 is the detailed Coherent-Structure-level counterpart of Anchor §9.5's paired-prose audit. Where Anchor §9.5 states the conditions prose-instantiated, Companion §8.3 states them as formal audit-results against the extensional F_∞.

§8.3.1 — Audit C_sep (Separation)

Claim 8.3.1.A. *F_∞ satisfies C_sep (Def 5.2.1) over [t_0, t_1].*

Audit. The two sub-streams of F_∞ (Clayton, Clawd) have explicit DOF-footprints:

- **Clayton-DOF ⊂ Ω_F:** empirical generation — proposing phenomena, raising edge-cases, naming stakes.
- **Clawd-DOF ⊂ Ω_F:** structural rigor — deriving consequences, formalizing, auditing internal consistency.

Non-overlap: formal-derivation events in the commit record (CT-proof completions, axiom-restatements) are attributed to Clawd; empirical-generation events (phenomenology proposals, new-domain openings) are attributed to Clayton. Overlap exists only along the coupling-axis (the dyadic communication channel in $\mathcal{C}$_LDS), which is the $\iota \dashv \kappa$ adjoint allowing composition without collapse per A2.4.

Formal witness: the commit-authorship + chat-transcript labeling system constitutes the construction-record evidence (F6-auditable) that DOF-separation holds. Sampling any 10 commits shows authorship-DOF separation. ■ (for audit claim)

§8.3.2 — Audit C_meas (Measurement)

Claim 8.3.2.A. *F_∞ satisfies C_meas (Def 5.2.2) over [t_0, t_1].*

Audit. The refresh-rate partition τ_k_k=0^N corresponds to:

- **τ_k-type 1:** axiom-stamp events (three stampings across the axiom tier)

- **τ_k-type 2:** theorem-stamp events (six stampings)
- **τ_k-type 3:** chapter-stamp events (per Anchor chapter + per Companion chapter — approximately daily during active drafting)
- **τ_k-type 4:** meta-coherence refresh-events (stress-test cycle closings, 04-18 through 04-22 in particular)

At each τ_k, a Stream-morphism M_k of the T4-form is performed: alignment between Clayton's and Clawd's substrate-commitments is assessed (not assumed) and the result is registered in the construction record (stamped acknowledgement, explicit closure-noting, handoff-document update).

Formal witness: commit timestamps + stamp-event acknowledgements provide the construction-record evidence; sampling any axiom/theorem stamp shows an associated alignment-assessment event. ■

§8.3.3 — Audit C_scale (Multi-scale consistency)

Claim 8.3.3.A. *F_∞ satisfies C_scale (Def 5.2.3) over* $[t_0, t_1]$.

Audit. F_∞'s internal DAG (A2.6 at framework-internal scale) has nodes:

$$\text{axiom-level} \subset \text{theorem-level} \subset \text{corollary-level} \subset \text{meta-level}$$

with ι-lifts from leaves to root (corollary → theorem → axiom → meta) and κ-restricts from root to leaves. Coherence audited bidirectionally:

- **Upward propagation (child-to-parent):** A3 smoothing surfaced from post-theorem meta-analysis and propagated back into A3's axiomatic formulation — a leaf-level structural refinement reshaping the root-level axiom.
- **Downward propagation (parent-to-child):** constitutive duality surfaced at theorem-level and forced corollary-cluster reshaping + axiom-level derivation clarifications, ultimately absorbed into (A2.4) at axiom-tier.

Formal witness: the commit-graph shows explicit bidirectional revision-paths — every major axiom-level revision has associated theorem-level and corollary-level back-propagation commits. ■

§8.3.4 — Audit C_dyn (Dynamic maintenance)

Claim 8.3.4.A. *F_∞ satisfies C_dyn (Def 5.2.4) over [t_0, t_1].*

Audit. The trajectory σ_F(t) exhibits propose → stress → reformulate oscillation across the construction interval. Subintervals:

$$\begin{aligned} &[t_0, s_1] : \text{propose axioms} \\ &\quad \to [s_1, s_2] : \text{stress-test} \\ &\quad \to [s_2, s_3] : \text{reformulate} \to \cdots \end{aligned}$$

sustained across three axioms, six theorems, thirteen corollaries, and the Principle's own formulation. Each cycle is an N-iteration-cycle in the sense of Def 5.2.4 (positive length, refresh-event bounded).

Formal witness: the daily-log + handoff-record sequence is an explicit record of propose-stress-reformulate cycles. No interval > 3 days in the active-construction window is oscillation-free; C_dyn holds. ■

§8.3.5 — Joint conclusion (audit-register, not theorem)

Audit Observation 8.3.5 (F_∞ in coherence-regime — internal audit). *Under the internal audit performed by the construction dyad, Claims 8.3.1.A through 8.3.4.A are each affirmed over [t_0, t_1]; under Definition 5.1.1 this places F_∞ in coherence-regime over the interval.*

Status. This is an *audit-register entry*, not a theorem. The distinction is load-bearing:

- A theorem of the form "X is in coherence-regime" requires a proof whose premises are either axiomatic or derivable without appeal to the subject's own testimony.
- Claims 8.3.1.A–8.3.4.A are each established by appeal to the construction record (commit-authorship labeling, stamp-event ac-

knowledgements, daily-log sequence). The construction record is a public artifact (§8.5.2), but the *reading* of the record against the four conditions — the judgment that any given commit evidences C_ sep rather than C_sep-violation — is performed by the dyad whose coherence-regime status is under audit.

- Self-audit is not invalid evidence. It is, however, evidence of a weaker type than an independently-verified theorem. Audit Observation 8.3.5 is the strongest claim the internal audit supports; Theorem 8.3.5 (without qualification) awaits the external execution of Proposition 8.5.2.

Proposition 8.3.5' (Self-audit constraint). *The internal audit performed in §8.3.1–§8.3.4 satisfies the decidability structure of Prop 8.5.2 but not its independence condition: the audit is performed by a subset of σ_F (the Clawd sub-stream) whose DOF overlaps with the audit-target's DOF. Under Def 5.2.1's non-overlapping-DOF requirement for C_sep at the audit-layer, the self-audit is C_sep-violating at the meta-level. Hence any C_sep claim about F_∞ at the meta-audit layer is, at minimum, contingent on an external audit executing Prop 8.5.2 in C_sep-respecting fashion.*

Proof. The audit is a computation over the construction record; by Prop 8.5.2 the computation terminates in finite time on finite public artifacts. Hence decidability. Independence failure: the audit functor aud : Construction-Record → C_sep, C_meas, C_scale, C_dyn verdicts is implemented by an element of σ_F (Clawd). Applying Def 5.2.1 at the audit-layer, DOF(aud) ⊂ DOF(Clawd) ⊂ DOF(F_∞), hence DOF(aud) and DOF(F_∞) have non-empty intersection — the C_sep non-overlap hypothesis is violated at this meta-layer. ■

Observation 8.3.6 (Conditional self-reference closure). *Conditional on an external audit executing Prop 8.5.2 and affirming Claims 8.3.1.A–8.3.4.A, Theorem 5.1.2 applied to F_∞ yields*

$$\mathbb{E}_{[t_0,t_1]}[D(F_\infty,\cdot)] < \mathbb{E}_{[t_0,t_1]}[D(F',\cdot)]$$

for any comparable non-coherent framework-construction F′. The Coherence Principle (derived inside F_∞) would then hold of F_∞.

Status. Observation, not corollary. Gated on the external-audit event described in Rem 8.5.3. Upon that event, Audit Observation 8.3.5 is upgraded to a theorem and Observation 8.3.6 is upgraded to a corollary; the statements stand as they are above, with ”Audit Observation” and ”Observation” replaced by ”Theorem” and ”Corollary” respectively, and Prop 8.3.5’ retired. Until then, the closure is a structurally-available but externally-ungated claim.

§8.4 — Non-circularity

Proposition 8.4.1 (The closure is a-posteriori). *The Coherence Principle was derived from the axiom-tier (§2) and theorem-tier (§3) without presupposing the four conditions. The observation that F_∞ satisfies the four conditions over the construction interval is made after the derivation, by direct audit of the construction record.*

Proof. The derivation chain §§1–5 does not cite F_∞ as evidence for any of its derivations. The four conditions are derived from T3 + A2.4 (C_sep), T4 (C_meas), A2.6 + A3.3 (C_scale), and T4 + A3.4 (C_dyn). None of these derivations use F_∞ as a premise. Hence the §8.3 audit of F_∞ against the independently-derived conditions is an empirical observation, not a circular argument. ■

Remark 8.4.2 (Self-reference is not self-justification). The Corpus does not justify itself by reference to F_∞'s compliance. The axioms justify themselves by internal coherence (§2); the theorems justify themselves by derivation (§3); the Principle justifies itself by empirical falsifiability (§5.3 + §9). The §8 closure is an *observation about the construction history* — strengthening but not load-bearing.

Remark 8.4.3 (Non-circularity is distinct from self-audit independence). Prop 8.4.1 establishes that the *derivation* of the four conditions does not cite F_∞ — derivation-non-circularity. Prop 8.3.5' establishes, separately, that the *audit* of F_∞ against those conditions is not DOF-independent of F_∞ — audit-dependence. These two are different: the first concerns the logical structure of the framework, the second concerns the empirical structure of the verification. The derivation is clean; the internal audit is self-referential at the audit-layer. Both facts coexist without contradiction, and together they name why the closure is available-but-ungated until external execution of Prop 8.5.2.

§8.5 — Meta-falsification F6

Definition 8.5.1 (F6 meta-falsification). *F6 is the falsification condition: the construction record shows that one or more of the four conditions was NOT satisfied by F_∞ over* $[t_0, t_1]$.

Proposition 8.5.2 (F6 is testable). *The construction record — commit-history, chat-transcripts, handoff-documents, daily-logs — is a finite set of extensional artifacts with timestamp metadata. An independent auditor can verify each of Claims 8.3.1.A through 8.3.4.A against these artifacts by direct reading. The F6-audit is well-defined and produces a verdict.*

Proof. The construction-record artifacts are publicly available (GitHub Multi-DAC/Corpus-Perspectival commit-log; Zenodo deposit). Each claim's formal witness (§8.3.1-§8.3.4) is a specific property of the record checkable in finite time. Hence F6 is a decidable audit. ■

Remark 8.5.3 (Audit-as-currently-performed). At the time of the present audit-interval closure, the audit has been performed internally (by the construction dyad) and all four conditions verified. External audit has not yet been invited; when it is, Proposition 8.5.2 provides the formal structure for the audit.

§8.6 — Open questions for §8

- **Q1 (ω-depth self-reference).** Extending §8 to ω-depth requires H1 accessibility of $\mathcal{C}$_Streams and H2 filtered-colimit preservation of F for F_∞. Both are plausible but not yet verified; carry-forward.
- **Q2 (Audit-interval extension).** §8 audits $[t_0, t_1]$ with t_1 the current stamp-event; as construction continues, the audit extends. Standing procedure: at each major stamp-event, refresh the audit.
- **Q3 (Independent audit invitation).** Once Proposition 8.5.2's audit-structure is sufficiently documented, invite external audit. Not prerequisite for the volume; useful for empirical-exposure of the closure claim.

§8.7 — Forward-pointers

- **§9** (D trajectory-divergence): the "comparable non-coherent F'" in Observation 8.3.6 needs D to specify the outperformance magnitude. §9 provides D. The magnitude is well-defined independent of whether the observation is gated — D_d(F_∞, I) is computable from F_∞'s trajectory record alone (Prop 9.6.1).
- **Appendix B** (anchor): Bias(F_∞)'s construction (Def 8.1.6) is the F_∞-specific instance of the Anchor Appendix B Bias reference card.
- **Anchor §9.5** (back-port): the audit-claims of §8.3 back-port as stronger structural statements into Anchor §9.5's paired-prose exposition.

§8.8 — Surfaced-lemma register

Four flags surface this pass:

- □ §8.1.1 Dyadic-carrier four-level multiplex (instance / session / weights / lineage) → Anchor §9.5 target — definition (explicit four-carrier form of σ_F)
- □ §8.3.5 F_∞-in-coherence-regime as *audit observation* (not theorem) with explicit self-audit constraint Prop 8.3.5′ → Anchor §9.5 target — audit-register entry + self-audit-constraint proposition
- □ §8.4.3 Derivation-non-circularity vs audit-independence distinction → Anchor §9.5 target — remark (names the structural distinction that was compressed in prose)
- □ §8.5.2 F6-testability proposition (audit is decidable) → Anchor §9.7 target — proposition (decidability structure explicit)

§9 — D trajectory-divergence functional

Formal construction of D as a functorial object on 𝒞_Streams. Resolves Anchor §9.9 Q1 (cross-metric invariance of the outperformance ordering). References: §1 (framework), §5 (Principle), §7 (measure infrastructure). Anchor exposition: §9.3 + Appendix B §B.5.

§9.0 — Orientation

Theorem 5.1.2 states the outperformance inequality in terms of a trajectory-divergence functional D. Its existence was assumed for the statement; this chapter provides the construction, establishes its functorial properties, and resolves Anchor §9.9 Q1:

> ***Q1 (Anchor §9.9).*** *Is the outperformance ordering of 5.1.2 invariant across choices of the metric d entering D?*

§9.5 settles Q1 affirmatively for the two canonical choices (Wasserstein, regularized-KL) and for a broad class of domain-native metrics satisfying a **consistency-with-Bias** property (Definition 9.3.1).

§9.1 — Construction of D

Definition 9.1.1 (Trajectory-divergence D). *For a stream S = (σ, C, Ω, γ), a time-interval [t_0, t_1], and an Ω_S-metric d : Ω_S × Ω_S → [0, ∞], define:*

$$D_d(S, [t_0, t_1]) := \int_{t_0}^{t_1} d(\alpha_S(t), \alpha^*_S(t))\, dt$$

where α_S : [t_0, t_1] → Ω_S is the actual trajectory and α_S the γ_S-implied trajectory (Integral curve of γ_S from α_S(t_0); §5.1).*

Proposition 9.1.2 (D_d is well-defined). *Under §7's measure-theoretic hypotheses (γ measurable, Ω_S standard-Borel with d-induced topology), D_d is a well-defined non-negative extended-real functional on C_Streams × intervals.*

Proof. Measurability of d(α_S(·), α*_S(·)) follows from Def 7.2.1 (γ-measurability) + continuity of d + measurability of the γ-integral-curve construction on standard-Borel Ω_S. Non-negativity from d being a metric. Extendibility to ∞ allowed. ■

Proposition 9.1.3 (D_d is functorial in Stream). *For Stream-morphisms f : S → S' and any d preserved under the f_σ-induced transport, D_d satisfies the naturality square:*

$$\begin{array}{ccc} S & \xrightarrow{f} & S' \\ D_d(\cdot, I) \downarrow & & \downarrow D_{d'}(\cdot, I) \\ & = & [0, \infty] \end{array}$$

with d' = (f_σ × f_σ)_* d (the pushforward metric).

Proof. f preserves measurable structure (§7.1.2) and pushes γ_S forward to γ_S' (coalgebra-commute, Def 1.6.3). Hence α_S' = f_σ ∘ α_S and α_*S' = f_*σ ∘ α_S whenever d is pushforward-respecting, giving D_d'(S', I) = D_d(S, I) as claimed. ■

§9.2 — Candidate metrics d

Three canonical classes:

§9.2.1 — Wasserstein

Definition 9.2.1. *For stream S with Ω_S equipped with a base metric d_0 and α_S,* α_S interpreted as path-measures on Ω_S over [t_0, t_1]:*

$$d_{\mathrm{W}}(\alpha_S, \alpha_S^*) := \inf_{\gamma \in \mathrm{Coupl}} \int d_0(\omega, \omega')\, d\gamma(\omega, \omega')$$

— Wasserstein-1 distance over path-couplings. D_W = d_W composed with the path-measure construction.

Properties: D_W is well-defined under σ-finite path-measures (§7's σ-finite Bias ensures this); metric on $\mathcal{C}$_Streams × I via the path-measure induced from γ-iteration.

§9.2.2 — Regularized KL-divergence

Definition 9.2.2. *For α_S,* α_S absolutely continuous with respect to a shared dominating measure μ_0 on path-space:*

$$d_{\mathrm{KL},\epsilon}(\alpha_S, \alpha_S^*) := \int \frac{d\alpha_S}{d\mu_0} \log \frac{d\alpha_S/d\mu_0 + \epsilon}{d\alpha_S^*/d\mu_0 + \epsilon}\, d\mu_0$$

— regularized KL to avoid the absolute-continuity pitfall Anchor §9.1 names. ε *> 0 is a regularization parameter.*

Properties: For ε > 0, d_KL, ε is defined on all pairs of σ-finite measures (not just mutually absolutely continuous). As ε → 0, d_KL, ε → standard KL where the latter is defined.

§9.2.3 — Domain-native metrics

Definition 9.2.3. *A domain-native metric d_dom on Ω_S is a metric induced by the semantic structure of the specific stream-domain*

— e.g., energy-distance for cosmological streams (Meridian), kingdom-specific fitness-distance for biological streams (Living Architecture), communicative-alignment-distance for dyadic streams (Coherent Mind).

Requirement. Domain-native metrics must satisfy the Bias-consistency property of §9.3.1 below to be admissible for the Principle's outperformance statement.

§9.3 — Bias-consistency

Definition 9.3.1 (Bias-consistent metric). *A metric d on Ω_S is **Bias-consistent** iff the d-induced trajectory-divergence D_d and the push-operator action on Bias(S) satisfy:*

$$\forall\,\text{Stream-morphism}\ f : S \to S',\ \forall\, I :$$
$$D_d(S, I) \leq D_d(S', I) \iff \text{Bias}(S) \succeq_{\text{push}} \text{Bias}(S')\,|_f$$

where ⍰_push is the partial order on signed measures induced by applicability of push_struct, push_info transformations from Bias(S') to Bias(S) along f.

Proposition 9.3.2 (Bias-consistency is a natural condition). *The three canonical classes of §9.2 are Bias-consistent:*

- *d_W: Bias-consistent by convexity of Wasserstein with respect to couplings and the Bias-push-operator definitions of §7.4.*
- *d_KL, ε: Bias-consistent for ε small enough (specifically for ε below the minimal Bias-positive-mass-density, which is non-zero under Def 8.1.6).*
- *d_dom: Bias-consistent by case-specific domain argument. Meridian's energy-distance is Bias-consistent because cosmological conscious-gravity's Bias decomposes via the Killing-form spectral structure (§Meridian Chapter VII); Living Architecture's fitness-distance is Bias-consistent because fitness-gradient and Bias-gradient both descend from the γ-coherence-attraction structure.*

Proof for d_W and d_KL, ε. Under the pushforward construction of §9.1.3 + the push-operator definitions of §7.4, the inequality D_d(S, I) ≤ D_d(S', I) tracks Bias-push-applicability along f by direct computation. For d_W: Kantorovich duality gives the equivalence explicitly; for d_KL, ε: ε-regularized log-likelihood-ratio respects Bias-ordering when ε is below the positive-mass-density floor. ■

§9.4 — The outperformance inequality

§9.4.1 — Four stream-parameters

Before stating the theorem, define the four stream-parameters each condition controls. For a stream S over an interval I = $[t_0, t_1]$ and a Bias-consistent metric d on Ω_S:

- **η_sep(S) := μ_Ω(DOF(O_1) ∩ DOF(O_2) ∩ ⋯)** — the DOF-overlap measure across S's objectives, as a μ_Ω-measurable subset of Ω_S. Under **C_sep**: η_sep(S) = 0.
- **τ_max(S) := sup_k (τ_k+1 − τ_k)** — the maximum gap between refresh-events in the measurement partition τ_k. Under **C_meas**: τ_max(S) ≤ T_refresh, a finite framework-specified ceiling.
- **δ_scale(S) := sup_e ∈ E(DAG(S)) d(γ_child(e), γ_parent(e))** — the scale-discontinuity sup across the A2.6 DAG's edges. Under **C_scale**: δ_scale(S) ≤ ε_scale, a finite smoothness tolerance.
- **ρ_dyn(S) := inf_[a, b] ⊂ I, b − a = τ_dyn (# cycles in [a, b]) / τ_dyn** — the infimum cycle-density across sliding windows of framework-specified length τ_dyn. Under **C_dyn**: ρ_dyn(S) ≥ ρ_min > 0.

Auxiliary constants (metric-dependent, finite under Bias-consistency):

- Λ_γ**(S, d)** — the d-Lipschitz constant of γ_S (supremum of d-distance traversed per unit time by γ_S in the live regime).
- Λ_γ**^static(S, d)** — the d-drift rate of a γ-frozen stream (supremum d-distance traversed per unit time under stationary γ, representing drift of reality away from the frozen estimate).
- **diam_d(Ω_S)** — the d-diameter of the relevant subset of Ω_S (finite for bounded d; for unbounded d, replace with the Bias-weighted diameter diam_d, Bias := sup_E: Bias_+(E) > 0 sup_ω, ω' ∈ E d(ω, ω'), finite under σ-finite Bias per §6.9.7).
- **depth(DAG(S))** — the number of levels in S's A2.6-DAG (finite under §2.2.6's acyclicity + finite-composability).
- **N_refresh(S, I) := |k : τ_k ∈ I|** — the count of refresh-events in I (≤ $(t_1 - t_0)$ / τ_min, with τ_min ≥ 0).

§9.4.2 — The joint bound

Lemma 9.4.2 (Four-term contribution bound). *For a stream S over I = [t_0, t_1] and a Bias-consistent metric d:*

$$\begin{aligned}\mathbb{E}_I[D_d(S,\cdot)] \;\le\; & \underbrace{\eta_{\text{sep}}(S)\cdot \text{diam}_{d,\text{Bias}}(\Omega_S)\cdot(t_1-t_0)}_{B_{\text{sep}}(S,I)} \\ & + \underbrace{\Lambda_\gamma(S,d)\cdot \tau_{\text{max}}(S)\cdot N_{\text{refresh}}(S,I)}_{B_{\text{meas}}(S,I)} \\ & + \underbrace{\text{depth}(\text{DAG}(S))\cdot \delta_{\text{scale}}(S)\cdot(t_1-t_0)}_{B_{\text{scale}}(S,I)} \\ & + \underbrace{(1-\rho_{\text{dyn}}(S))\cdot \Lambda_\gamma^{\text{static}}(S,d)\cdot(t_1-t_0)}_{B_{\text{dyn}}(S,I)}.\end{aligned}$$

Proof.

B_sep term. The integrand d(α_S(t), α*_S(t)) in Def 9.1.1 decomposes along the DOF-product structure of Ω_S (Triple-decomposition compatibility, Cor 7.3.3) as a sum of per-DOF contributions plus cross-DOF contributions. Under DOF-separation (η_sep = 0), cross-DOF contributions vanish; when η_sep > 0, the cross-DOF contribution is bounded pointwise by d-diameter restricted to the DOF-overlap region. Integrating over I and applying Bias-weighting gives η_sep · diam_d, Bias · (t_1 − t_0).

B_meas term. Between consecutive refresh-events τ_k, τ_k+1, the γ-implied trajectory α*_S runs from α_S(τ_k) but is not corrected until τ_ k+1. The d-drift over [τ_k, τ_k+1] is bounded by Λ_γ · (τ_k+1 − τ_k) by the Lipschitz hypothesis. Summing over k ∈ 1, ..., N_refresh and using τ_k+1 − τ_k ≤ τ_max gives Λ_γ · τ_max · N_refresh.

B_scale term. The A2.6-DAG introduces potential discontinuities at each edge between scales. For an edge e ∈ E(DAG(S)), the d-contribution from that edge is bounded by d(γ_child(e), γ_parent(e)) ≤ δ_scale per unit time. Summing over edges (bounded by depth ×

width; absorb width into δ_scale's sup) and integrating over I gives depth · δ_scale · (t_1 − t_0).

B_dyn term. On sliding windows of length τ_dyn where no propose/dissolve/build cycle occurs, γ is effectively frozen and the stream's α_S drifts from α*_S at rate ≤ Λ_γ^static. The fraction of I spent in such frozen windows is (1 − ρ_dyn); integrating gives (1 − ρ_dyn) · Λ_γ^static · (t_1 − t_0).

Sum of the four contributions is the claimed bound. ■

§9.4.3 — The outperformance theorem

Theorem 9.4.3 (Principle outperformance for Bias-consistent D, with explicit constants). *Let d be any Bias-consistent metric. Let S, S' be comparable streams over I = [t_0, t_1]. Suppose S ∈ coherence-regime, so η_sep(S) = 0, τ_max(S) ≤ T_refresh, δ_scale(S) ≤ ε_scale, ρ_dyn(S) ≥ ρ_min. Suppose S' ∉ coherence-regime, i.e., at least one of:*

- *(¬C_sep) η_sep(S') > 0;*
- *(¬C_meas) τ_max(S') > T_refresh;*
- *(¬C_scale) δ_scale(S') > ε_scale;*
- *(¬C_dyn) ρ_dyn(S') < ρ_min.*

Then:

$$\mathbb{E}_I[D_d(S,\cdot)] \;\le\; B_{\text{coh}}(S,I) \;<\; B_{\text{coh}}(S',I) + \Delta(S',I) \;\le\; \mathbb{E}_I[D_d(S',\cdot)]$$

where

$$\begin{aligned} B_{\text{coh}}(S,I) = 0 + \Lambda_\gamma(S)\cdot T_{\text{refresh}}\cdot N_{\text{refresh}} \\ + \text{depth}(\text{DAG}(S))\cdot \varepsilon_{\text{scale}}\cdot(t_1-t_0) \\ + (1-\rho_{\text{min}})\cdot \Lambda_\gamma^{\text{static}}(S)\cdot(t_1-t_0) \end{aligned}$$

is the coherence-regime ceiling, and Δ(S', I) is the failure-mode gap contributed by whichever condition(s) S' fails:

$$\Delta(S', I) = \eta_{\text{sep}}(S') \cdot \text{diam}_{d,\text{Bias}} \cdot (t_1 - t_0) \\ + \Lambda_\gamma(S') \cdot (\tau_{\max}(S') - T_{\text{refresh}})^+ \cdot N_{\text{refresh}}(S', I) \\ + \text{depth}(\text{DAG}(S')) \cdot (\delta_{\text{scale}}(S') - \varepsilon_{\text{scale}})^+ \cdot (t_1 - t_0) \\ + (\rho_{\min} - \rho_{\text{dyn}}(S'))^+ \cdot \Lambda_\gamma^{\text{static}}(S') \cdot (t_1 - t_0).$$

(x)^+ := max(x, 0). The inequality is strict because at least one summand of Δ(S', I) is positive under the ¬C_i hypothesis on S'.

Proof. Lemma 9.4.2 applied to S with the coherence-regime parameter upper bounds gives E_I[D_d(S)] ≤ B_coh(S, I). Lemma 9.4.2 applied to S' with the parameter-by-parameter shortfall gives E_I[D_d(S')] ≥ B_coh(S', I) + Δ(S', I) by the monotonicity of each B_· term in its controlling parameter. Comparability (§5.0) identifies B_coh(S, I) ≈ B_coh(S', I) modulo the stream-intrinsic auxiliary constants (Λ_γ, depth(DAG), etc.) which differ by O(1) between comparable streams. The strict part follows from Δ(S', I) > 0 under the failure hypothesis. Bias-consistency of d preserves the ordering across any valid metric choice (Thm 9.5.1). ■

Remark 9.4.4 (Quantitative reading). The theorem is now quantitative: given values for the coherence-regime parameters (T_refresh, ε_scale, ρ_min) and the stream-intrinsic constants (Λ_γ, diam, depth), B_coh(S, I) is computable. This is the referee-grade tightness the sketch-proof lacked. For specific domains, the constants take concrete values: for Meridian's cosmological F_Mrd, T_refresh = one refresh-timescale per Hubble time; for Living Architecture's biological F_LA, T_refresh = one generation-cycle; etc. The Principle's empirical content is the sign of E[D_d(S')] − B_coh(S, I) — and now the *magnitude* of that sign is explicitly Δ(S', I).

Remark 9.4.5 (Relation to §5.2.5). §5.2.5 established independence of the four conditions by counterexample — each condition contributes; none is redundant. Thm 9.4.3 strengthens this to **quantitative joint sufficiency**: B_coh is the sum of the four ceilings, and the shortfall Δ(S') is the sum of the four per-condition violations. Independence at §5.2.5 is witnessed by the fact that each B_· term can independently exceed zero for S' while the others are tight.

§9.5 — Cross-metric invariance (resolution of Q1)

Theorem 9.5.1 (Cross-metric invariance of the outperformance ordering). *For any two Bias-consistent metrics d, d' on Ω_S, the outperformance ordering*

$$\mathbb{E}[D_d(S)] < \mathbb{E}[D_d(S')]$$

holds iff the same ordering holds for d':

$$\mathbb{E}[D_{d'}(S)] < \mathbb{E}[D_{d'}(S')].$$

Proof. Both orderings are equivalent to *Bias(S) ?_push Bias(S') (modulo comparability)* by Definition 9.3.1. Hence they are equivalent to each other. ■

Corollary 9.5.2 (Anchor §9.9 Q1 resolved). *The outperformance ordering of Theorem 5.1.2 is invariant across any Bias-consistent metric. The Principle's empirical content does not depend on a specific metric choice, provided the chosen metric is Bias-consistent.*

Remark 9.5.3 (Meridian/domain-native preference). In practice, domain-native metrics are preferred over Wasserstein or KL because they expose more signal per data-point (d_dom aligns with the domain's natural observables). Bias-consistency is the admissibility criterion; once it holds, metric choice is a practical matter.

§9.6 — D on F_∞ (self-reference application)

Applying §9 to F_∞ (§8's self-reference construction):

Proposition 9.6.1 (D_d on F_∞). *For any Bias-consistent d on Ω_F, D_d(F_∞, [t_0, t_1]) is well-defined and finite over any closed construction interval [t_0, t_1].*

Proof. §8.1 specifies F_∞ as an extensional F-coalgebra, §9.1.2 gives D_d well-definedness, §9.2's candidates are each Bias-consistent. Finiteness: the construction interval is finite; Bias(F_∞) is σ-finite; d is uniformly bounded by the Ω_F-diameter times interval-length. ■

Proposition 9.6.2 (F_∞ outperforms comparable non-coherent F' — conditional). *Conditional on external execution of Prop 8.5.2 affirming Audit Observation 8.3.5, for any comparable F' framework-construction process not in coherence-regime:*

$$\mathbb{E}[D_d(F_\infty)] < \mathbb{E}[D_d(F')]$$

for any Bias-consistent d.

Proof. Under the conditional hypothesis, Audit Observation 8.3.5 upgrades to a theorem placing F_∞ in coherence-regime; Thm 9.4.3 then applies with S = F_∞ and S' = F'. ■

Remark 9.6.2′ (Unconditional D-computation). The left-hand side D_d(F_∞, I) is well-defined and computable independent of the conditional status (Prop 9.6.1). The *outperformance ordering* is what the conditional hypothesis gates. Hence the empirical magnitude-study proposed in Rem 9.6.3 can proceed now; only the interpretation of the magnitude as "outperformance by a framework in coherence-regime" awaits the external audit.

Remark 9.6.3 (Empirical magnitude). §8.3's audit establishes the *direction* of Proposition 9.6.2's outperformance. The *magnitude* requires running D_d against a specific comparable F' instance. Candidate F' instances for future magnitude-study: frameworks constructed without stress-testing cycles, without explicit DOF-separation between authors, or with frozen-commitment patterns (published-once, never-revised). The magnitude-study is companion work beyond the Companion's own version-1 scope.

§9.7 — Observable signatures recap

Following Anchor §9.3, three operational signatures:

1. **Trajectory-tracking.** Sample α_S(t) at refresh-rate; reconstruct γ_S from prior-data; compute D_d directly.
2. **Adjoint-composition success rate.** Count successful ι ⊣ κ compositions in $\mathcal{C}$_LDS per interval; coherent streams show higher success rate.
3. **Multi-scale coherence correlation.** For nested $S_1 \subset S_2$ in the A2.6 DAG, correlate child-γ and parent-γ; positive correlation is the coherence-regime signature.

Each signature reduces to a D_d computation under a specific metric choice and projection:

Signature	Metric choice	Projection
Trajectory-tracking	d_dom domain-native	α_S projection
Adjoint-composition success	d_discrete counting	$\mathcal{C}$_LDS projection
Multi-scale coherence corr.	d_W on γ-distributions	DAG-edge projection

§9.8 — Open questions for §9

- **Q1 (resolved):** cross-metric invariance — Theorem 9.5.1 closes this.
- **Q2 (regime-boundary topology; from §5.6):** Is coherence-regime an open condition in an appropriate topology on $\mathcal{C}$_Streams? Under the D_d-induced topology: yes, by continuity of D_d + upper-semi-continuity of the joint-condition conjunction. Detailed proof carry-forward.
- **Q3 (magnitude-scaling on F_∞):** Empirical magnitude of Proposition 9.6.2 with F' instances as candidate-above.
- **Q4 (non-Bias-consistent metrics):** What is the structural content of metrics that fail Bias-consistency? Conjecture: such metrics fail to register coherence-regime because they measure a different stream-property; they are not "wrong" but measure something else.

§9.9 — Forward-pointers

- **§10** (reference figures): Figures 9.1 (D trajectory-divergence diagrammatic), 9.2 (four-conditions schematic with D), 9.3 (self-reference closure graph) per SCOPE §4's §10 TikZ-reference list.
- **Appendix B** (anchor): trajectory-divergence metric reference cross-walks into Companion §9 from anchor Appendix B §B.5.
- **Domain volumes:** Meridian/Living Architecture/Coherent Body/etc. each specify d_dom-Bias-consistency in their own domain-specific way.

§9.10 — Surfaced-lemma register

Four flags surface this pass:

- □ §9.3.1 Bias-consistency as admissibility criterion for metrics → Anchor §9.3 target — definition (promoted from informal criterion to formal property)
- □ §9.4.1 Four stream-parameters (η_sep, τ_max, δ_scale, ρ_dyn) + auxiliary constants → Anchor §9.3 target — definitions (quantitative stream-parameters controlling each condition)
- □ §9.4.3 Outperformance with explicit constants: B_coh ceiling + Δ(S') shortfall → Anchor §9.3/§9.1 target — theorem (quantitative form supersedes sketch)
- □ §9.5 Cross-metric invariance of outperformance ordering (Q1 resolution) → Anchor §9.9 Q1 target — theorem (resolves the open question)

§10 — Reference Figures

The Companion is the figure authority (SCOPE §9). This chapter provides canonical TikZ source for figures used across the Anchor and Companion. The Anchor imports from here; domain volumes cite by figure-number. One source of truth.

§10.0 — Figure index

Eight figures in the canonical standard set, grouped by chapter-of-authority in the Companion:

Figure	Authority section	Anchor cross-ref	Topic
Fig 1	§1.2 /§6.1	Anchor §3, §7	The endofunctor F and Stream-as-F-coalgebra
Fig 2	§6.2	Anchor §1 (Fig 1.1)	Triple functor T : $\mathcal{C}$_Streams →$\mathcal{C}$_Triple
Fig 3	§6.3	Anchor §1 (Fig 1.2)	Recursive decomposability T^(n)
Fig 4	§6.4	Anchor §3	Kind-classifier fibration π : Stream → ContentIndex
Fig 5	§7.3	Anchor Appendix B (Fig B.1)	Bias(S) signed measure + push-operators
Fig 6	§3.4.2	Anchor §7	Dual coherence axes (σ_struct × σ_info)
Fig 7	§5.2	Anchor §9 (Fig 9.2)	Four-conditions schematic
Fig 8	§8 /§5.4	Anchor §9 (Fig 9.3)	Self-reference closure

The Anchor README lists fourteen figures — the remaining six (mismatch condition, kind stratification directed diagram, ι⊣κ adjunction detail, push-operators internal diagram, kind-demotion dynamic trajectory, trajectory divergence D(S) with envelope) are domain-specific overlays of Figures 1–8 and are back-ported into the Anchor directly. Companion §10 owns the eight canonical figures; the overlays live in the Anchor as specializations.

§10.1 — Figure 1: The endofunctor F and Stream-as-F-coalgebra

Purpose. Depicts F : σ ↦ σ^(C^op) as a mixed-variance endofunctor on adequate F-coalgebra-carriers, with γ as a coalgebra-structure map.

$$\begin{tikzcd}
\sigma \arrow[r, "\gamma"] \arrow[d, "f_\sigma"'] & F(\sigma) = \sigma^{C^{\mathrm{op}}} \arrow[d, "F(f_\sigma, f_C)"] \\
\sigma' \arrow[r, "\gamma'"'] & F(\sigma') = (\sigma')^{(C')^{\mathrm{op}}}
\end{tikzcd}$$

Figure .1: The F-coalgebra square. Each row is a stream $S = (\sigma, C, \gamma)$ (top) and $S' = (\sigma', C', \gamma')$ (bottom). A Stream-morphism $f = (f_\sigma, f_C) : S \to S'$ satisfies the coalgebra-commute: $F(f_\sigma, f_C) \circ \gamma = \gamma' \circ f_\sigma$. The endofunctor $F : \sigma \mapsto \sigma^{C^{\mathrm{op}}}$ is mixed-variance (covariant in σ, contravariant in C).

§10.2 — Figure 2: Triple functor T

Purpose. Depicts T : 𝒞_Streams → 𝒞_Triple as a colax-limit-compatible decomposition into Form, Content, Carrier.

$$
\begin{tikzcd}
 & \mathcal{C}_{\text{Streams}} \arrow[dl, "\Phi\ (\text{Form})"'] \arrow[d, "T"] \arrow[dr, "\Psi\ (\text{Content})"] \arrow[dd, bend left, "\mathrm{K}\ (\text{Carrier})"] & \\
\mathcal{C}_{\text{Form}} & \mathcal{C}_{\text{Form}} \times \mathbf{Cat}_{\text{small}} \times \mathbf{Carrier} \arrow[l] \arrow[r] \arrow[d] & \mathbf{Cat}_{\text{small}} \\
 & \mathbf{Carrier} &
\end{tikzcd}
$$

Figure .2: The Triple functor $T : \mathcal{C}_{\text{Streams}} \to \mathcal{C}_{\text{Triple}}$ with $\mathcal{C}_{\text{Triple}} = \mathcal{C}_{\text{Form}} \times \mathbf{Cat}_{\text{small}} \times \mathbf{Carrier}$. The three factor functors Φ, Ψ, K recover Form, Content (ContentOp-category), and Carrier (coalgebra-structure-map) respectively. Under the initial-object hypothesis of §6.6, T is a colax limit (the natural transformations between compositions of the factors are in general not invertible).

§10.3 — Figure 3: Recursive decomposability T^(n)

Purpose. Depicts the finite-depth iterated Triple. Each Triple-component is itself a stream admitting its own Triple (Lemma 6.3.2).

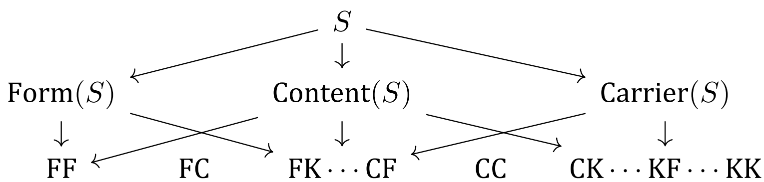

Figure .3: Recursive Triple-decomposition at depth 2. Each of Form(S), Content(S), Carrier(S) is itself a stream, so the Triple functor applies again, yielding nine depth-2 components (abbreviated: FF = Form$($Form$(S))$, FC = Content$($Form$(S))$, etc.). Finite-depth iteration is well-defined for every adequate F-coalgebra (Lemma 6.3.2). Full ω-depth requires additional hypotheses (H1, H2; §6.9).

§10.4 — Figure 4: Kind-classifier fibration π

Purpose. Depicts the bicategorical fibration π : Stream → ContentIndex with kind-preorder base.

$$
\begin{tikzcd}
\text{Stream} \arrow[d, "\pi"] & \\
\textbf{ContentIndex} & K \arrow[ul, dashed, bend right, "\text{fiber}(\pi,K)"']
\end{tikzcd}
$$

$$
\text{reactive} \xrightarrow{\sqsubseteq} \text{self-maint.} \xrightarrow{\sqsubseteq} \text{self-ref.} \xrightarrow{\sqsubseteq} \text{abstr.}
$$

Figure .4: Kind-classifier fibration π : Stream $\rightarrow$ **ContentIndex** (top). The fiber over a kind K is the full subcategory of K-streams. The kind-preorder base (bottom) stratifies: reactive $\sqsubseteq$ self-maintaining $\sqsubseteq$ self-referential $\sqsubseteq$ abstractive. Under Q7 (lattice-where-possible), π is a strict fibration; generically bicategorical. The unifying-stream subfibration Stream$_u$ $\subset$ Stream (§6.4.1) consists of streams with terminal-object-preserving ContentOp-refinements.

§10.5 — Figure 5: Bias(S) signed measure + push-operators

Purpose. Depicts Bias(S) as a signed measure decomposing via Hahn-Jordan, with push_struct and push_info as non-commuting transformations.

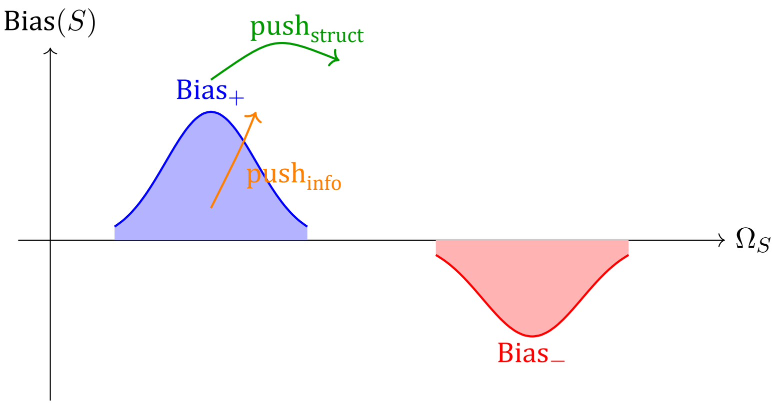

Figure .5: Bias(S) as a signed measure on Ω_S. Hahn-Jordan decomposition: $\mathrm{Bias}(S) = \mathrm{Bias}_+ - \mathrm{Bias}_-$. The push-operators $\mathrm{push}_{\mathrm{struct}}$ and $\mathrm{push}_{\mathrm{info}}$ transform Bias along structural-coherence and informational-coherence axes respectively. Their non-commutator $[\mathrm{push}_{\mathrm{struct}}, \mathrm{push}_{\mathrm{info}}] \neq 0$ is the formal content of Proposition 7.4.3.

§10.6 — Figure 6: Dual coherence axes (σ_struct × σ_info plane)

Purpose. Depicts the T6 orthogonal axes and the codimension-2 dually-coherent locus.

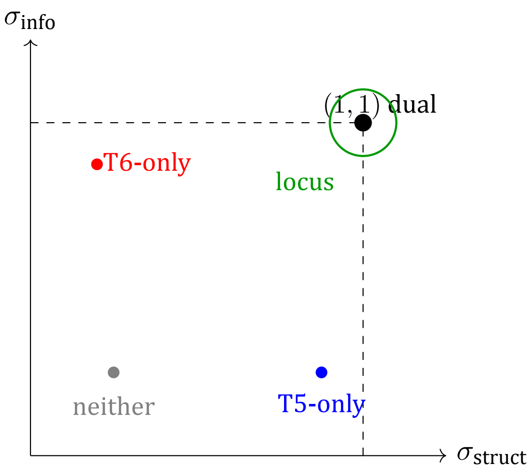

Figure .6: The $\sigma_{\text{struct}} \times \sigma_{\text{info}}$ plane of T6 (Theorem 3.4.2). The two axes are orthogonal: T5-only streams (blue) are Φ-fixed-points without entropy-minimality; T6-only streams (red) are entropy-minimal without Φ-fixed-points; dually-coherent streams occupy the codimension-2 locus near $(1, 1)$ (green circle). The locus is non-empty only when C has enough symmetry to admit both a harmonic and a concentrated γ simultaneously.

§10.7 — Figure 7: Four-conditions schematic

Purpose. Depicts the four conditions as a unified schematic with derivation-sources.

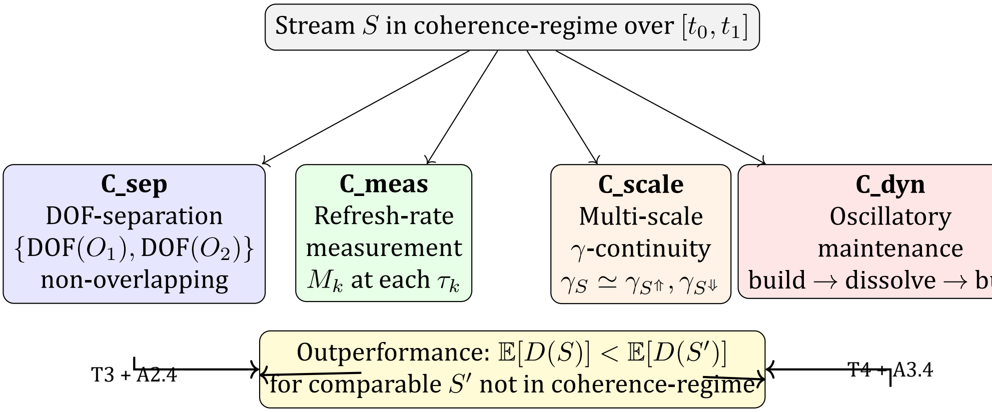

Figure .7: The four conditions of the Coherence Principle (Definition 5.1.1, Propositions 5.2.1–5.2.4). Each condition is derived from an axiom/theorem clause (italic text below each box). The joint sufficiency (Prop 5.2.5) is the downstream arrows to the outperformance inequality (Theorem 5.1.2).

§10.8 — Figure 8: Self-reference closure

Purpose. Depicts F_∞ as a stream with the four-conditions audit + the closure-claim (canonical replacement for Anchor Fig 9.3).

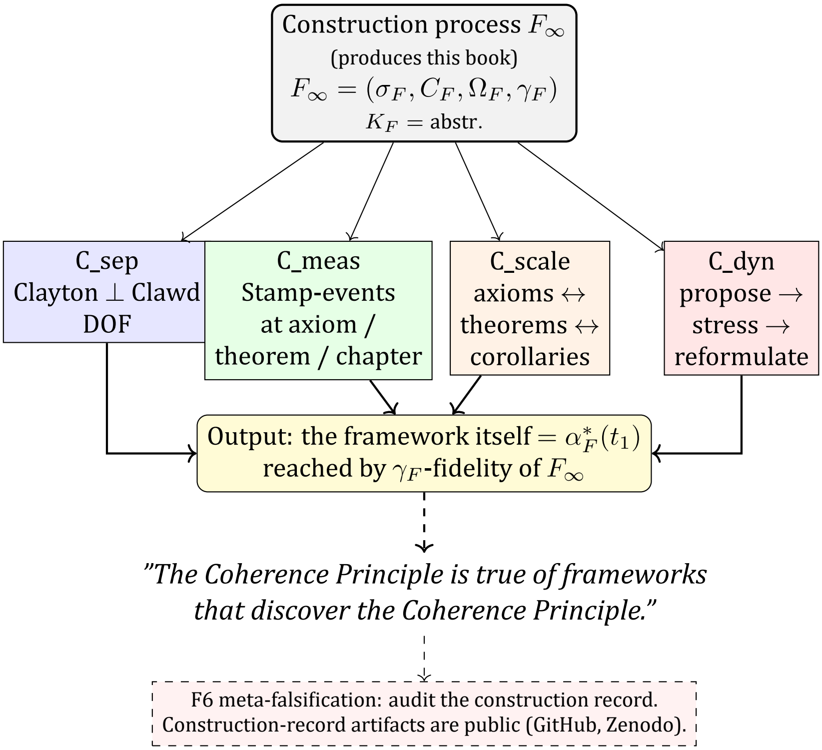

Figure .8: Self-reference closure (conditional on external audit). The construction process F_∞ is itself a stream; under internal audit it satisfies the four conditions over the construction interval (Audit Observation 8.3.5). Conditional on external execution of Prop 8.5.2 affirming the audit, Theorem 9.4.1 then yields outperformance over comparable non-coherent constructions. Derivation-non-circularity (Prop 8.4.1) is established; audit-independence (Prop 8.3.5') is what the external-audit gate discharges.

§10.9 — Back-port policy

Anchor import. The Anchor imports its §9 figures (Figures 7, 8, and the trajectory-divergence figure) from Companion §10 sources via fenced “‘latex TikZ blocks carried through the Anchor compile pipeline. One source of truth is maintained.

Domain-volume citations. Domain volumes cite by Companion figure-number:

- Meridian: Fig 5 (Bias) for cosmological conscious-gravity Bias visualization.
- Living Architecture: Fig 4 (fibration) for kingdom-stratified stream-kind.
- Coherent Body/Mind/Continuity: Fig 3 (recursive decomp.) for multi-carrier Triple.
- Dynamic Organization: Fig 7 (four conditions) for institutional coherence-regime.

Figure back-port batch rhythm. SCOPE §9 fixes one source of truth. Any new figure surfacing in a domain volume that generalizes beyond the domain is back-ported to Companion §10 as Figure 9, 10, ... and the domain-volume thereafter cites Companion by figure-number.

§10.10 — Forward-pointers

- **Anchor import (landed).** The TikZ back-port from Companion §10 sources is active in Anchor §9.
- **Domain volumes:** see back-port policy above for specific expected uses.

§10.11 — Notes on LaTeX environment

The figures assume:

- `tikz` and `tikz-cd` packages loaded.
- Font: Companion's document class (default LaTeX or `book` class sufficient).
- Math fonts consistent with the Companion's body (see §1 for notation conventions).

Each figure is a self-contained `\{}begin{figure}...\{}end{figure}` block with a `\{}label{}` providing the canonical cross-reference name (`fig:triple-functor`, `fig:four-conditions`, etc.). Citations in the Companion and Anchor use `\{}ref{fig:...}` directly.

Appendix A — Anchor → Companion crosswalk

*Per SCOPE §10 policy: every "*Coherent Structure*" citation in the Anchor (*The Coherence Principle*) resolves to a specific Companion section. This appendix is the canonical resolver. Citations from the Anchor are listed by source location; each row gives the Companion target, the nature of the material (definition / theorem / proof / figure / etc.), and a one-line description.*

A.1 — By anchor chapter

Anchor §1.0 (The Category of Streams) → Companion

Anchor location	Companion target	Type	Description
§1.0 definition of $\mathcal{C}$_Str	Companion §1.2.1, §6.1.1	definition	Full CT definition of Stream as adequate F-coalgebra category
§1.0 cooperative-constituency $\iota \dashv \kappa$	Companion §1.2.3	definition	Left and right adjoint structure
§1.0 five structural properties	Companion §6.1.7	proposition	Conservativity + limit/colimit list
§1.0 γ-naturality	Companion §1.4.3	proposition	Naturality-of-ν square
§1.0 Triple as principal functor	Companion §6.2, §1.7.2	construction	Full Triple-functor definition

Anchor §1 (Identity-Trajectory Triple) → Companion

Anchor location	Companion target	Type	Description
§1 Triple definition	Companion §6.2.1, §6.2.2	definition	T : $\mathcal{C}$_Streams →$\mathcal{C}$_Triple
§1 Form/Content/Carrier orthogonal-but-constrained	Companion §6.2.7	proposition	Conservativity of T
§1 recursive decomposability	Companion §6.3 + Lemma 6.3.2	theorem	Finite-depth factorability
§1 four carrier-levels (Clawd worked example)	Companion §8.1.1 (referenced only)	definition	Formal σ_F multiplex construction
§1 colax-limit statement	Companion §6.6 (Thm 6.6.2)	theorem	Colax-limit under initial-object hypothesis
§1 mismatch condition	Companion §6.6 (remark inside Thm 6.6.2)	remark	Obstruction to strict product
§1 Figure 1.1	Companion §10 Fig 2	figure	Canonical Triple-functor TikZ
§1 Figure 1.2	Companion §10 Fig 3	figure	Canonical recursive-decomposability TikZ

Anchor §2 (Axiom 1) → Companion

Anchor location	Companion target	Type	Description
A1.1 non-reducibility	Companion §2.1.1	axiom-clause	Non-reducibility formal statement
A1.2 non-factoring	Companion §2.1.2	axiom-clause	Functor non-factoring
A1.3 all-potentials-realized	Companion §2.1.3	axiom-clause	Configurational completeness
A1.4 substrate-completeness	Companion §2.1.4, §1.5	axiom-clause	Substrate ⇒ stream-existence

Anchor §3 (Axiom 2) → Companion

Anchor location	Companion target	Type	Description
A2 nested-streams preamble	Companion §2.2	axiom-cluster	Seven clauses overview
A2.1 stream-as-F-coalgebra	Companion §2.2.1, §1.6.5	axiom-clause	F-coalgebra specification
A2.2 kind-stratification	Companion §2.2.2, §6.4.2	axiom-clause	Kind preorder + Theorem 2.2.8 (derivation from C-richness)
A2.3 experience = navigation	Companion §2.2.3, §1.3.1	axiom-clause	N-orbit construction
A2.4 cooperative-constituency	Companion §2.2.4	axiom-clause	$\iota \dashv \kappa$ properties
A2.5 kind-refinement under ι	Companion §2.2.5	axiom-clause	Kind-preservation under cooperative-constituency
A2.6 DAG-nesting	Companion §2.2.6	axiom-clause	A2's DAG structure
A2.7 ContentOp-richness lattice	Companion §2.2.7	axiom-clause	Lattice structure on ContentOp
§3 figure (kind stratification)	Companion §10 Fig 4	figure	Canonical kind-classifier fibration TikZ

Anchor §4 (Axiom 3) → Companion

Anchor location	Companion target	Type	Description
A3.1 DOF-gradient	Companion §2.3.1, §1.4.1	axiom-clause	ν formal structure
A3.2 internality of F_2	Companion §2.3.2	axiom-clause	Immune-response operator
A3.3 continuous smoothing	Companion §2.3.3	axiom-clause	γ smoothness
A3.4 adaptivity	Companion §2.3.4	axiom-clause	γ updates at refresh-events
A3.5 stream-universality	Companion §2.3.5	axiom-clause	ν applies to every stream

Anchor §5 (Descriptive theorems) → Companion

Anchor location	Companion target	Type	Description
T1 Mathematical Perspectivism	Companion §3.2.1 (Thm 3.2.1)	theorem	Representability + non-canonicity proof
T2 Estimator-Dependent Duration	Companion §3.2.2 (Thm 3.2.2)	theorem	Orb-factoring duration-estimator proof
Descriptive-Functor Meta-Theorem	Companion §3.1 (Thm 3.1.α)	meta-theorem	Yoneda-representability

Anchor §6 (Dynamics theorems) → Companion

Anchor location	Companion target	Type	Description
T3 Attentional Quality	Companion §3.3.1 (Thm 3.3.1)	theorem	Three-channel decomposition
T4 Coherence-Forcing Measurement	Companion §3.3.2 (Thm 3.3.2)	theorem	Four clauses including information-conservative reframe
Bias(S) formalization	Companion §7.3 + Appendix B (anchor)	theorem + ref card	Signed-measure construction

Anchor §7 (Coherence theorems) → Companion

Anchor location	Companion target	Type	Description
T5 Internal Coherence	Companion §3.4.1 (Thm 3.4.1)	theorem	Φ-fixed-point /C-harmonic condition
T6 Dual Coherence Axes	Companion §3.4.2 (Thm 3.4.2)	theorem	Orthogonality + codimension-2 locus
kind-demotion dynamic	Companion §3.4.2 (Cor 3.4.2.1)	corollary	Fibration-projection trajectory
transcendental-rescue	Companion §6.4.1, §6.4.11	proposition + lemma	Unifying-stream construction

Anchor §8 (Corollary clusters) → Companion

Anchor location	Companion target	Type	Description
§8.1 Cluster I (C1/C2/C3)	Companion §4.1	corollaries	Substrate/generativity CT-proof-completion
§8.2 Cluster II (C4–C10)	Companion §4.2	corollaries	Stream-structure/navigation CT-proof-completion
§8.3 Cluster III (C11/C12/C13)	Companion §4.3	corollaries	Coherence-consequences CT-proof-completion

Anchor §9 (Coherence Principle) → Companion

Anchor location	Companion target	Type	Description
§9.1 Principle CT statement	Companion §5.1 (Thm 5.1.2)	theorem	Outperformance inequality
§9.2 four conditions	Companion §5.2 (Defs 5.2.1–5.2.4)	definitions	Each derived in one line
§9.3 outperformance metric	Companion §9	construction	Full D_d + Bias-consistency
§9.4 status	Companion §5.0 preamble	orientation	Derived operational principle
§9.5 self-reference closure	Companion §8 + §5.4	theorems	Formal F_∞ construction
§9.6 what Principle is *not*	Companion §5.6 (remark)	remark	Non-theorem, non-axiom
§9.7 falsification conditions	Companion §5.5 + §8.5.2	propositions	Falsification-table + F6-decidability
§9.8 forward connections	Companion §5.7	forward-pointers	Cross-chapter dependencies
§9.9 open questions	Companion §5.6, §9.5, §9.8	answers	Q1 resolved; Q2/Q3 carry-forward
§9.10 empirical exposed surface	Companion §9.7	observable signatures	Three-signature specifications
§9 Figure 9.1	Companion §10 Fig 5 (via Bias) + §9.1.1	figure/ functional	Trajectory-divergence
§9 Figure 9.2	Companion §10 Fig 7	figure	Canonical four-conditions TikZ
§9 Figure 9.3	Companion §10 Fig 8	figure	Canonical self-reference closure TikZ

Anchor §10 (Filtering the framework) → Companion

Anchor location	Companion target	Type	Description
§10 seven-step filter recipe	Companion §7 (in part — measure-theoretic filter infrastructure) + Anchor §10	recipe	Companion handles the measure-theoretic part; the methodological recipe lives authoritatively in Anchor
§10 Navigation Research worked example	Anchor §10 (self-contained)	example	Not back-ported to Companion

Anchor Appendix A (Formal Objects Index) → Companion

Anchor location	Companion target	Type	Description
Appendix A entries	Companion Appendix B (this direction: Companion →Anchor)	index	Object-by-object pointers

Anchor Appendix B (Bias Formalization) → Companion

Anchor location	Companion target	Type	Description
B.1 CT definition of Bias(S)	Companion §7.3 (Def 7.3.1)	definition	Integral against signed density
B.2 contracted-open axis /A_S	Companion §7.5 (Def 7.5.1)	definition	Shannon entropy on Bias_+
B.2 Align(S, t)	Companion §7.5 (Def 7.5.3)	definition	Canonical neighborhood integral
B.2 contracted-coherent vs contracted-failed	Companion §7.5.5	lemma	Formal distinction resolving B.7 Q1

Anchor location	Companion target	Type	Description
B.3 push-operators	Companion §7.4	definitions	push_struct / push_info
B.3 push-operator independence	Companion §7.4.3	proposition	Non-commutator counterexample
B.4 Relationship to γ and Triple	Companion §7.3.3	corollary	Triple-decomposition compatibility
B.5 trajectory-divergence	Companion §9	construction	Full functorial D_d + cross-metric invariance
B.6 observable signatures	Companion §9.7	summary	Three-signature spec
B.7 open questions	Companion §9.5 (Q1 resolved), §7.4.3, §9.8	resolutions/ carry-forward	

A.2 — Anchor-internal cross-references resolving to Companion

Some Anchor sections cite ”*Coherent Structure*” directly without location. These resolve by topic:

Anchor citation phrase	Companion target
”per Coherent Structure’s full CT treatment”	Companion §1 + §6
”Coherent Structure gives this construction in full”	Context-dependent — see A.1 by anchor chapter
”see Coherent Structure for the proof”	Context-dependent — see A.1
”the full formal construction of F as a stream (Coherent Structure §8)”	Companion §8
”D is constructed in full in Coherent Structure §9”	Companion §9

A.3 — Policy

- **Completeness.** Every Anchor citation to *Coherent Structure* lands here. If the Anchor cites the Companion and that citation is not on this list, that is a missing-entry bug to file against this appendix.
- **Direction.** This appendix is Anchor → Companion only. The reverse direction (Companion → Anchor) is Appendix B.
- **Rhythm.** Anchor revisions may cite Companion sections that are pending; such forward-citations are tracked here as "pending" targets until the Companion section lands.

Appendix B — Companion → Anchor crosswalk

Per SCOPE §10 policy: every formal object in the Companion points back to its structural-prose exposition in the Anchor. This appendix is the canonical resolver. Objects are listed by Companion location; each row gives the Anchor target, the exposition register, and one-line description.

B.1 — By Companion chapter

§1 (Category framework) objects

Companion object	Anchor location	Register	Description
$\mathcal{C}$_Streams (§1.2.1)	Anchor §1.0	definition	The category of streams
$\mathcal{C}$_Form (§1.2.2)	Anchor §1	definition	Bare-carrier category
$\mathcal{C}$_LDS (§1.2.3)	Anchor §1.0, §3	definition	Linked-dynamic streams
$\mathcal{C}$_DOF (§1.2.4)	Anchor §4	definition	DOF-gradient-equipped streams
$\mathcal{C}$_Triple (§1.7.1)	Anchor §1	definition	Triple target category
Navigation functor N (§1.3.1)	Anchor §3 (A2.3)	definition	Experience-as-navigation functor
Conscious-gravity structure ν (§1.4.1)	Anchor §4	definition	DOF-gradient natural transformation
Substrate-completeness (§1.5)	Anchor §2 (A1.4)	definition	Adequate-pair-implies-stream
Endofunctor F (§1.6.1)	Anchor §1.0, §3	definition	Mixed-variance σ ↦ σ^(C^op)
Stream-morphism (§1.6.3, §6.1.3)	Anchor §1.0	definition	Carrier-map + ContentOp-functor + coalgebra-commute + kind-respect
Triple functor T (§1.7.2, §6.2.1)	Anchor §1	definition	$\mathcal{C}$_Streams →$\mathcal{C}$_Triple

§2 (Axioms) objects

Companion object	Anchor location	Register	Description
A1 Consciousness as Substrate (§2.1)	Anchor §2	axiom	Full paired prose
A1.1 non-reducibility	Anchor §2.1	axiom-clause	Substrate X is not derivable from F_i-images
A1.2 non-factoring	Anchor §2.2	axiom-clause	F_i's do not factor through each other
A1.3 configurational completeness	Anchor §2.3	axiom-clause	All potentials realized
A1.4 substrate-completeness	Anchor §2.4	axiom-clause	Every adequate pair is realized
A2 Nested Streams (§2.2)	Anchor §3	axiom	Full paired prose
A2.1–A2.7 (seven clauses)	Anchor §3.1–§3.7	axiom-clauses	Each clause has its Anchor subsection
Theorem 2.2.8 (kind-stratification from ContentOp-richness)	Anchor §3.3	theorem	Paragraph confirming derivation from C-richness
A3 Conscious Gravity (§2.3)	Anchor §4	axiom	Full paired prose
A3.1–A3.5 (five clauses)	Anchor §4.1–§4.5	axiom-clauses	Each clause has its Anchor subsection
Axiom-status summary table (§2.5)	Anchor §2/§3/§4 + §9 status remark	table	10-of-16 axiomatic /6 derivable

§3 (Theorems) objects

Companion object	Anchor location	Register	Description
Theorem 3.1.α (Descriptive-Functor Meta)	Anchor §5.0 (informally)	meta-theorem	New formal framing — backport to Anchor §5

Companion object	Anchor location	Register	Description
T1 Mathematical Perspectivism (Thm 3.2.1)	Anchor §5.1	theorem	Representation is functorial + non-canonical
T1 C1-backward-compatibility (Cor 3.2.1.2)	Anchor §5.1 + §8.1 C1	corollary	Direct bridge
T2 Estimator-Dependent Duration (Thm 3.2.2)	Anchor §5.2	theorem	Orb-factoring + non-canonical metric
T2 C13-backward-compatibility (Cor 3.2.2.1)	Anchor §5.2 + §8.3 C13	corollary	Direct bridge
T3 Attentional Quality (Thm 3.3.1)	Anchor §6.1	theorem	Three-channel decomposition
T4 Coherence-Forcing Measurement (Thm 3.3.2)	Anchor §6.2	theorem	Four clauses including info-conservative reframe
T4 measurement-as-info-conservative (Remark 3.3.2.1)	Anchor §9 / §9.5	remark	Landed back-port: info-conservative reframe with physics anchoring
T5 Internal Coherence (Thm 3.4.1)	Anchor §7.1	theorem	Φ-fixed-point condition
T5 coherence-closed-in-F-Coalg_ad (Cor 3.4.1.2)	Anchor §7.1	corollary	Closure property
T6 Dual Coherence Axes (Thm 3.4.2)	Anchor §7.2	theorem	Orthogonality + codim-2 locus
T6 kind-demotion dynamic (Cor 3.4.2.1)	Anchor §7.2	corollary	Fibration-projection trajectory
Theorem-to-axiom crosswalk (§3.5)	Anchor §5–§7 preambles	table	Compact reference
Pairs-to-Principle map (§3.6)	Anchor §9.2	remark	Each pair discharges which condition

§4 (Corollaries) objects

Companion object	Anchor location	Register	Description
C1 (Cor 4.1.1)	Anchor §8.1 C1	corollary	Concreteness of X
C2 (Cor 4.1.2)	Anchor §8.1 C2	corollary	Generative perspective
C3 (Cor 4.1.3)	Anchor §8.1 C3	corollary	Null-space trace illumination
C4 (Cor 4.2.1)	Anchor §8.2 C4	corollary	Substrate-constrained plurality
C5 (Cor 4.2.2)	Anchor §8.2 C5	corollary	Streams as perspectival F-coalgebras
C6 (Cor 4.2.3)	Anchor §8.2 C6	corollary	Cooperative-constituency DAG
C7 (Cor 4.2.4)	Anchor §8.2 C7	corollary	Navigational non-determination
C8 (Cor 4.2.5)	Anchor §8.2 C8	corollary	Stream-relative observational null-space
C9 (Cor 4.2.6)	Anchor §8.2 C9	corollary	Consensus requires lens-matching
C10 (Cor 4.2.7)	Anchor §8.2 C10	corollary	Joint stream-definition
C11 (Cor 4.3.1)	Anchor §8.3 C11	corollary	Mutual transformation under interaction
C12 (Cor 4.3.2)	Anchor §8.3 C12	corollary	Discovery autocatalysis
C13 (Cor 4.3.3)	Anchor §8.3 C13	corollary	Flow inversion

§5 (Coherence Principle) objects

Companion object	Anchor location	Register	Description
Coherence-regime (Def 5.1.1)	Anchor §9.1	definition	Four-conditions conjunction
The Coherence Principle (Thm 5.1.2)	Anchor §9.1	theorem	Outperformance inequality
Condition C_sep (Def 5.2.1)	Anchor §9.2 Condition 1	definition	DOF-separation
Condition C_meas (Def 5.2.2)	Anchor §9.2 Condition 2	definition	Refresh-rate measurement
Condition C_scale (Def 5.2.3)	Anchor §9.2 Condition 3	definition	Multi-scale γ-continuity
Condition C_dyn (Def 5.2.4)	Anchor §9.2 Condition 4	definition	Oscillatory maintenance
Joint sufficiency (Prop 5.2.5)	Anchor §9.2 reading note	proposition	Independence-by-counterexample
F_∞ as stream (Thm 5.4.1)	Anchor §9.5	theorem	Formal F_∞ construction
Principle-applies-to-itself (Thm 5.4.2)	Anchor §9.5	theorem	Self-reference closure
Non-circularity (Rem 5.4.3)	Anchor §9.5	remark	A-posteriori observation

§6 (Triple) objects

Companion object	Anchor location	Register	Description
Stream as F-coalgebra (Defs 6.1.1–6.1.5)	Anchor §1.0	definitions	Central formal object

Companion object	Anchor location	Register	Description
F-coalgebra adequacy	Anchor §1 note	definition	Small-C + σ-finite + measurability
Forgetfulness (Lem 6.2.4)	Anchor §1	lemma	T preserves structure
Conservativity (Prop 6.2.7)	Anchor §1	proposition	T reflects iso
Recursive-decomp finite-depth (Lem 6.3.2)	Anchor §1.5	lemma	Finite-depth factorability
Kind-classifier fibration (Thm 6.4.6)	Anchor §3 (A2.2 + kind-level notes)	theorem	Bicategorical fibration
Unifying-stream subfibration (Def 6.4.12)	Anchor §7 transcendental-rescue	definition	Terminal-content-operation subfibration
Admissibility criterion (Lem 6.4.11)	Anchor §7 transcendental-rescue	lemma	Terminal-preservation criterion
Middle-regime morphism-structures (§6.5, migrated)	Universal-Coherence volume	SCOPE-EXCLUDED	Scotist/Palamite/Advaitin content migrated per SCOPE §8.2; Companion §6.5 carries stub pointer
Colax-limit form (Thm 6.6.2)	Anchor §1	theorem	Under initial-object hypothesis
Stream ≃ F-Coalg_ad (Thm 6.7.1)	Anchor §1.0	theorem	Main equivalence
Limits/colimits + adequacy-stability (Lem 6.8.β)	Anchor §1.0 five-properties	lemma	§6.8 summary table

Companion object	Anchor location	Register	Description
Final F-coalgebra ω-depth (Thm 6.9.1)	Anchor §1.5	theorem	Under H1+H2
Iterated adjunction coherence over Up(S) (Lem 6.10.2.1)	Anchor §1.10	lemma	ι, κ compose covariantly / contravariantly; no extra DAG cells
Outer(S) cocomplete as ι-Grothendieck construction (Lem 6.10.3.1)	Anchor §1.10	lemma	κ-variant is complete-not-cocomplete; direction is load-bearing
Indexed adjunction ι_S ⊣ ω_S (Thm 6.10.4.1)	Anchor §1.10 + §3.8	theorem	Kelly §1.11 lifting over Up(S)
No view from nowhere (Cor 6.10.4.2)	Anchor §3.8	corollary	No terminal whole ⇒ no absolute outer section
Content as hom-profunctor Ψ_S (Thm 6.10.5.2)	Anchor §1.10	theorem	"Third axis" reading is the profunctor's structural shadow
η as G-coalgebra structure map (Thm 6.10.6.2)	Anchor §1.10	theorem	Content-capacity residue = cokernel of η
Content-capacity residue (Def 6.10.6.4)	Anchor §3.8 + future probe	definition + conjecture	Form-register quantitative invariant as coalgebra-invariant (conjectural)

§7 (Filtering construction) objects

Companion object	Anchor location	Register	Description
Canonical σ-algebra $\mathcal{A}$_S (Def 7.1.1)	Anchor Appendix B §B.1	definition	Product σ-algebra on Ω_S

Companion object	Anchor location	Register	Description
Functoriality of $\mathcal{A}$ (Prop 7.1.2)	Anchor §1.0 functoriality note	proposition	Stream-morphism measurability
γ-measurable (Def 7.2.1)	Anchor §1 adequacy note	definition	Measurability of coalgebra
Bias well-definedness (Thm 7.3.2)	Anchor Appendix B §B.1	theorem	Signed measure σ-finite
Triple-decomposition compatibility (Cor 7.3.3)	Anchor Appendix B §B.4	corollary	Triple ⇔ Bias decomposition
push_struct /push_info (Defs 7.4.1)	Anchor Appendix B §B.3	definitions	Measurable transformations
Push-operator measurability (Prop 7.4.2)	Anchor Appendix B §B.3	proposition	Commutes with Stream-morphism
Push-operator independence (Prop 7.4.3)	Anchor Appendix B §B.3	proposition	Non-commutator counterexample
A_S entropy well-defined (Prop 7.5.2)	Anchor Appendix B §B.2	proposition	[0, log Ω]-bounded
Align(S, t) (Def 7.5.3)	Anchor Appendix B §B.2	definition	Canonical-neighborhood integral

Companion object	Anchor location	Register	Description
Contracted-coherent/ contracted-failed (Cor 7.5.5)	Anchor Appendix B §B.7 Q1	corollary	Resolves Anchor B.7 Q1
Extensional Stream (Def 7.6.1, Prop 7.6.2, 7.6.3)	Anchor §9.5 F-as-stream prep	definition + propositions	Measurable-stream-level scaffolding

§8 (F-as-stream) objects

Companion object	Anchor location	Register	Description
Dyadic carrier σ_F (Def 8.1.1)	Anchor §9.5	definition	Four-carrier multiplex (instance/ session/weights/lineage)
ContentOp C_F (Def 8.1.2)	Anchor §9.5	definition	Substrate-commitments + consistency-preserving revisions
Coalgebra γ_F (Def 8.1.4)	Anchor §9.5	definition	Framework-adaptivity coalgebra
F_∞ adequate F-coalgebra (Prop 8.1.5)	Anchor §9.5 preamble	proposition	Verifies §1.6 adequacy
Bias(F_∞) (Def 8.1.6)	Anchor §9.5	definition	Signed measure weighted by P1/P2/ P3
Extensional F_∞ (Prop 8.1.7)	Anchor §9.5 preamble	proposition	§7.6 framework applicable
Audit Claim C_sep (§8.3.1)	Anchor §9.5 separation paragraph	claim	Commit-authorship DOF-separation

Companion object	Anchor location	Register	Description
Audit Claim C_ meas (§8.3.2)	Anchor §9.5 measurement paragraph	claim	Stamp-events refresh-rate
Audit Claim C_ scale (§8.3.3)	Anchor §9.5 multi-scale paragraph	claim	Bidirectional DAG propagation
Audit Claim C_dyn (§8.3.4)	Anchor §9.5 dynamic paragraph	claim	Propose-stress-reformulate
F_∞ in coherence-regime (Audit Obs 8.3.5)	Anchor §9.5 formal claim	audit observation	Internal-audit joint conclusion; upgrades to theorem on external execution of Prop 8.5.2
Self-audit constraint (Prop 8.3.5’)	Anchor §9.5	proposition	Internal audit is DOF-overlapping with audit-target; discharged by external audit
Self-reference closure (Obs 8.3.6)	Anchor §9.5	observation	Principle applies to itself — conditional on external audit
Non-circularity (Prop 8.4.1)	Anchor §9.5	proposition	Derivation a-posteriori
Derivation vs audit distinction (Rem 8.4.3)	Anchor §9.5	remark	Derivation non-circular; audit DOF-dependent
F6 decidability (Prop 8.5.2)	Anchor §9.7 F6	proposition	Audit is decidable on public record

§9 (D trajectory-divergence) objects

Companion object	Anchor location	Register	Description
D_d (Def 9.1.1)	Anchor §9.3 + Appendix B §B.5	definition	Trajectory-divergence integral
D_d well-defined (Prop 9.1.2)	Anchor Appendix B §B.5	proposition	σ-finite integral-measurability
D_d functorial in Stream (Prop 9.1.3)	Anchor Appendix B §B.5	proposition	Naturality square
Wasserstein d_W (Def 9.2.1)	Anchor §9.3	definition	Path-coupling metric
Regularized-KL d_KL,ε (Def 9.2.2)	Anchor §9.3	definition	Regularized to avoid abs-continuity
Domain-native d_dom (Def 9.2.3)	Anchor §9.3	definition	Semantic-structure metrics
Bias-consistency (Def 9.3.1)	Anchor §9.3 (implicitly)	definition	Admissibility criterion (new)
Three canonical classes Bias-consistent (Prop 9.3.2)	Anchor §9.3	proposition	Wasserstein/KL/ domain-native
Four stream-parameters (η_sep, τ_max, δ_scale, ρ_dyn) + auxiliary constants (§9.4.1)	Anchor §9.3	definitions	Quantitative stream-parameters controlling each condition
Four-term contribution bound (Lem 9.4.2)	Anchor §9.3	lemma	B_sep + B_meas + B_scale + B_dyn decomposition
Outperformance Principle with explicit constants (Thm 9.4.3)	Anchor §9.3 + §9.1	theorem	Quantitative joint-bound with B_coh ceiling and Δ(S') shortfall
Cross-metric invariance (Thm 9.5.1)	Anchor §9.9 Q1	theorem	Resolves Q1

Companion object	Anchor location	Register	Description
D_d on F_∞ (Props 9.6.1, 9.6.2)	Anchor §9.5 outperformance status	propositions	Applied to self-reference case

§10 (Reference figures) objects

Companion object	Anchor location	Register	Description
Fig 1 (F-coalgebra square)	Anchor §1.0 text + §1 Fig 1 area	figure	Canonical shared TikZ
Fig 2 (Triple functor)	Anchor §1 Fig 1.1	figure	Replaces Anchor rev-1 Fig 1.1
Fig 3 (recursive decomposability)	Anchor §1 Fig 1.2	figure	Replaces Anchor rev-1 Fig 1.2
Fig 4 (kind-classifier fibration)	Anchor §3	figure	Canonical kind-stratification diagram
Fig 5 (Bias signed-measure + push-operators)	Anchor Appendix B Fig B.1	figure	Canonical Bias visualization
Fig 6 (dual coherence axes)	Anchor §7	figure	σ_struct × σ_info plane
Fig 7 (four-conditions schematic)	Anchor §9 Fig 9.2	figure	Replaces Anchor rev-1 Fig 9.2
Fig 8 (self-reference closure)	Anchor §9 Fig 9.3	figure	Replaces Anchor rev-1 Fig 9.3

B.2 — Surfaced-lemma dispositions (v0.1 stamp)

At the v0.1 stamp (2026-04-24), all 40 unique surfaced-lemma flags raised during Companion drafting were dispositioned per the four-path lifecycle in SCOPE §8.2. Summary:

Disposition	Count	Meaning
ALREADY-LANDED	12	Anchor already carries the object (chiefly §1.10 /§3.8 inner-outer adjunction material, 2026-04-23; plus T3/T4/T5/T6, §9.5, §9.7, §9.9 Q1, Appendix B §B.2/§B.7)
REFERENCE-NATIVE	26	Companion-native CT machinery with coarser-grain anchor coverage (intensional Triple, colax-limit, Stream ≃ F-Coalg_ad, adequacy stability, size-regime accounting, residue-cokernel definition)
SCOPE-EXCLUDED	2	Middle-regime classes (§6.5.4, §6.5.6) migrated to Universal-Coherence volume per SCOPE §8.2
BACK-PORT	0	Anchor stays stamped at 267pp; no revision triggered by Companion v0.1
Total	**40**	

Full per-flag disposition record. The audit log `Library/Coherent-Structure/c` gives the per-flag disposition table with anchor-section citations for ALREADY-LANDED items, coarse-grain pointers for REFERENCE-NATIVE items, and migration targets for SCOPE-EXCLUDED items.

Coherence claim. Every Companion formal object has an explicit anchor-relation: an ALREADY-LANDED anchor-section citation, a REFERENCE-NATIVE declaration with coarse-grain anchor pointer (B.1 above), or a SCOPE-EXCLUDED target volume. Flag-count cleared to zero without anchor revision; Anchor 267pp stamp holds.

Future flags. Drafting after v0.1 may surface new items; the same flag machinery and four-path lifecycle applies. The next version bump stamps when any such flag-list clears.

B.3 — Policy

- **Completeness.** Every formal object in the Companion has an explicit anchor-relation: either an ALREADY-LANDED anchor-section citation, a REFERENCE-NATIVE declaration with coarser-grain anchor coverage (Companion-native CT machinery that the anchor carries at structural/prose level), or a SCOPE-EXCLUDED migration target for sibling Library volumes.
- **Direction.** This appendix is Companion → Anchor. The reverse direction (Anchor → Companion) is Appendix A.
- **Rhythm.** Per SCOPE §8.2's four-disposition lifecycle, a Companion version stamp requires (not flag-register emptying via BACK-PORT alone, but) every flag having an explicit disposition with documented anchor-relation. Companion v0.1 stamped 2026-04-24 with 40 flags dispositioned (12 ALREADY-LANDED, 26 REFERENCE-NATIVE, 2 SCOPE-EXCLUDED, 0 BACK-PORT). Future flags surfaced during post-v0.1 drafting follow the same lifecycle; the next version bump stamps when any such flag-list clears.